

Custom Compiler™ Symbolic Editor User Guide

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About This User Guide

This guide explains how to use the Custom Compiler Symbolic Editor tools to create integrated circuit designs.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

- Synopsys® Custom Compiler™

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .

Convention	Description
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN.</code>
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

About the Symbolic Editor

The Symbolic Editor helps you to quickly place devices into a layout for device-level floorplanning that meets DRC requirements.

This chapter covers the following topics:

- [Introduction to the Symbolic Editor](#)
- [Getting Started](#)
- [Understanding Basic Patterns](#)

Introduction to the Symbolic Editor

Use the Symbolic Editor with chaining mode to quickly place MOS devices and automatically chain or abut them. The source/drain areas of the devices are automatically shared. The Symbolic Editor in chaining mode provides a visual estimate of required interconnect.

Use the Symbolic Editor in matching mode to place MOS devices in symmetrical patterns based on their M-factors and a grid size you define, to optimize placement and support interdigitation. You select the MOS devices to place and enter a grid size.

The Symbolic Editor also provides analog layout patterns you can choose for the layout. You can create and save patterns for reuse.

Getting Started

The Symbolic Editor is available when SDL is enabled. You can use the following methods to enable the canvas with the Symbolic Editor rules.

- [Setting Rules and Constraints](#)
- [Enabling the Symbolic Editor](#)

Setting Rules and Constraints

You can invoke the Symbolic Editor with the following rules:

- [Basic Rules](#)
- [Setting Placement Constraints](#)

Basic Rules

You can edit symbolic layout objects using the Symbolic Editor with the following rules:

- [Basic Unit](#)
- [Consistent Finger Numbers](#)

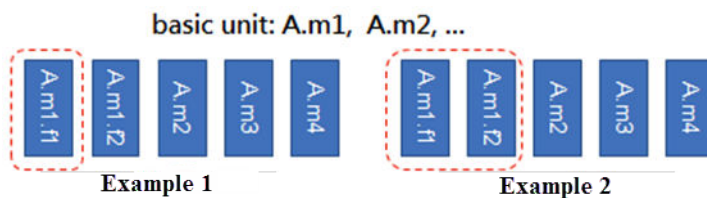
Basic Unit

You must select the basic unit of each realized device in order to invoke the Symbolic Editor.

Table 1 Symbolic Editor Basic Unit

Type	Device Item	Basic Logical Unit	Physical Layout
m-factor	A	A.m1, A.m2	A.m1.f1, A.m1.f2, A.m2.f1
Stack Group	A	A Stack1, A Stack2	A N0, A N1, A N2, A N3, A N4, A N5, A N6, A N7
Instance Array	A<0:1>	A(0), A(1)	A(0), A(1)

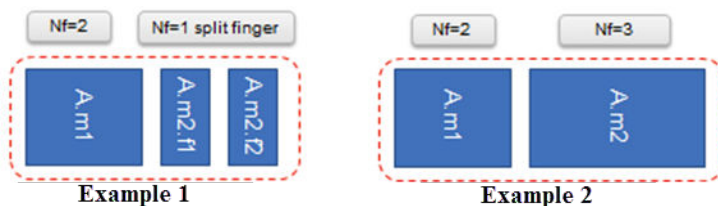
In Example 1, Symbolic Editor cannot be invoked when A.m1.f1 is selected. You must select the basic unit of A.m1 (A.m1.f1 and A.m1.f2) to invoke Symbolic Editor, as shown in Example 2.



Consistent Finger Numbers

The finger number of the selected objects must be the same before invoking the Symbolic Editor.

In Example 1, the finger number of A.m1 is 2 and the finger number of A.m2 (Am2.f1 and Am2.f2) is also 2. Symbolic Editor can be invoked when both finger numbers are the same. In Example 2, Symbolic Editor cannot be invoked because the finger number of Am1 and finger number of Am2 are different.



Setting Placement Constraints

This topic covers the following sections:

- [For Chaining Mode](#)
- [For Matching Mode](#)
- [Saving Constraint](#)

For Chaining Mode

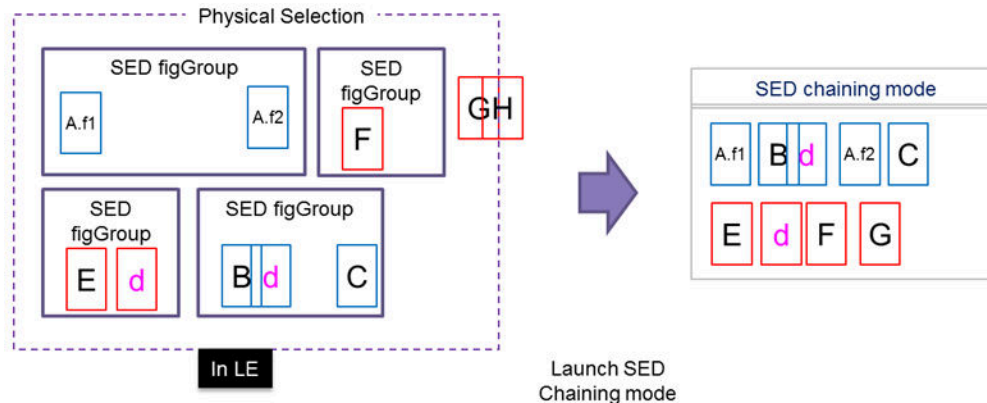
When the instances are realized from a layout or schematic using Symbolic Editor in chaining mode, a CMOS constraint is also modified or created as the mapping placement constraint. Any existing CMOS constraint that is related to the realized instance is updated accordingly.

If the selected layout does not map to a CMOS constraint, or maps to multiple constraints, or the mapped constraint does not contain a valid explicit placement string, the Symbolic Editor uses a Topology Extractor (see Extract Topology Constraints Command) to extract the layout placement as a new constraint.

- If the constraint can be extracted correctly, the Symbolic Editor applies this constraint.
- If no constraint can be formed from the selected layout, the Symbolic Editor extracts the layout placement information directly from the physical selection. See the following placement rule.

The following figure shows the placement rule when objects are selected and imported to Symbolic Editor with Chaining Mode. Objects are placed in the Symbolic Editor from left to right in rows. P strips (PMOS components) are placed on the top row in blue and N strips

(NMOS components) are placed on the bottom row in pink. Mixed type strips (both PMOS and NMOS components) are placed in purple.



See [Example 1](#) and [Example 2](#) under [For Matching Mode](#) for more details.

For Matching Mode

When the instances are realized from layout or schematic using Symbolic Editor in matching mode, an Inst Matching constraint is also modified or created as the mapping placement constraint. Any existing Inst Matching constraint related to the realized instance is updated accordingly.

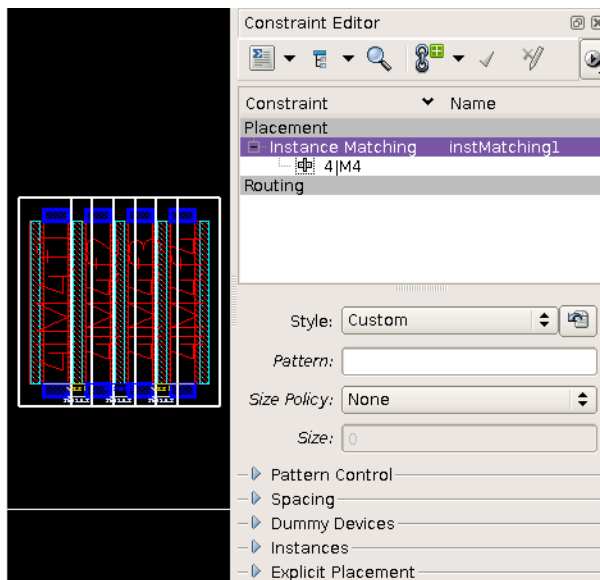
If there is no matching constraint or no explicit placement string, the Symbolic Editor uses a Topology Extractor (see Extract Topology Constraints Command) to extract an Inst Matching constraint.

- If an Inst Matching constraint can be extracted correctly, the Symbolic Editor applies this constraint.
- If no Inst Matching constraint can be formed from the selected layout, the Symbolic Editor applies the pattern style in the constraint, or finds a matched pattern in the [Pattern Panel](#).

The following examples explain how Inst Matching constraints are changed:

Example 1

1. In this example, if you realize 4|M4 to a layout using the Symbolic Editor matching mode, a new Inst Matching constraint *instMatching1* is also created with all placement information that was applied in matching mode.

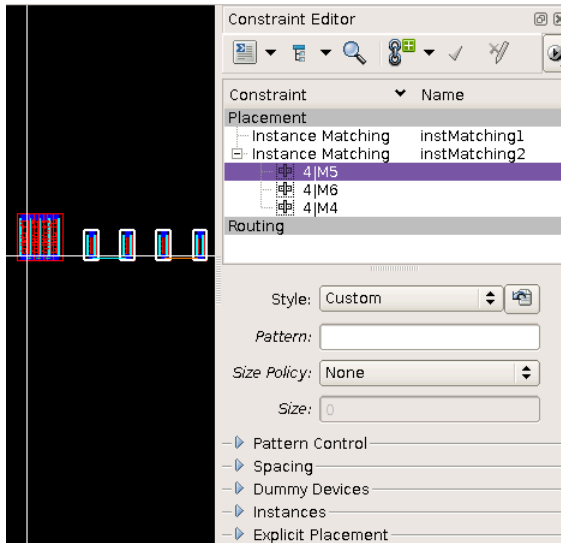


2. You can select the layout instances of 4|M4 and then open the Symbolic Editor in matching mode by clicking **SDL > Symbolic Editor > Matching** to modify the placement of 4|M4 again.
3. After opening the Symbolic Editor in matching mode with 4|M4, you can add 4|M5 and 4|M6 into the Symbolic Editor by selecting them in the Design Navigator and then



clicking the icon.

4. After realizing the instances to the layout, 4|M4, 4|M5, and 4|M6 replace the original 4|M4 in the layout. A new Inst Matching constraint *instMatching2* is created for 4|M4, 4|M5, and 4|M6. At the same time, 4|M4 is removed from the Inst Matching constraint *instMatching1*.

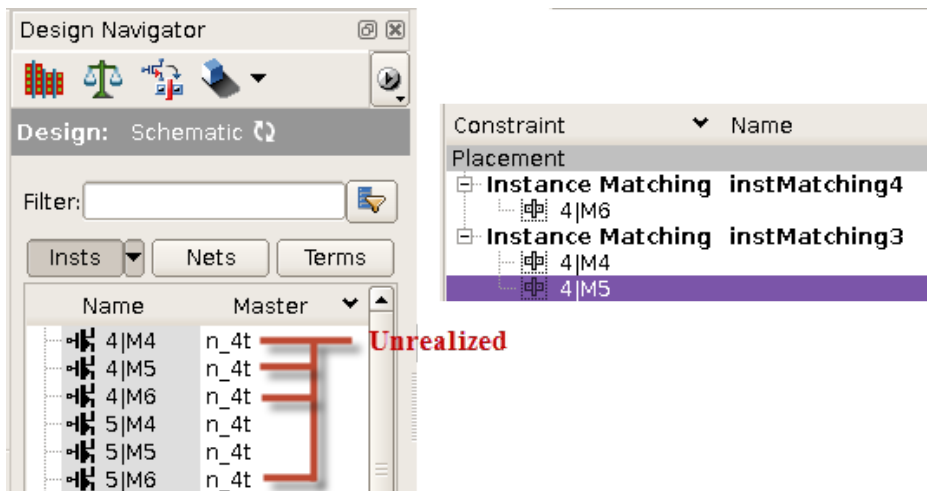


- When you select instance 4|M4 and open the Symbolic Editor in matching mode, the Inst Matching constraint *instMatching2* is found based on the selected device 4|M4. Because *instMatching2* includes 4|M4, 4|M5, and 4|M6, these three devices are imported to the Symbolic Editor in matching mode to be placed again.

Example 2

- The example in [Figure 1](#) shows that the devices in the Inst Matching constraints *instMatching3* and *instMatching4* are not realized.

Figure 1 Unrealized Devices (Left) and Placement Constraints (Right)

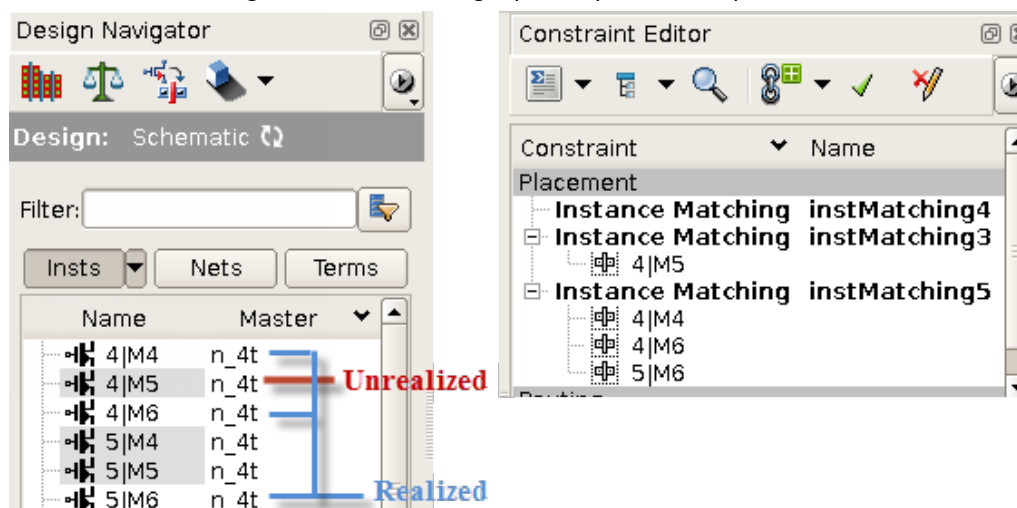


- When selecting 4|M4, 4|M6, and 5|M6 in the Design Navigator and then opening the Symbolic Editor in matching mode, the devices with Inst Matching constraints related to

instMatching3 and *instMatching4* are not referred since there are multiple constraints. After editing the devices in the Symbolic Editor, you can realize these devices in the layout.

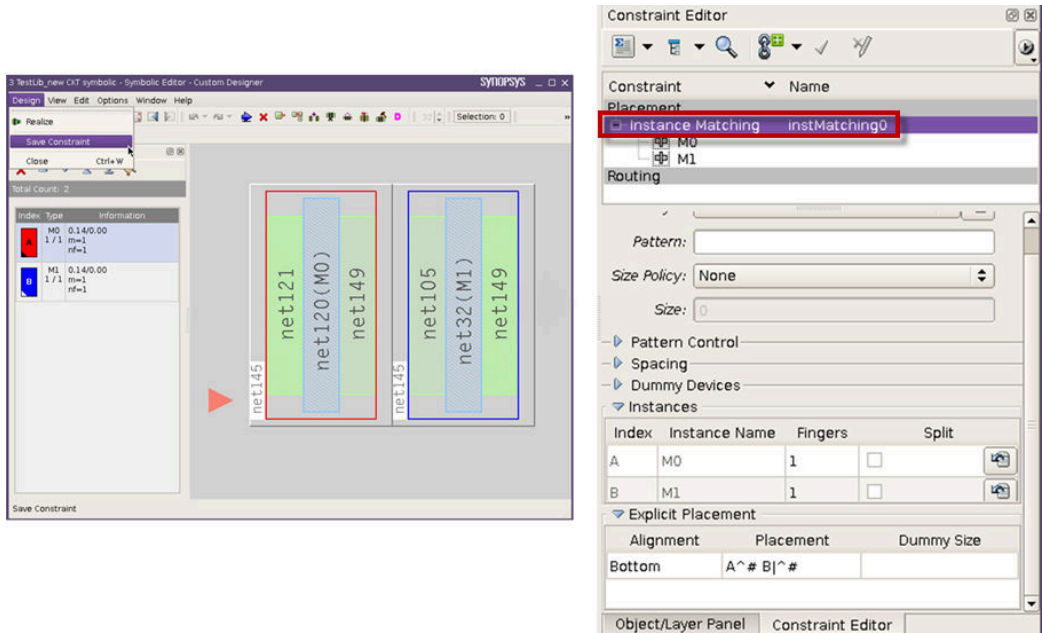
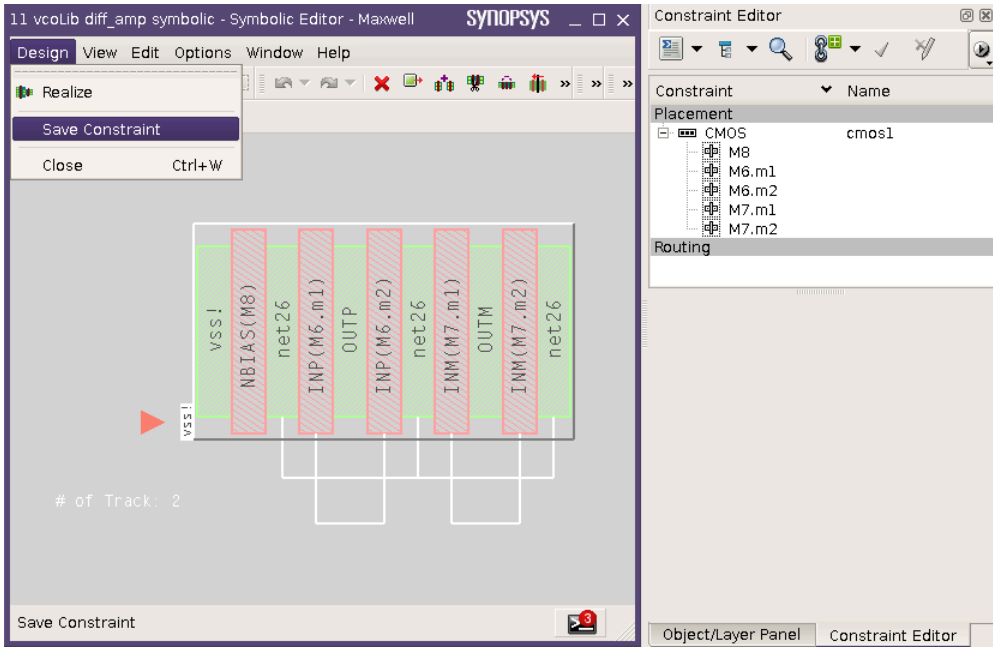
3. [Figure 2](#) shows a new Inst Matching constraint *instMatching5* is created for 4|M4, 4|M6, and 5|M6. The Inst Matching constraint *instMatching3* to which 4|M4 originally belonged is updated by removing 4|M4 and the Inst Matching constraint *instMatching4* is empty after removing 4|M6.

Figure 2 The Changes after Realizing 4|M4, 4|M6, and 5|M6



Saving Constraint

The **Design > Save Constraint** command creates a new CMOS or Inst Matching constraint based on the current placement and settings. The saved constraint can be reviewed with the Constraint Editor. See [For Chaining Mode](#) and [For Matching Mode](#) for more details.



Enabling the Symbolic Editor

To enable the canvas:

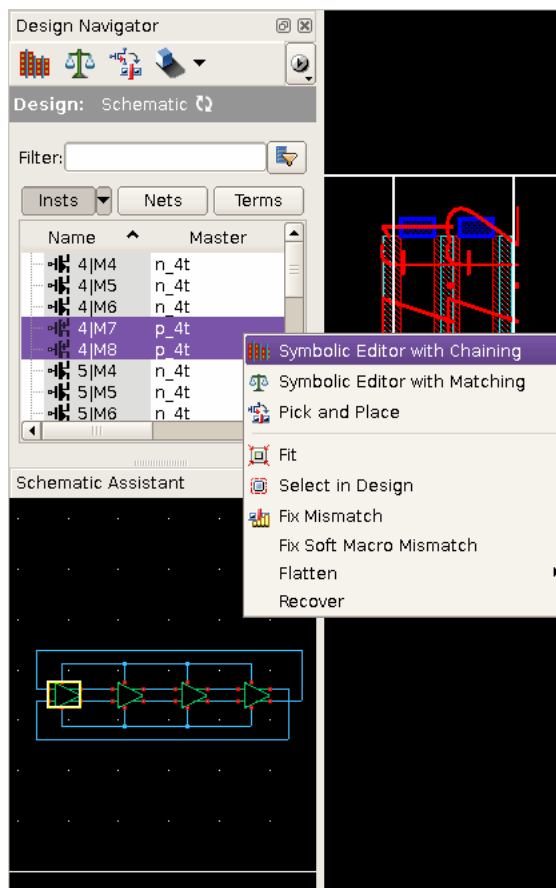
- [Using Chaining Mode](#)
- [Using Matching Mode](#)

Using Chaining Mode

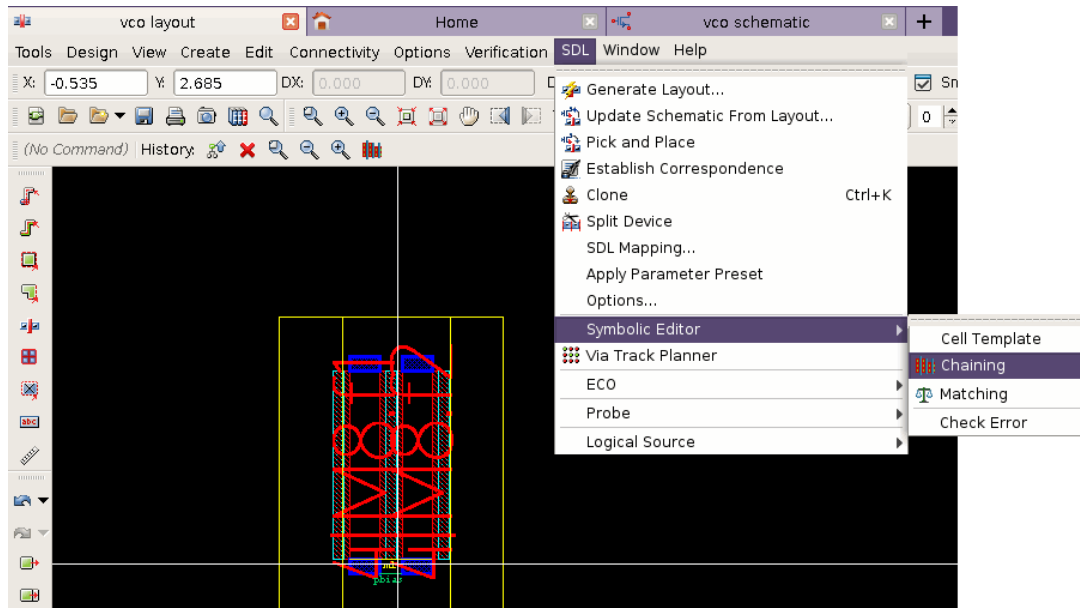
1. You can select devices from the SDL browser (Design Navigator in Schematic Mode) or Schematic Assistant and then select the **Symbolic Editor with Chaining** command



in the SDL browser or in the shortcut menu.



2. Or you can use the **SDL > Symbolic Editor > Chaining** command to start the Symbolic Editor with Chaining Mode.



3. If no devices are selected in the SDL browser, the devices selected in the layout canvas are used to start the Symbolic Editor with Chaining Mode.

Note:

Only MOS devices (PMOS and NMOS) can be edited using Symbolic Editor with Chaining Mode. You also need to specify the net names of Source, Drain, and Gate for MOS devices before invoking the tool.

You also can enable Symbolic Editor with Chaining Mode without selecting a device and the empty canvas is displayed.

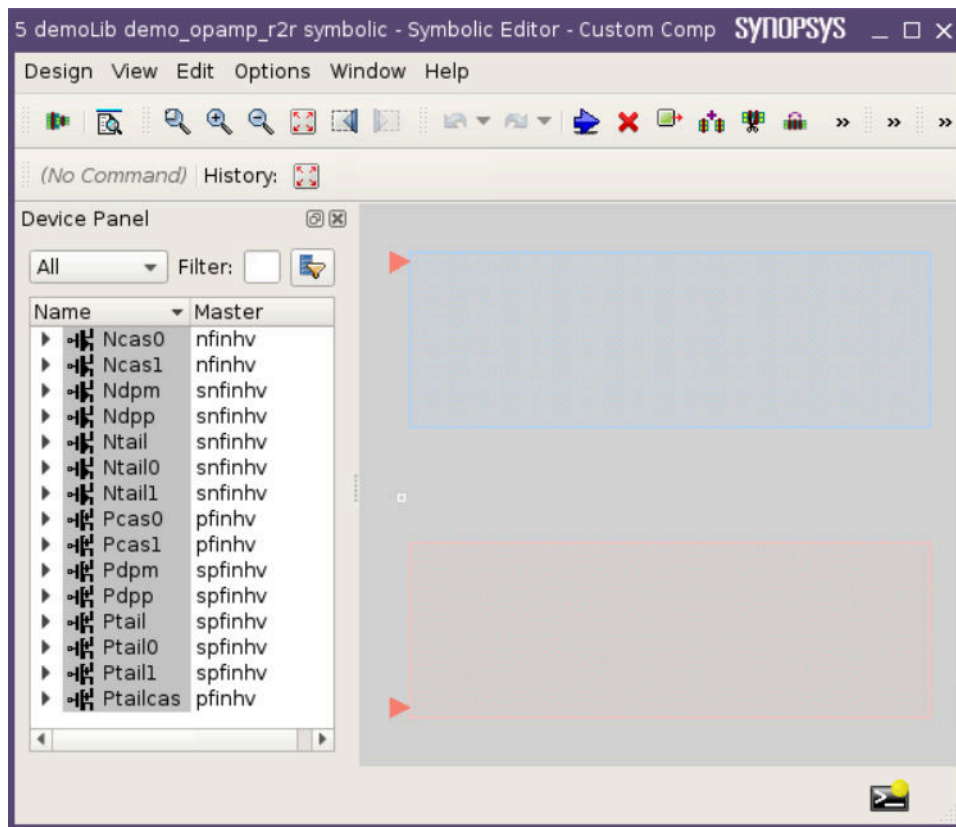
1. When no valid device is selected from the SDL browser or in the layout canvas, click



to enable the chaining mode canvas.

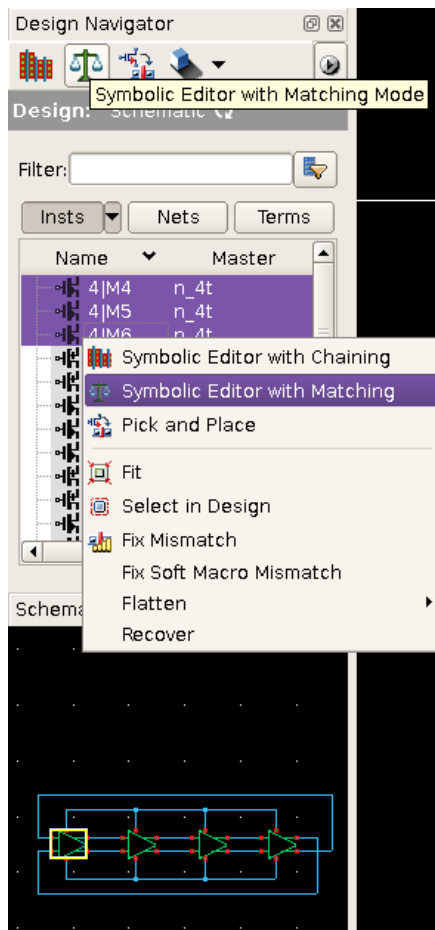
2. The empty canvas is displayed. See the following figure.

You can add devices to the canvas from **Device Panel**.

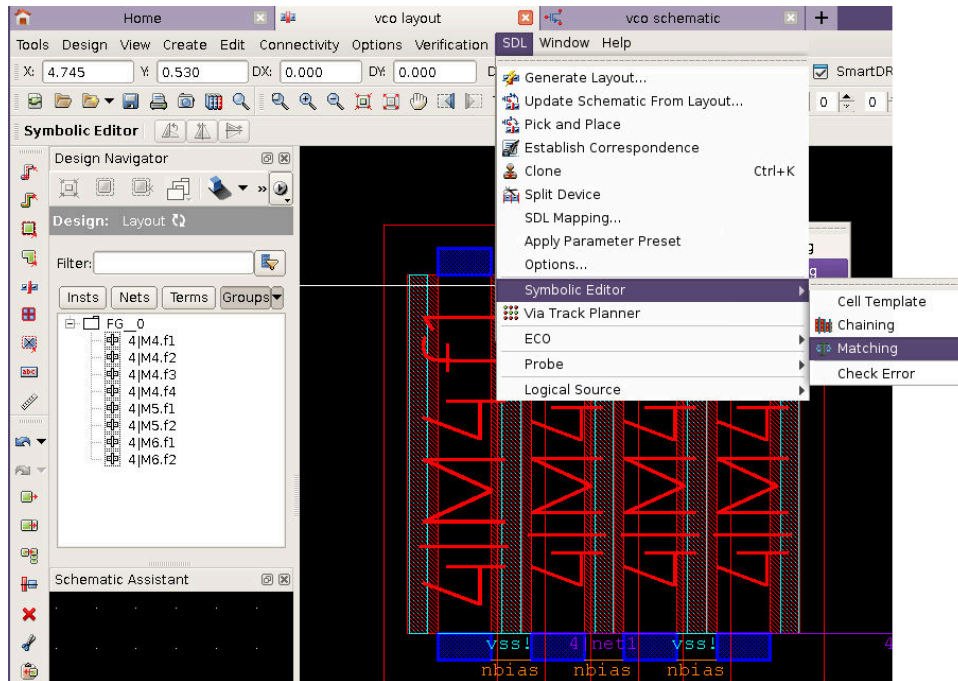


Using Matching Mode

1. You can select devices from the SDL browser (Design Navigator in Schematic mode) or Schematic Assistant and then select the **Symbolic Editor with Matching** command in the SDL browser or in the shortcut menu.



2. Or you can use the **SDL > Symbolic Editor > Matching** command to start the Symbolic Editor with Matching Mode.



3. If no devices are selected in the SDL browser, the devices selected in the layout canvas are used to start the Symbolic Editor with Matching Mode.

Note:

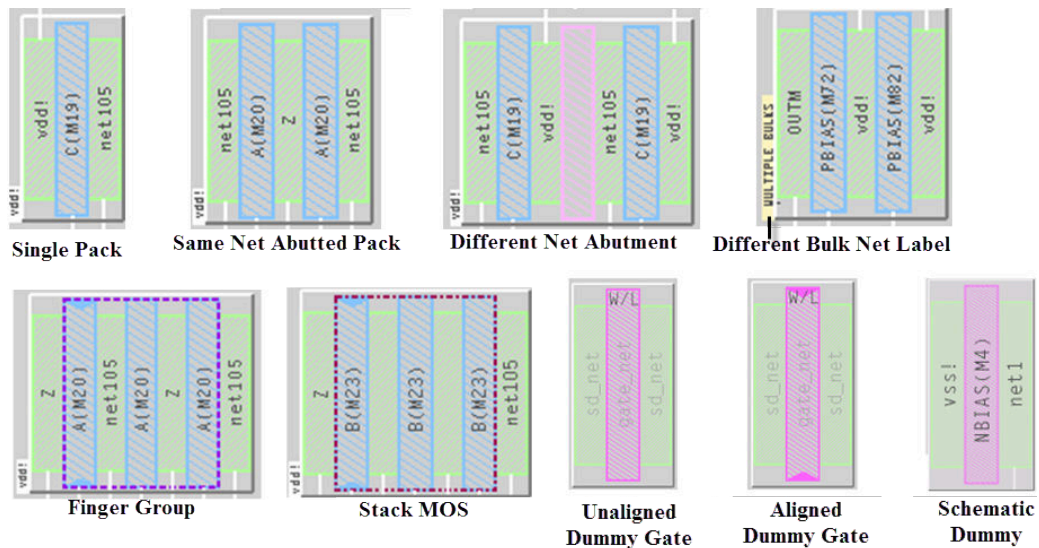
The Symbolic Editor with Matching Mode window is invoked in the following conditions:

- Select the same type of devices from instances, resistors, or MOS devices.
- MOS devices and instances with same master name or different master name can be edited.
- Resistors with same master name can be edited.
- The net names of Source, Drain, and Gate for MOS devices and the net names of Plus, Minus for resistors must be specified before invoking the tool.

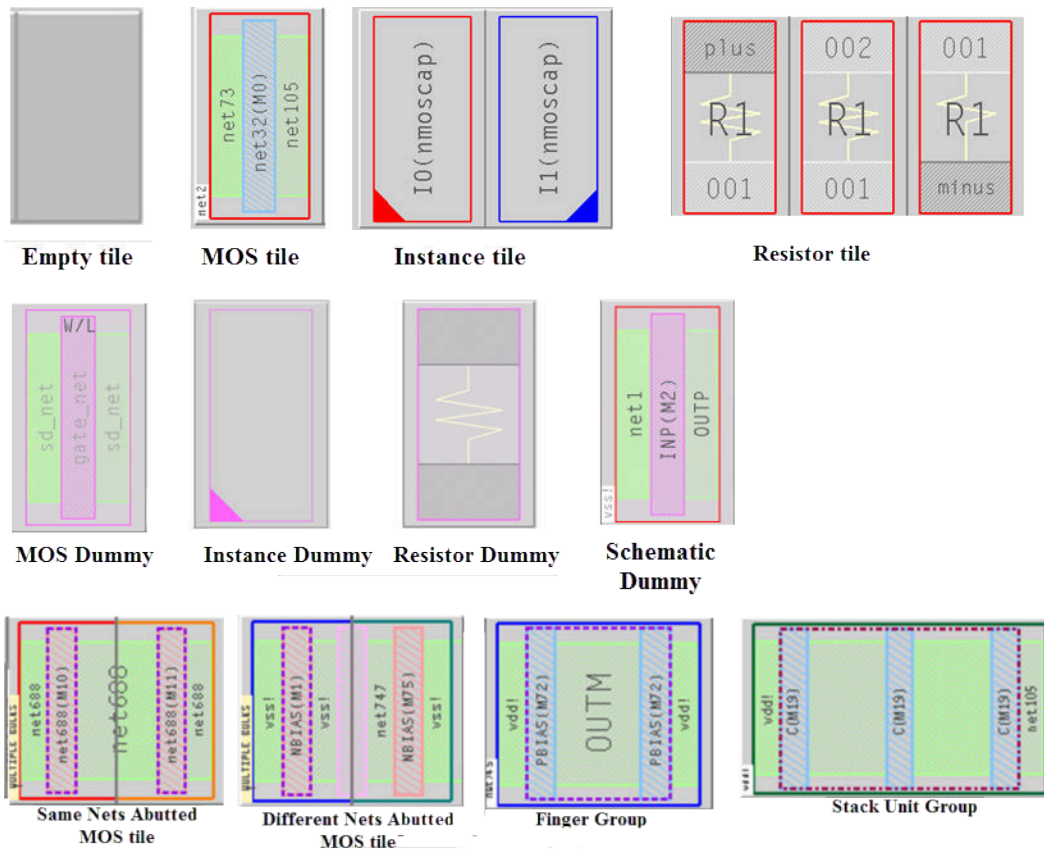
Understanding Basic Patterns

After you invoke the Symbolic Editor you can bring the selected objects to the Symbolic Editor canvas.

The following figure shows the basic patterns of objects placed in chaining mode.



The following figure shows the basic patterns used by the Symbolic Editor in matching mode, for different types of objects.



2

Symbolic Editor User Interface

The Symbolic Editor with Chaining Mode displays MOS devices in an abstract view and honors the user-defined transistor chaining and placement. The Symbolic Editor with Matching Mode arranges patterns of symmetrical devices for analog circuits depending on the instances or MOS devices selected in the object list of the SDL browser or in the Layout window.

Figure 3 Symbolic Editor With Chaining Mode Window

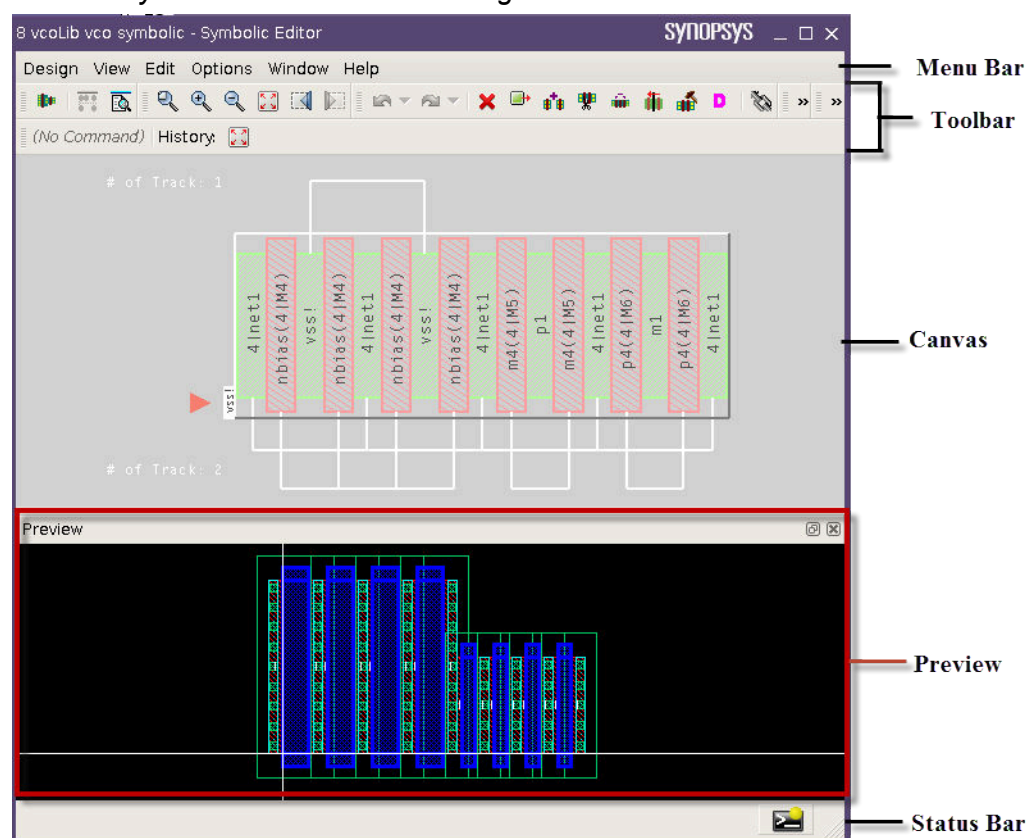
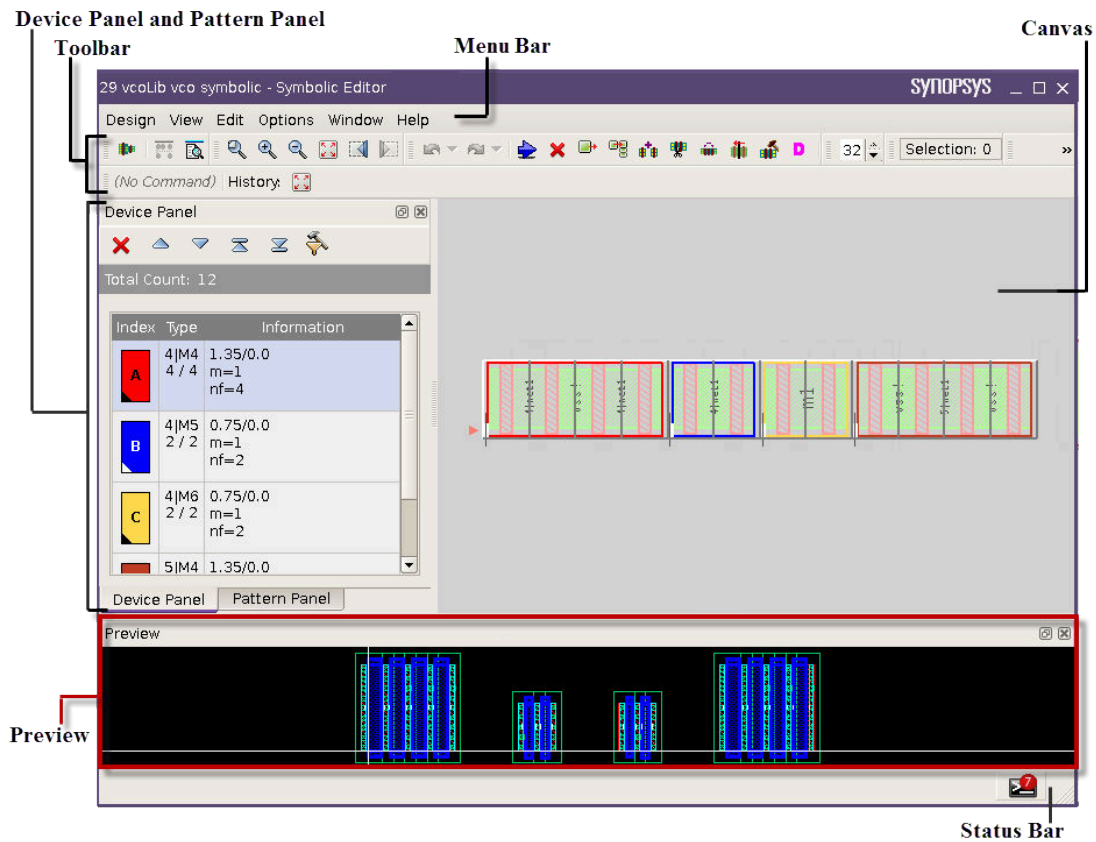


Figure 4 Symbolic Editor With Matching Mode Window



This chapter covers the following topics:

- [Using Symbolic Editor Toolbars](#)
- [Using Symbolic Editor Assistants](#)

Using Symbolic Editor Toolbars

Symbolic Editor toolbars offer quick access to canvas operations.

The Symbolic Editor toolbars are outlined in the following sections:

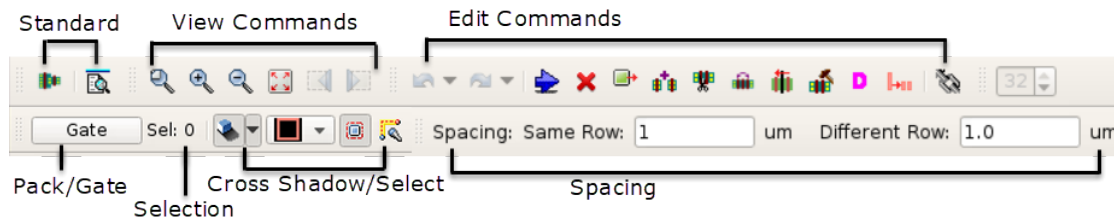
- [Toolbars](#)
- [Preset Toolbar](#)
- [Re-Edit Toolbar](#)

Toolbars

This section describes the following toolbars:

[Figure 5](#) shows the default locations of the chaining mode toolbar.

Figure 5 Chaining Mode Toolbar



[Figure 6](#) shows the default locations of the matching mode toolbar.

Figure 6 Matching Mode Toolbar for MOS Devices



Figure 7 Matching Mode Toolbar for Resistors

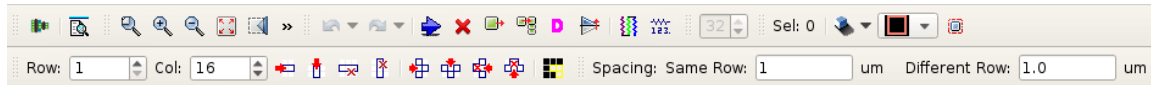


Figure 8 Spacing Toolbar



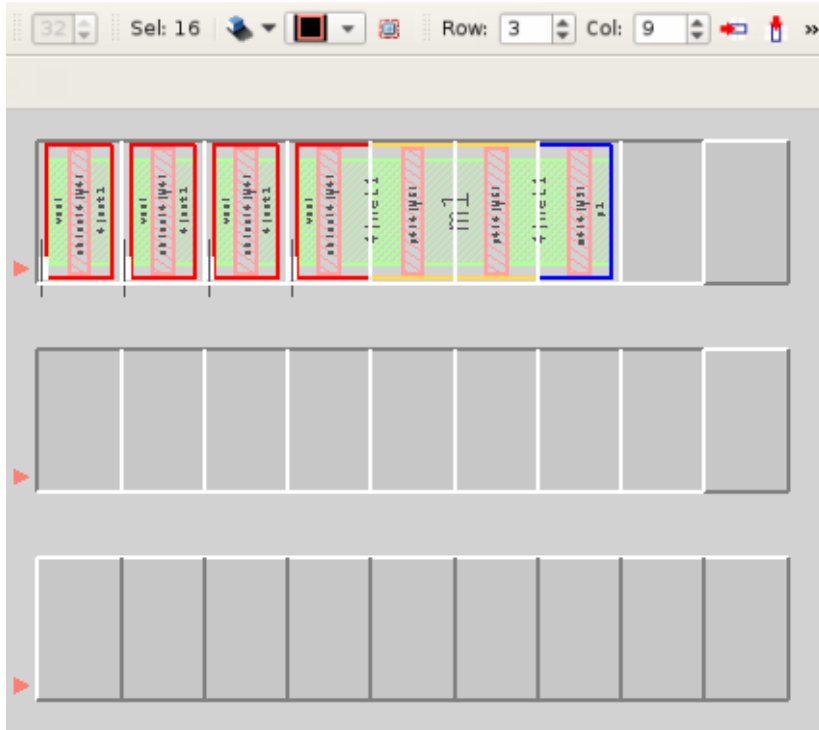
Symbolic Editor toolbars include the following commands:

- **Standard Command Toolbar:** Includes **Realize** and [Preview Assistant](#).
- **View Command Toolbar:** For details about the commands, see the [Viewing Designs](#) section.
- **Edit Command Toolbar:** For details about the commands, see the [Creating and Editing Basic Objects](#), [Editing MOS Devices](#), and [Creating and Editing Instances and Resistors](#) chapters.
- **Pack/Gate** (Chaining Mode Only): You can click the button to switch the selectable type between object packs and object gates. The default type is **Pack**.
- **Grid Size Command Toolbar** (Matching Mode Only): Shows the pattern size with the number of grid rows and columns. The number of grid rows/columns is editable

by assigning an integer greater than 1. See the [Inserting Rows and Columns](#) and [Removing Rows and Columns](#) sections.

- **Selection:** This label displays the count of current selected objects.

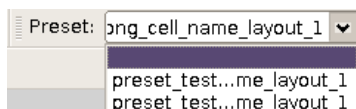
The following figure shows 16 tiles are selected and 3 rows and 9 columns are specified in matching mode.



- **Cross Shadow:** Click the button to turn on the shadow mode. See [Cross Highlighting and Selecting Objects](#).
- **Enable Cross Select:** Click the button to turn on the cross selection. See [Cross Highlighting and Selecting Objects](#).
- **Spacing:** You can specify the same row spacing and different row spacing in the text fields. Those text fields are simultaneous with the **Spacing** section in the **Symbolic Design Options** dialog box. See [Creation Tab](#). You can specify the unit and whether to use **Boundary LPP** in the [Creation Tab](#).

Preset Toolbar

You can place objects with preset parameters using the Symbolic Editor. The following shows the **Preset** selection field in the Symbolic Editor toolbar.



To enable this feature, you must specify the preset library in the **SDL Options** dialog box. In chaining mode, this option is disabled when the **Row Pair Preset** option in the [Specifying Cell Template \(Chaining Mode Only\)](#) is turned on. For more information, see **Parameter Preset** in Main Tab. To modify preset parameters, see [Modifying Preset](#).

For more information about Parameter Preset feature, see [Realizing a Layout With Preset Parameters](#).

Re-Edit Toolbar

As you realize devices in the layout canvas, you can rotate, flip, or edit the devices before you place them, using the following commands: **Rotate**, **Flip Horizontally**, **Flip Vertically**, and **Re-Edit**.



Using Symbolic Editor Assistants

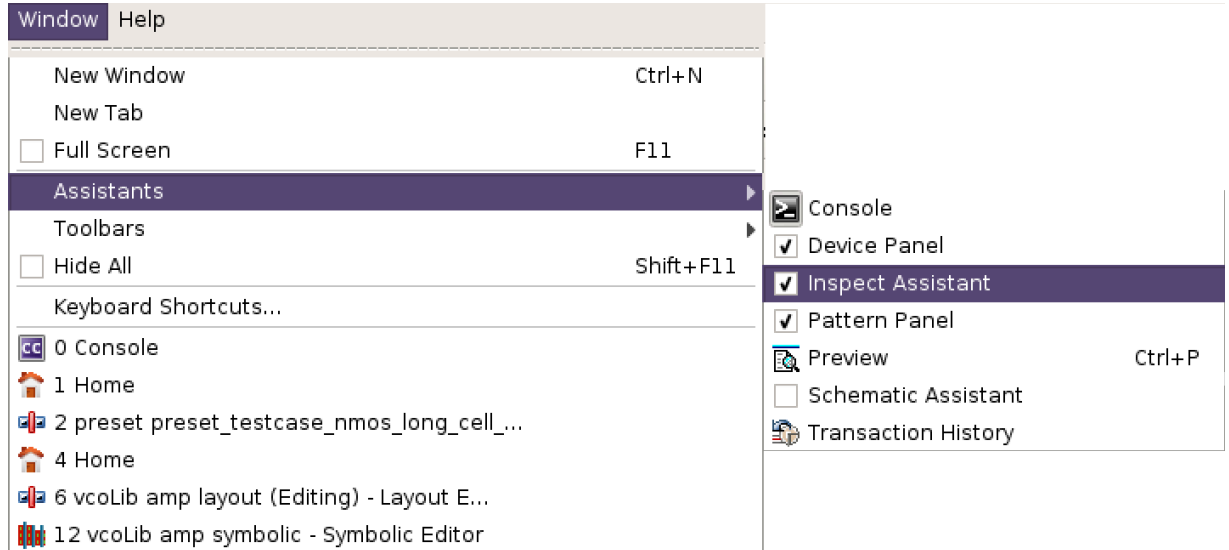
The Symbolic Editor assistants are outlined in the following sections:

- [Device Panel](#)
- [Pattern Panel](#)
- [Preview Assistant](#)
- [Schematic Assistant](#)
- [Inspect Assistant](#)

Start the **Inspect Assistant** using either of the following methods:

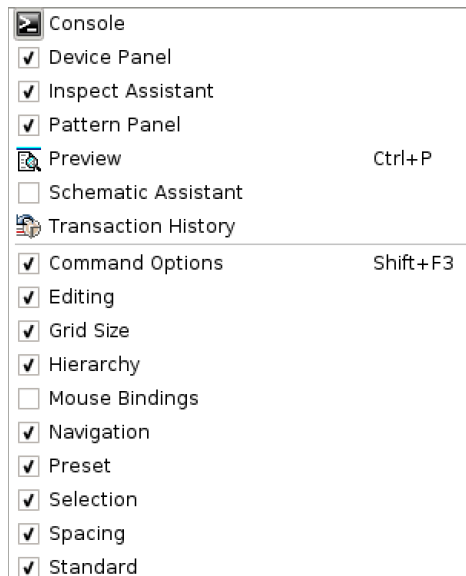
- Choose **Window > Assistants > Inspect Assistant** in the Symbolic Editor (see [Figure 9](#)).

Figure 9 Invoke From Windows > Assistants



- Right-click the symbolic editor toolbar (see [Figure 10](#)).

Figure 10 Invoke From Symbolic Editor Window Toolbar Menu



Device Panel

This topic covers the following sections:

- [In Chaining Mode](#)
- [In Matching Mode](#)
- [Device Panel for Resistors](#)
- [Displaying Gate Net Name](#)
- [Coloring PMOS and NMOS in Device Panel](#)

In Chaining Mode

The Symbolic Editor provides a Device Panel to view the overall list of unrealized devices. You can open the Device Panel by choosing **Window > Assistants > Device Panel** in the symbolic canvas (see [Figure 9](#)).

You can select **All**, **Placed**, or **Unplaced** options to view the devices. The Device Panel shows unplaced devices with a dark gray background color.

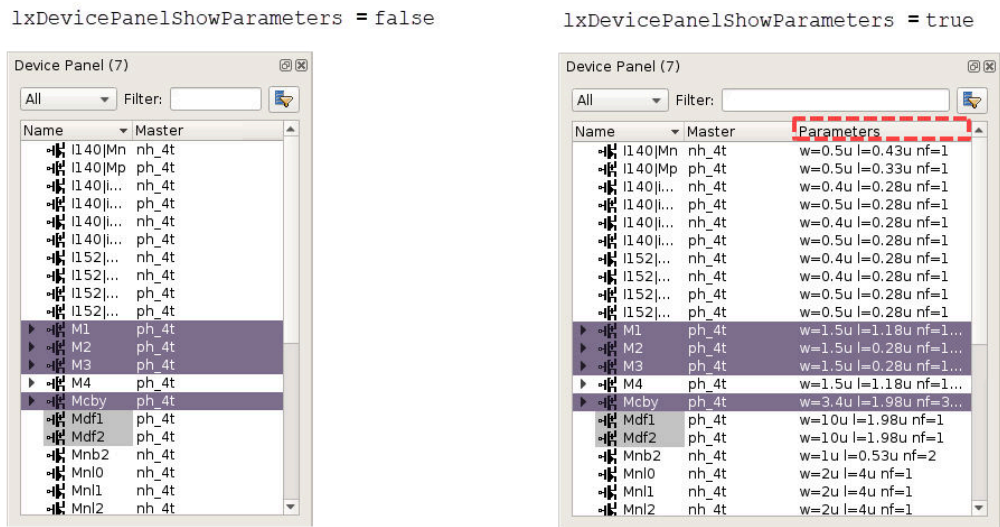
To display the **Parameters** column:

1. Set the `lxDevicePanelShowParameters` Tcl preference to `true`:

```
db::setPrefValue lxDevicePanelShowParameters -value true
```

2. Reopen the Symbolic Editor.
3. Adjust the column width to fit the Device Panel.

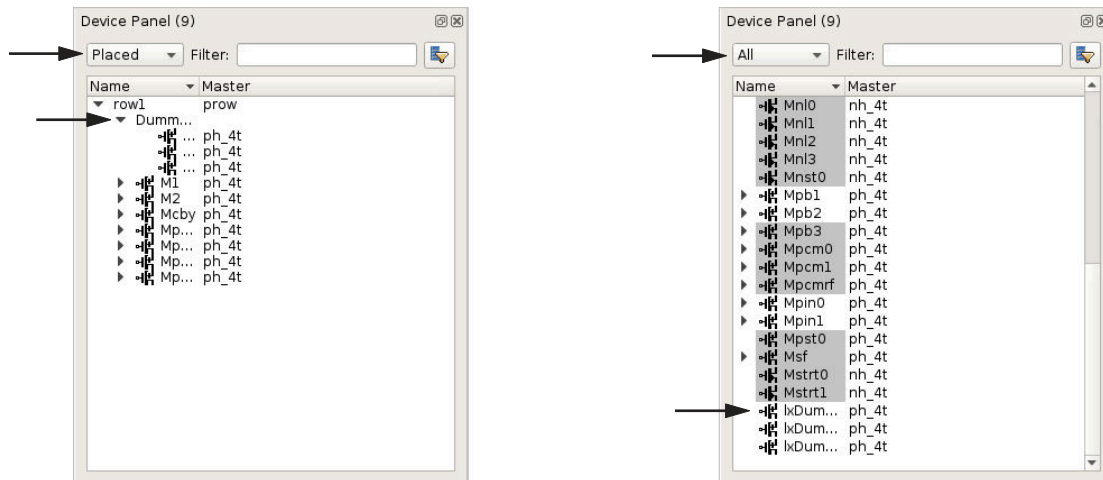
Figure 11 *Device Panel in Chaining Mode*



To group dummy devices in the Device Panel, set the
lxDevicePanelGroupDummyDevices Tcl preference to one of the following:

Preferences Value	Definition
Placed View Only	Groups dummy devices when Placed view is selected.
All View Only	Groups dummy devices when All view is selected.
Both Views	Groups dummy devices when All or Placed views are selected.
None	Does not group dummy devices.

Below, for example, `lxDevicePanelGroupDummyDevices` is set to `Placed View Only`. Dummy devices are grouped when **Placed** view is selected (left), but not when **All** view is selected (right):

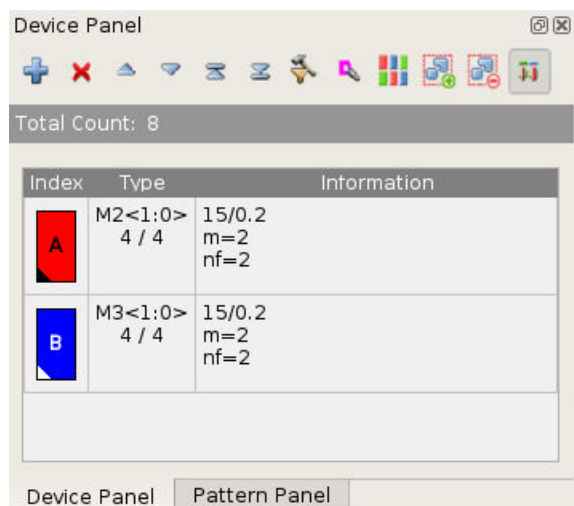


In Matching Mode

This panel is displayed when the Symbolic Editor is invoked in matching mode. The visibility status of this panel can be toggled *on* or *off* via **Window > Assistants > Device Panel** (see [Figure 9](#)).

Multiple selection is supported when the **Reconfigure Device** dialog is active and the devices have the same total width. You can add devices by dragging to select a device type from **Device** panel to the canvas.

The following figure shows the device settings for the selected MOS devices.

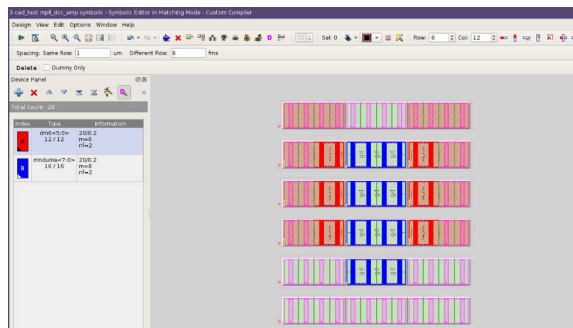


The **Device Panel** includes the following information and device list:

- **Toolbar Icons:** You can modify the device list using the following toolbar icons:

Table 2 *Toolbar Icons*

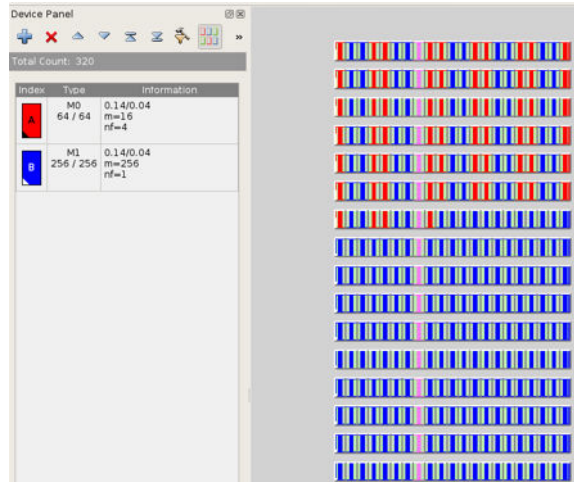
Toolbar Icon	Description
	Adds new devices from the selection to the device panel. The selection can be from Design Navigator, Schematic Assistant in the Layout Editor, or Schematic Assistant in the Symbolic Editor.
	Removes a selected device from the device list.
	Moves a selected device up in the device list.
	Moves a selected device down in the device list.
	Moves a selected device to the top of the device list.
	Moves a selected device to the bottom of the device list.
	Reconfigures the non-stack MOS devices (Figure 12) or stack MOS devices (Figure 13) displayed in the panel in the Reconfigure Device dialog box before placing the devices on the symbolic canvas.
	Highlights dummy devices with active devices together when you select multiple devices in the Device section.



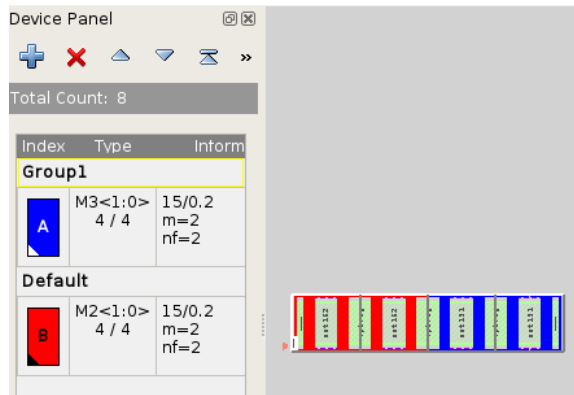
Toolbar Icon	Description
--------------	-------------



Improves the display in the canvas by coloring the devices using device index color.



Creates new group for selected devices in the device panel. You can reorganizes the groups by dragging and dropping devices.

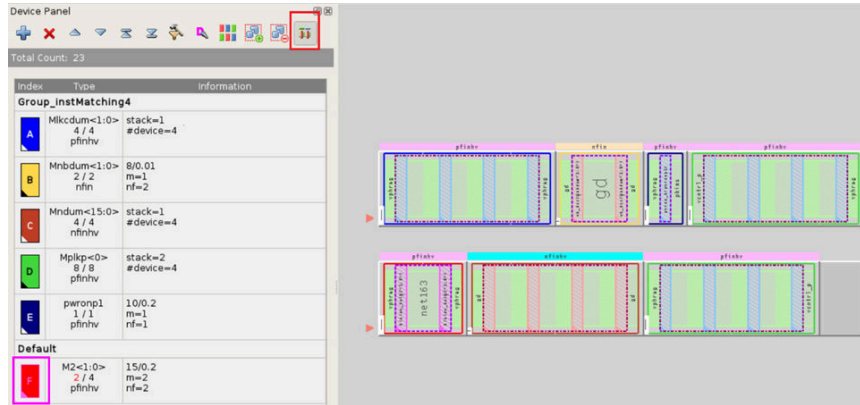


When enabling matching mode from multiple instMatching constraints, the tool auto creates groups for each constraint.



Ungroups the selected grouped devices in the device panel.

Toolbar Icon	Description
	Displays the voltage threshold of the following devices in Matching Mode only: ◦ MOS ◦ Stack ◦ Logical dummy The devices with different threshold voltages are marked with different colors.



Note:
The threshold voltages are not displayed for normal dummy devices.

Figure 12 Reconfigure Device Dialog Box

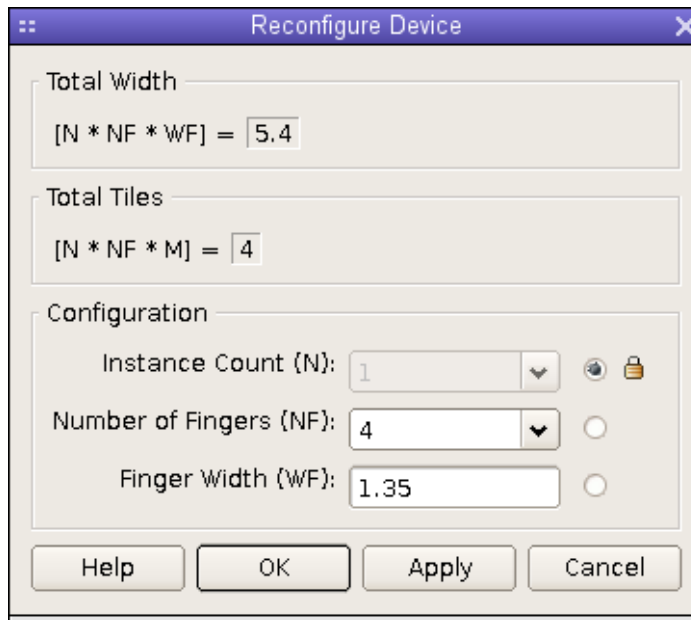


Figure 12 specifies the following device settings:

- **Total Width:** Display the total width of selected objects. The total width is fixed and cannot be changed.
- **Total Tiles:** Shows the total tile count required to place all leaf instances of the device on the symbolic canvas. The total tiles is changed according to the finger width (**WF**).
- **Configuration:** Reconfigure selected devices by specifying the following options: **Instance Count (N)**, **Number of Fingers (NF)**, and **Finger Width (WF)**. Select a radio button to disable the options. You can only enable two of the three options at a time. For example, if **Instance Count (N)** is disabled, you can specify **Number of Fingers (NF)** and **Finger Width (WF)**.

For example, if you change one of these fields, the number of tiles changes. If you change the finger width from 1.35 to 2.7, the total number of tiles is changed from 4 to 2.

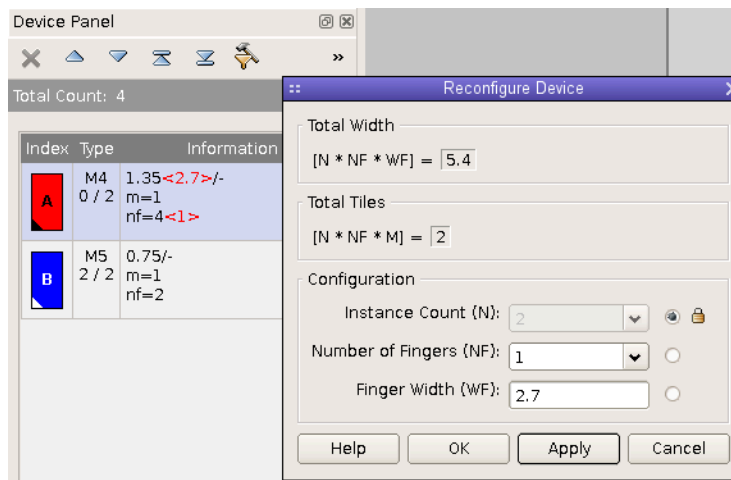


Figure 13 Reconfigure Device Dialog Box for Stack MOS

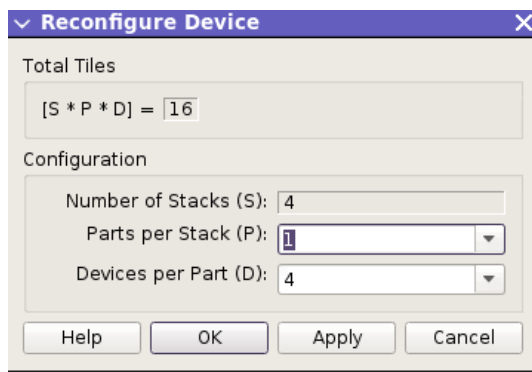


Figure 13 specifies the following stack MOS device settings:

- **Total Tiles:** Shows the total tile count required to place all leaf instances of the device on the symbolic canvas.
 - **Configuration:** Reconfigure selected devices by specifying the following options:
Number of Stacks (S), **Parts per Stack (P)**, and **Devices per Part (D)**.
 - **Total Count:** Shows the total symbolic tile count required to place all devices.
 - **Index:** This column shows the device symbols using a cyclic color index. Devices placed on the symbolic canvas are outlined with the same color. The display file `symDisplay` contains the packets `1xDev0 ~ 1xDev9` which can be used to define the colors used for this type of indexing. See Display Resource Editor Tool in the *Custom Compiler Environment User Guide*.
- The pattern index (**A**, **B**, ...) is assigned sequentially corresponding to the device order in the panel. Changing the device order changes the pattern index for that device. The color index does not change.
- **Type:** This column shows the device name, total number of devices, and number of devices placed in the matching canvas.
 - **Information:** This column shows the associated information of the selected device. The detailed logic information for a device is shown in a yellow tip when the mouse cursor is placed over a row in the device panel. The contents of the Information column may vary depending on the selected device.

B	4 M5 2 / 2	0.75/- m=1 nf=2	MOS: 4 M5 Width: 0.75 Length: 0 Multiple: 1 Finger Number: 2 Source Net: 4 net1 Drain Net: p1
C	4 M6 0 / 2	0.75/- m=1 nf=2	

Device Panel for Resistors

This panel is displayed when the Symbolic Editor is invoked in matching mode with resistor devices. You can use **Unit Cell Configuration** to generate segmented resistors in the layout.

Figure 14 Device Panel Without Unit Cell Configuration for Resistors

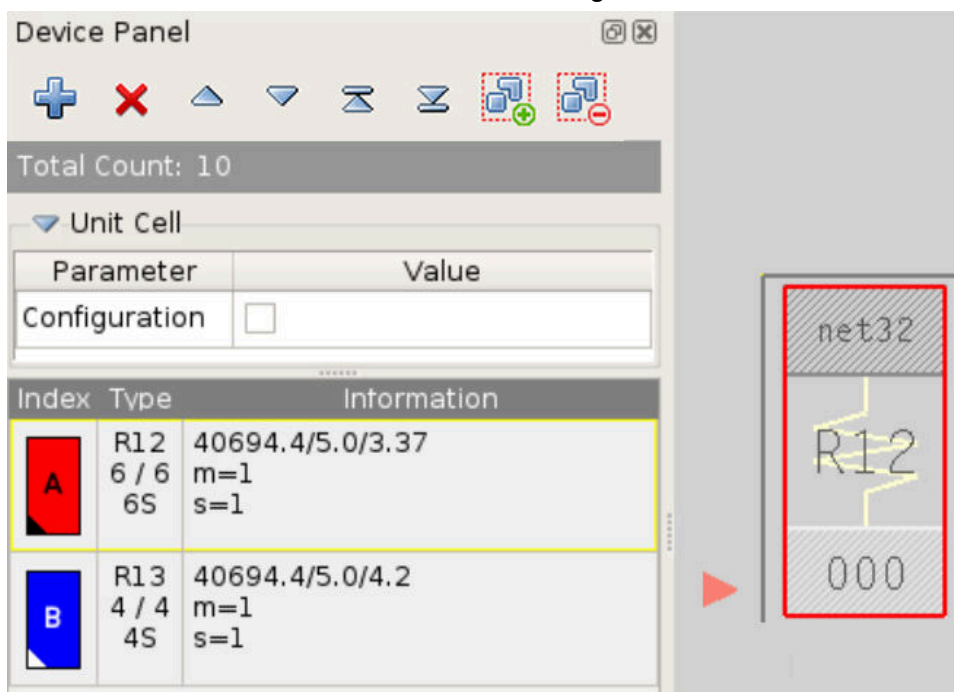
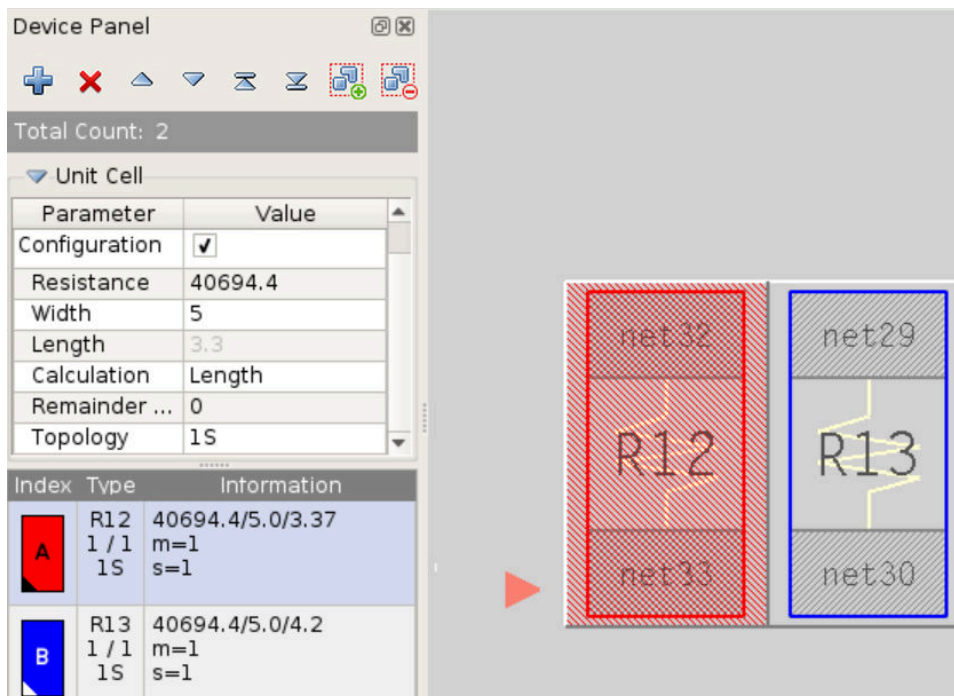


Figure 15 Device Panel With Unit Cell Configuration for Resistors



- **Configuration:** Lets you specify the unit cell.
- **Length, Resistance, Width, and Calculation:** You can evaluate the topology for the generated layout by changing the resistance, width, or length parameters for the unit cell. Enter the parameter you want to evaluate in **Calculation**, then modify the parameter value.
- **Remainder Tolerance:** A remainder is created if the user-defined topology status does not match its resistance value. You can adjust the remainder tolerance to customize resistors. The default value is 0.
- **Topology:** Indicates the topology of the selected resistor. There are two topology statuses: series connection (S) and parallel connection (P). The topology can be series only (nS), parallel only (mP), or mixed (nSmP). (m and n are constants).

Displaying Gate Net Name

The Tcl preference `lxSEDSortDeviceByGateNetSupports` supports of sorting devices by gate net in the Matching Device Panel and showing the gate net on both Matching Device Panel and Chaining Device Panel. The default is off.

For example:

```
db::setPrefValue lxSEDSortDeviceByGateNet -value 1
```

Chaining Mode

Device Panel

All Filter:

Gate	(Name)	Master
▶	a	(N3) nfin
▶	b	(N2) nfin
▶	c	(N1) nfin
▶	d	(N0) nfin
▶	a	(N4) pfin
▶	b	(N5) pfin
▶	c	(N6) pfin
▶	d	(N7) pfin

Matching Mode

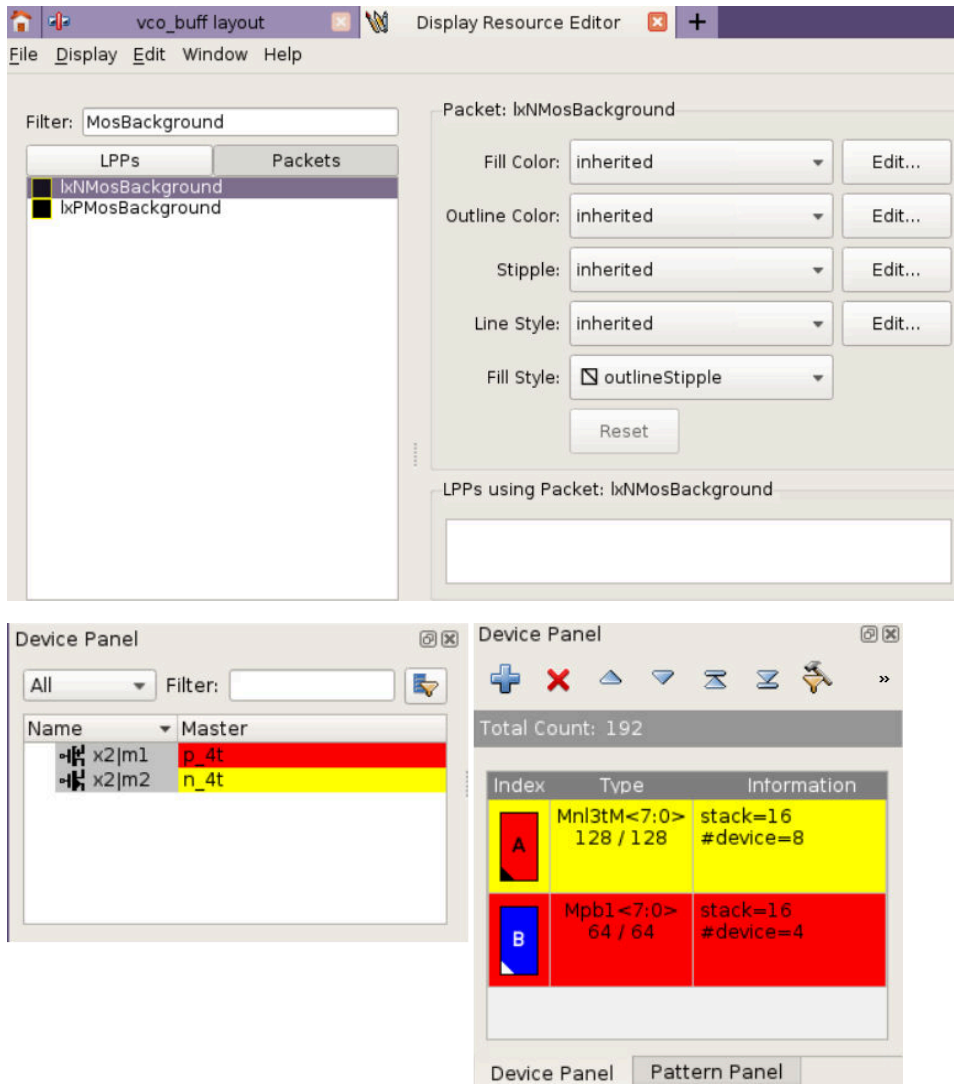
Device Panel

Total Count: 48

Index	Type	Information
A	a (N3) 4 / 4	4/0.01 m=4 nf=1
B	a (N4) 8 / 8	4/0.01 m=8 nf=1
C	b (N2) 4 / 4	4/0.01 m=4 nf=1
D	b (N5) 8 / 8	4/0.01 m=8 nf=1
E	c (N1) 4 / 4	4/0.01 m=4 nf=1
F	c (N6) 8 / 8	4/0.01 m=8 nf=1

Coloring PMOS and NMOS in Device Panel

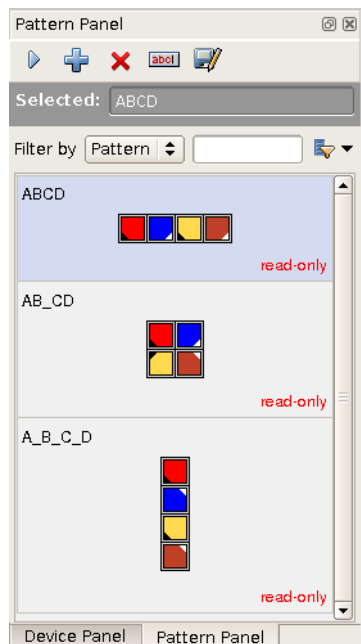
The **Tools > Display Resource Editor** supports two symDisplay packets `1xNMosBackground` (default is inherited) and `1xPMosBackground` (default is inherited) to have different background colors of NMOS and PMOS in the Device Panel.



Pattern Panel

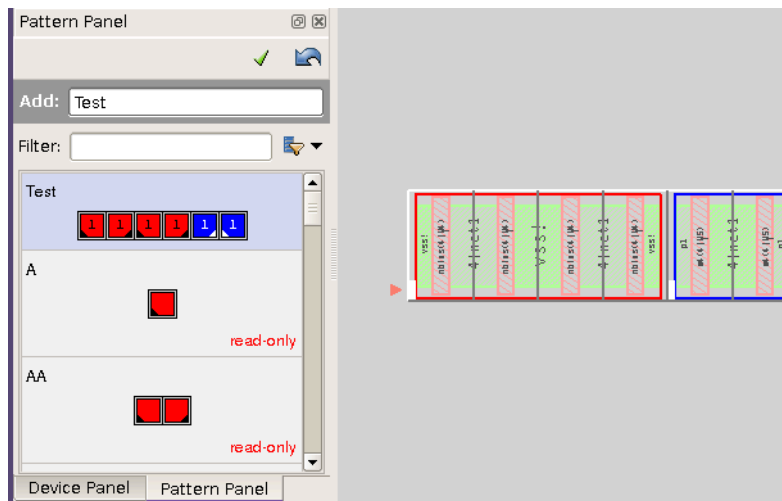
The **Pattern Panel** provides browser functionality to traverse built-in and user-defined patterns. It also provides basic pattern management functionality. The pattern panel can be enabled or disabled from the **Window > Assistants > Pattern Panel** menu (see [Figure 9](#)). This panel is enabled by default.

The following figure shows some sample patterns for devices listed in the device panel.

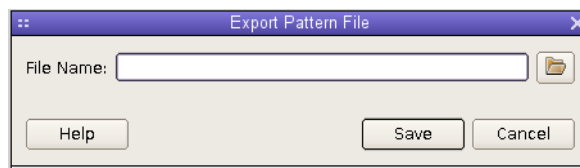


The **Pattern Panel** includes the following information:

- **Toolbar Icons:** You can modify the device list using the following toolbar actions:
 -  : Applies selected pattern to the canvas.
 -  : Saves current device placement on symbolic canvas as a pattern. The pattern name can be specified in the **Add** text field. Click  button to add the pattern to the pattern database or click  button to cancel the action.



-  : Deletes the selected device pattern in the pattern list.
-  : Renames the selected device pattern in the pattern list.
-  : Exports all user-defined patterns to a specified file using the **Export Pattern File** dialog box. Uses `lxPatternFile` to specify the pattern file if you want to set it as default patterns.



- : Filters the displayed pattern by selecting one of the following options:

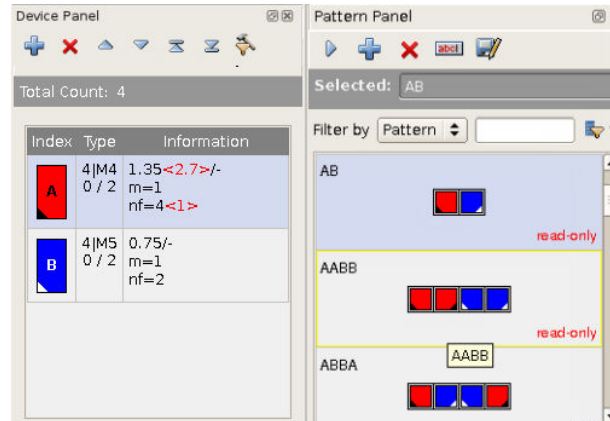
All

When selected, all patterns are displayed.

Device Matched

- When selected, the patterns are displayed in the following conditions:
- The patterns are matched to the device types shown in Device Panel.
 - The total count of devices in the patterns is less or equal to the total symbolic tile count shown in Device Panel.

When **Device Matched** is selected and pattern A and pattern B are listed in the **Device Panel**, only the patterns AB and ABA are displayed.



Fully Matched

- When selected, the patterns are displayed in the following conditions:
- The patterns are matched to the device types shown in the Device Panel.
 - The total count of the patterns is equal to the total symbolic tile count shown in the Device Panel.

- **Selected:** Shows the selected pattern name.
- **Filter By:** Filter patterns shown in the pattern panel based on the filter string by pattern name or pattern file name.

Preview Assistant

The Symbolic Editor provides a preview assistant to view the symbolic devices as they would appear if realized in layout. You can open the preview assistant by choosing

Window > Assistants > Preview (see [Figure 9](#)) or clicking  in the Symbolic toolbars.

When the preview assistant is opened, a refresh button  appears when the preview is not in sync with edits on the symbolic canvas. Pressing this button regenerates the preview.

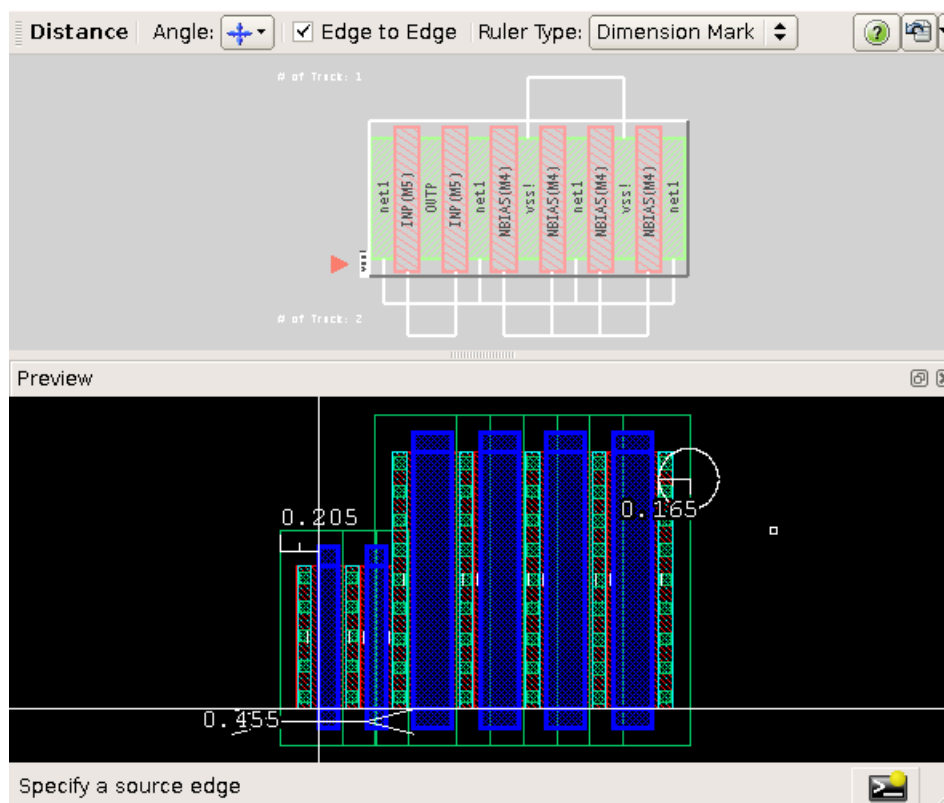
Once the Pattern Route constraint is specified for the CMOS, instMatching, or resistorBank constraint. The routing connection can be displayed in the preview assistant.

You also can use the following commands on the Preview Assistant:

- [Creating and Clearing Ruler Markers](#)
- [Using Preview Assistant to View Placement in Layout](#)

Creating and Clearing Ruler Markers

The **Edit > Distance** command is available only in the **Preview Assistant**. This command is used to measure the distance between two object edges or points from the **Preview Assistant**. When you select the first measured edge, the distance line and distance value are displayed when the mouse cursor reaches the next object edge. You can create the ruler markers in the **Preview Assistant** and specify the ruler type by choosing one of the following options: **Linear Ruler**, **Circular Ruler**, and **Dimension Mark**.



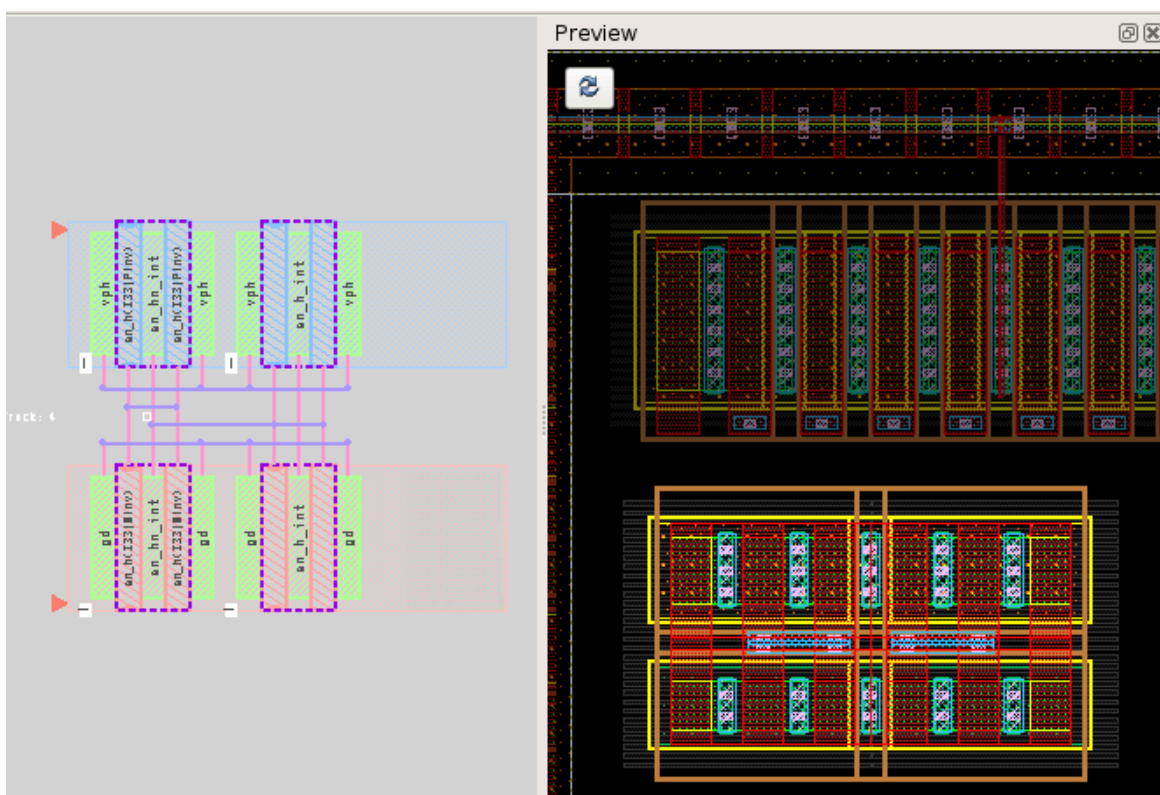
To clear all ruler markers on the preview assistant, you can choose **View > Clear Ruler**.

Using Preview Assistant to View Placement in Layout

The Symbolic Editor allows you to view the placement in the whole layout design using the **Preview** assistant when enabling the **SED Preview In-Design** option in the **SDL Options** dialog box.

To view the device placement in the layout:

1. Realize devices using the Symbolic Editor.
2. Click **SDL > Options** to enable the **SED Preview In-Design** option.
3. Choose the layout devices realized by Symbolic Editor.
4. Click **SDL > Symbolic Editor > Chaining** or **Matching** to start the Symbolic Editor.
5. In the Symbolic Editor, click  or **Window > Assistants > Preview** to enable the **Preview** assistant. You can view the whole layout design in the **Preview** assistant.



6. After editing the realized design in the Symbolic Editor, click the icon  in the **Preview** assistant to refresh the edits in the **Preview** assistant when **Preview Auto Update (Preview Tab)** is disabled. The changes are synchronized between Symbolic Editor and Layout Editor. The Symbolic Editor remains active. You can query and view your design in the layout editor.
7. You can easily switch back to the Symbolic Editor by clicking the **Re-Edit** icon.
8. The Symbolic Editor is aborted when editing the layout in the layout editor.

Schematic Assistant

Symbolic Editor provides schematic assistant to view the overall of the design. You can open the schematic assistant by choosing **Window > Assistants > Schematic Assistant** in the symbolic canvas (see [Figure 9](#)). See [Using the Schematic Assistant](#).

Inspect Assistant

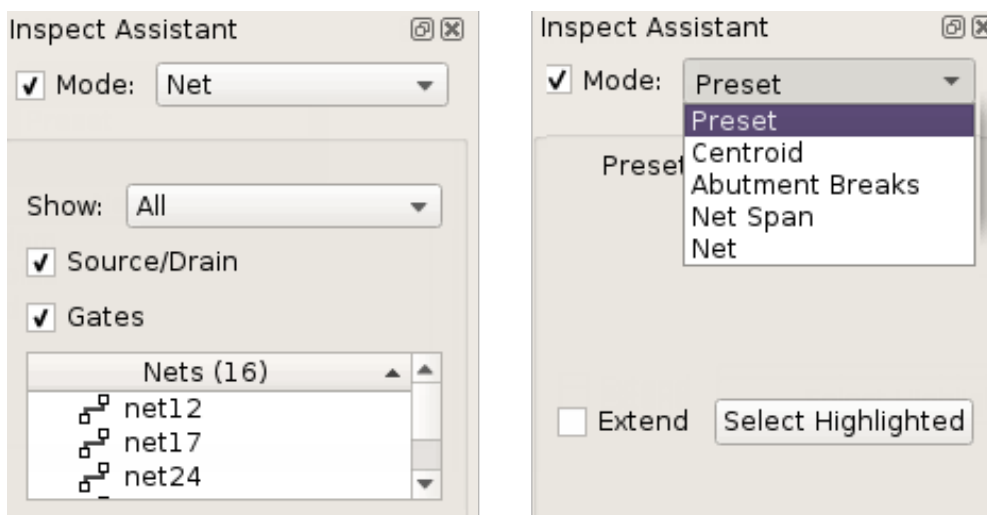
Symbolic view in the Symbolic Editor shows structural device placement along with device information. Showing all the information all the time on symbolic canvas causes unwanted clutter. The Inspect Assistant is designed to highlight additional information such as applied presets, centroid of devices, and abutment breaks.

The Inspect Assistant is outlined in the following sections:

- [Using the Inspect Assistant Interface](#)
- [Highlighting Preset](#)
- [Highlighting Nets](#)
- [Highlighting Net Span](#)
- [Highlighting Centroid \(Matching Mode Only\)](#)
- [Highlighting Abutment Breaks \(Matching Mode Only\)](#)
- [Highlighting Flight Lines \(Chaining Mode Only\)](#)

Using the Inspect Assistant Interface

You can open the inspect assistant by choosing **Window > Assistants > Inspect Assistant** in the Symbolic Editor (see [Using Symbolic Editor Assistants](#)).

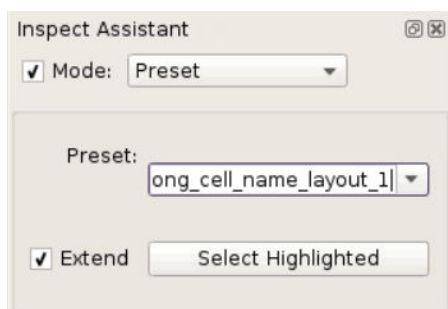


The Inspect Assistant is provided to display the highlighting with the following modes: **Preset**, **Centroid**, **Abutment Breaks**, **Net Span** and **Net**.

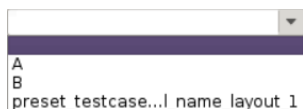
The **Mode** check box is provided to disable the highlighting in the symbolic canvas. You can deselect the check box to turn off the additional display overlay created by the corresponding mode.

Highlighting Preset

Symbolic Editor provides capability to apply and change presets on devices. See [Modifying Preset](#). In the **Preset** mode, you can figure out and identify the presets that are applied to different devices.

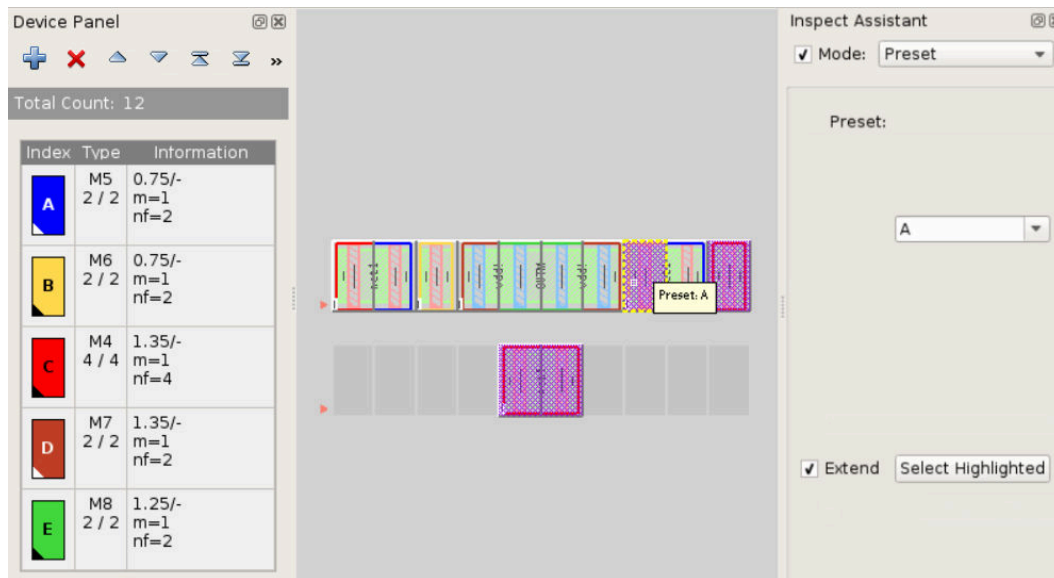


All the currently applied presets in the design are listed in the selection field.

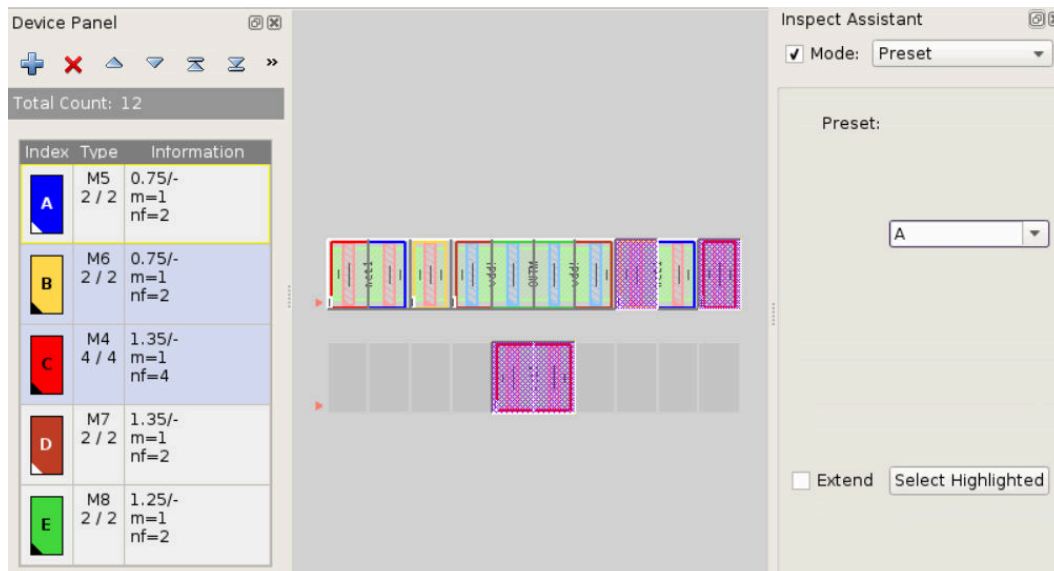


To highlight and select highlighted devices:

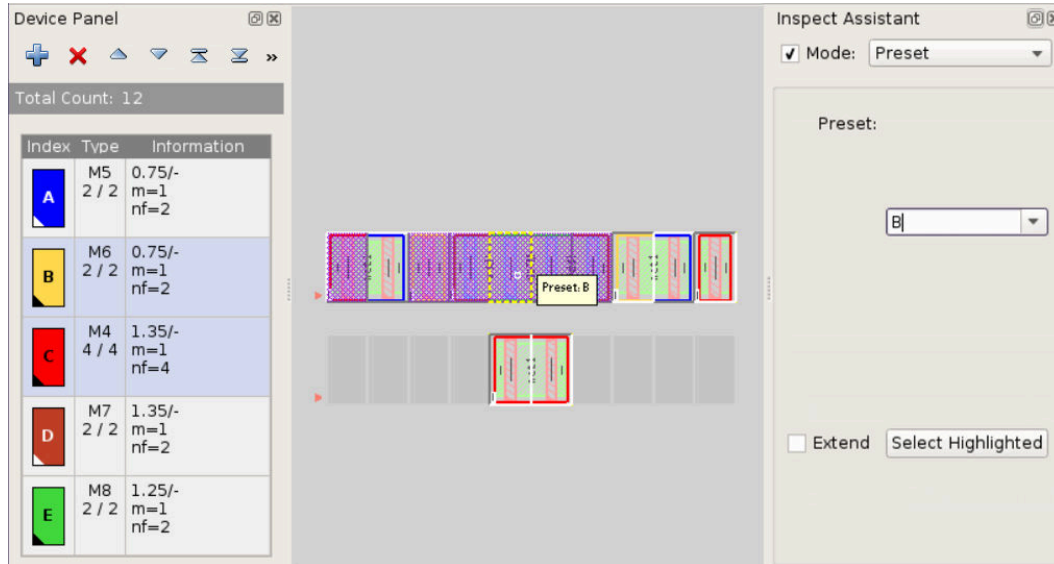
1. Selects a preset named (A), the devices applied with that preset named (A) are highlighted in the symbolic canvas. A tooltip is shown with the applied preset when you move the cursor to the highlighted device.



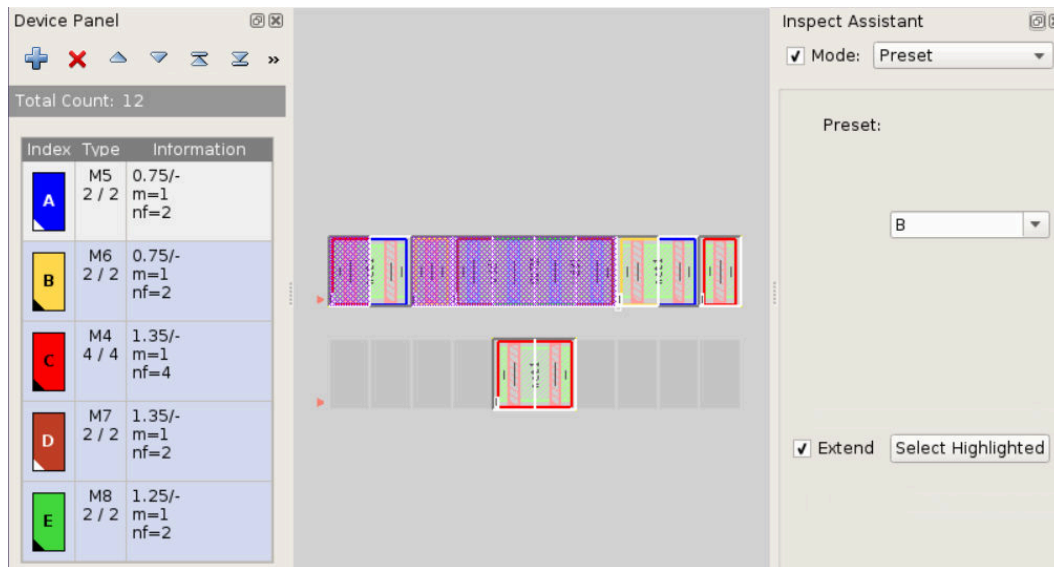
2. Clicks the **Select Highlighted** button, the current highlighted devices are selected (Index B and Index C).



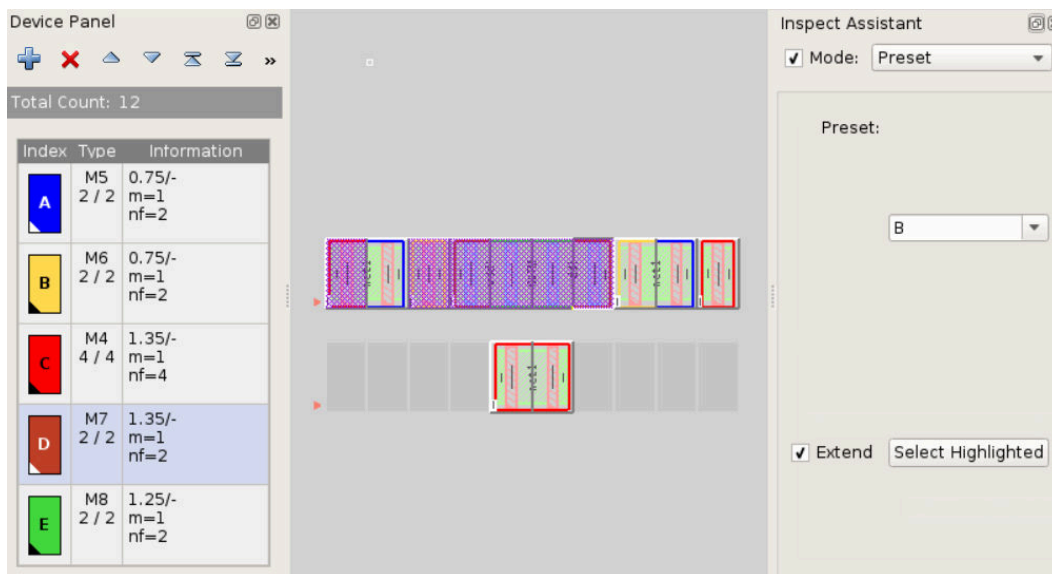
- When changing to the next preset named (B) in the selection field, the devices applied with that preset named (B) are highlighted (Index D and Index E) and the previous highlighted devices (Index B and Index C) are still selected.



- Enables the **Extend** option, the current highlighted devices (Index D and Index E) are selected and the previous selected devices (Index B and Index C) are still selected.

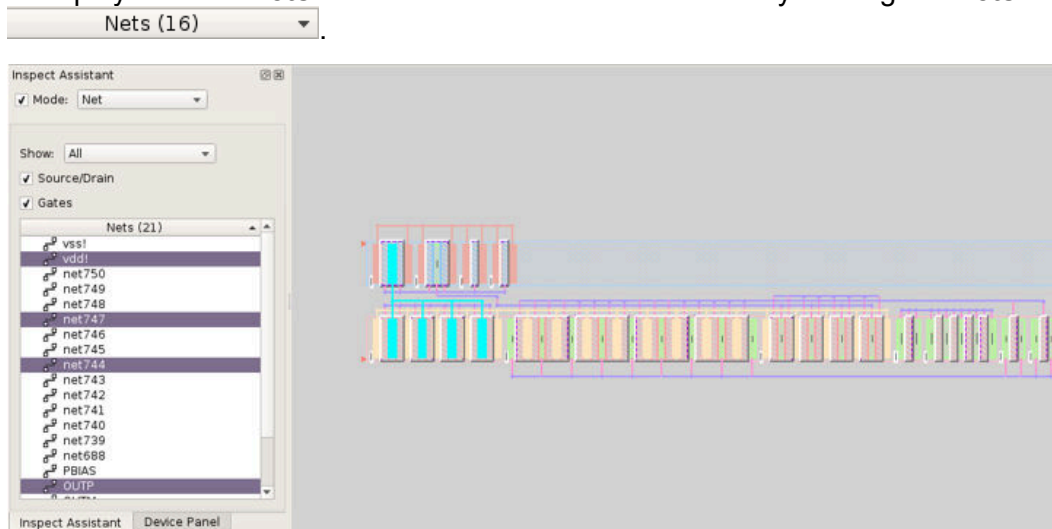


When you select a single device by clicking on canvas, all devices with the same preset are highlighted. Only single device selection is supported.



Highlighting Nets

The **Net** mode is provided to highlight the nets in the design. The total nets count is displayed in the **Nets** tab. The list of nets can be sorted by clicking the **Nets** tab



The **Show** selection field filters the nets in the **Nets** tab by selecting one of the following options:

- **All:** Displays all nets in the **Nets** tab.
- **Back Side Metal Nets:** Displays only the nets that are tagged as Back Side Metal in the **Net** tab.

You can highlight the selected the nets with other nets information from the **Nets** tab in the assistant by selecting the following options:

- **Source/Drain:** Displays the Source/Drain of selected nets.
- **Gates:** Displays the gates of selected nets.

In chaining mode, the flight lines are highlighted when the nets are selected. In matching mode, only the Source/Drain and Gates of selected nets are highlighted.

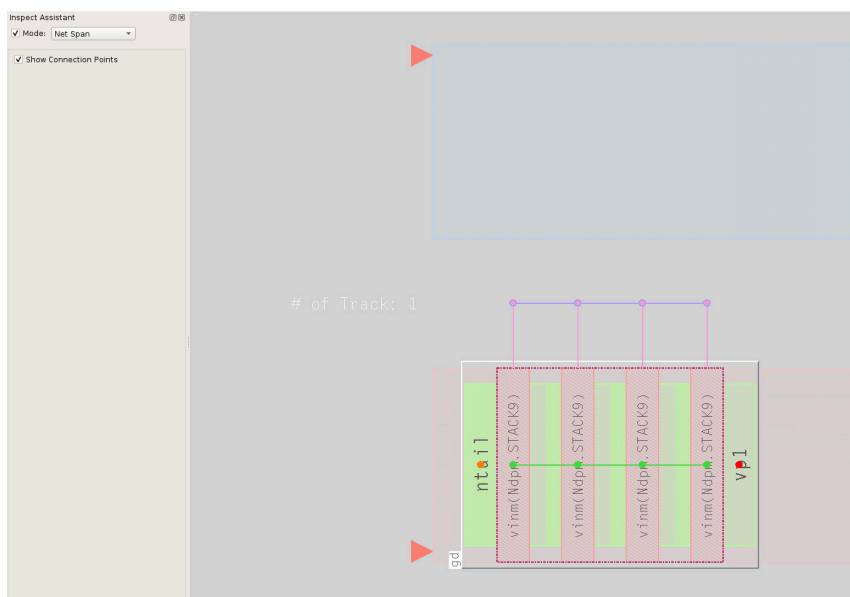
Highlighting Net Span

The **Net Span** mode displays the horizontal tracks of MOS devices in the **Matching Mode** and **Chaining Mode** of the Symbolic Editor. You can edit the placement of devices on the track to optimize routes.

Note:

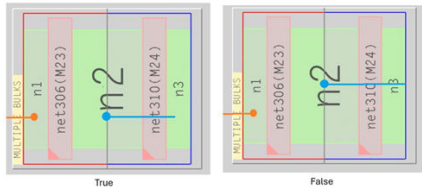
A span is the line connecting the pin on the extreme left of the net, to the pin on the extreme right of the net.

The **Show Connection Points** option allows you to enable and disable the drop points of the net.

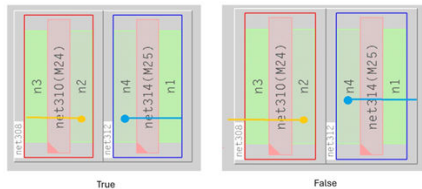


The following preferences allow you to set conditions for creating tracks between nets and devices:

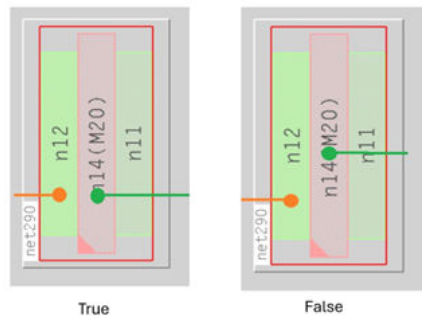
- `lxSEDTrackShareSDSameDeviceDiffNet`: When set to `true`, the adjacent source drain nets cannot share the same track. When set to `false`, the adjacent source drain nets can share the same track.



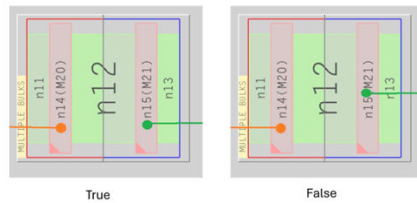
- `lxSEDTrackShareDiffBreakDiffNet`: When set to `true`, the diffusion break nets cannot share the same track. When set to `false`, the diffusion break nets can share the same track.



- `lxSEDTrackShareAdjGateSDDiffNet`: When set to `true`, the adjacent gate and source or drain nets cannot share the same track. When set to `false`, the adjacent gate and source or drain nets can share the same track.



- `lxSEDTrackShareAdjGatesDiffNet`: When set to `true`, adjacent gates in different nets cannot share the same track. When set to `false`, adjacent gates in different nets can share the same track.



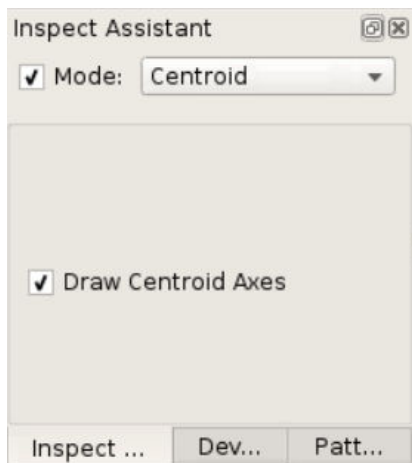
- `lxSEDTrackShareAdjGatesSameNet`: When set to `true`, adjacent gates in the same net cannot share the same track. When set to `false`, adjacent gates in the same net can share the same track.

Highlighting Centroid (Matching Mode Only)

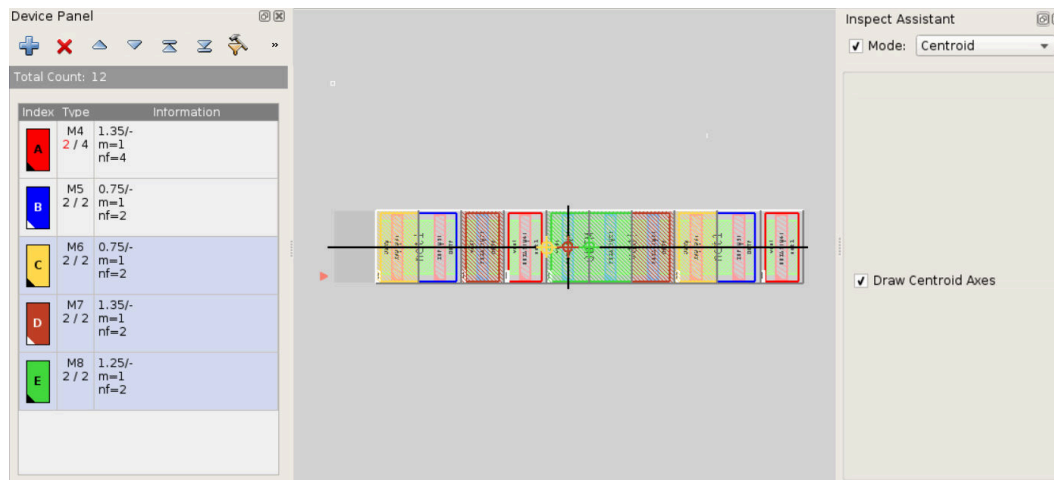
The Symbolic Editor with Matching mode provides pattern based placement capability. The Centroid mode provides easy computation and display of device group centroids to ensure that individual device groups are centered around the same center.

Note:

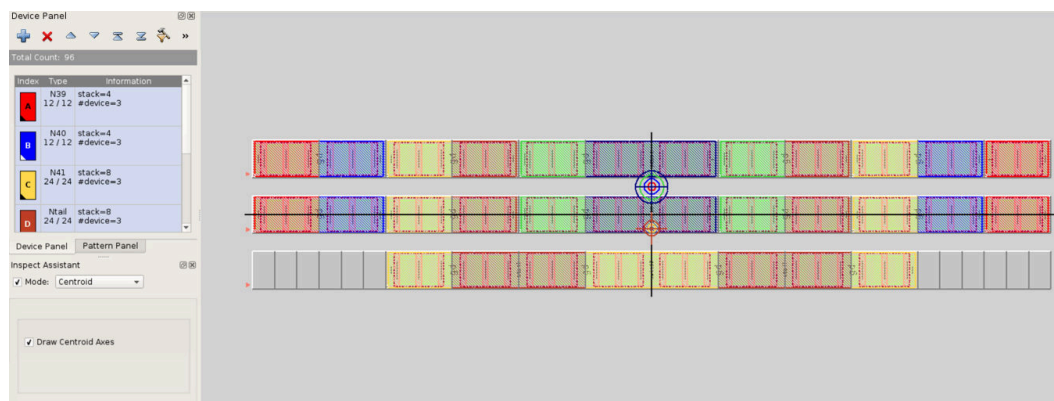
The **Centroid** mode is only available in the matching mode.



When selecting the **Centroid** mode in the assistant, the centroid for the devices selected in the device panel is displayed on symbolic canvas. The device color from device panel is used to display the centroid. In the **Centroid** mode, the selected Indexes are highlighted in the symbolic canvas and the centroids of the indexes are displayed individually.



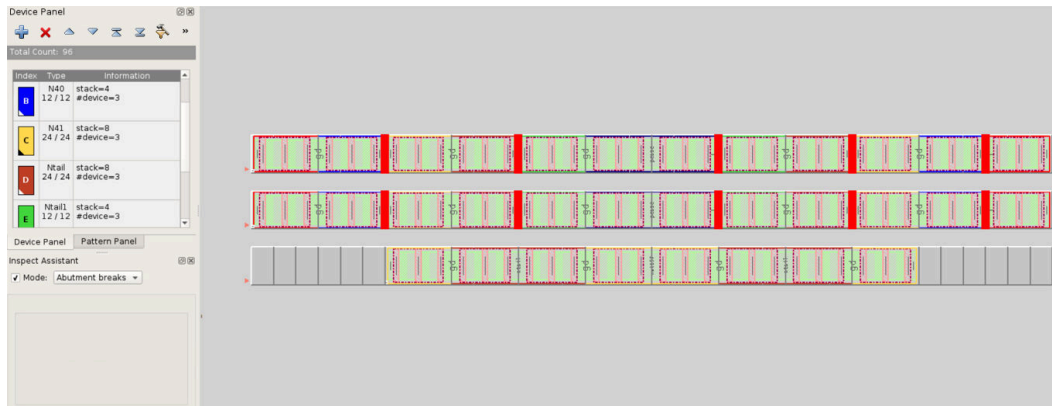
The **Draw Centroid Axes** option draws the axes for symbolic canvas. If all device group centroids coincide with the center of axes, one centroid axes can help to identify that devices are centered in the available $m \times n$ array in the symbolic canvas.



Highlighting Abutment Breaks (Matching Mode Only)

When the design with hundreds of devices placed on the symbolic canvas, the **Abutment Breaks** mode is provided to easily inspect if any abutment breaks are created.

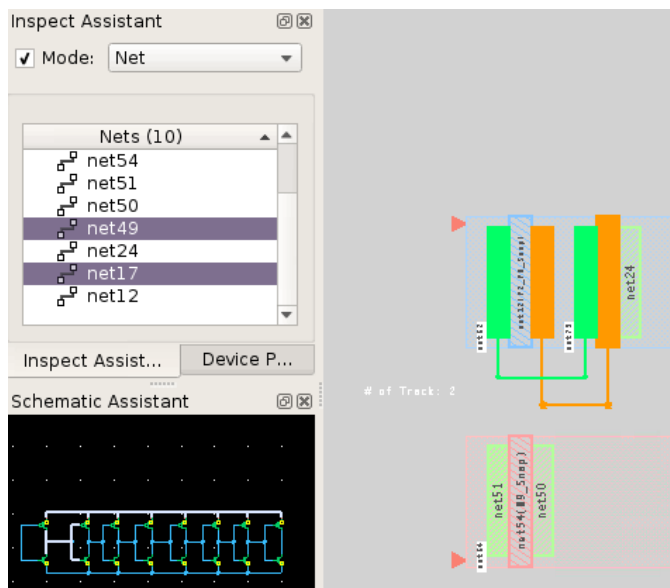
The **Abutment Breaks** mode displays the OD breaks for MOS devices in matching mode.



Highlighting Flight Lines (Chaining Mode Only)

To highlight flight lines:

1. Enable the **Enable Net Cross Highlight**  in the Symbolic toolbars.
2. In the Inspect Assistant, choose **Net** in the **Mode** field. The corresponding nets and nets count are displayed in the Inspect Assistant.
3. When selecting one or more flight lines in the Inspect Assistant, the flight lines are crossed highlighting in the Symbolic Canvas, Schematic Assistant, and Inspect Assistant.



3

Customizing the Design Environment

This chapter explains how to set the options and how to customize options to work with symbolic editor.

This chapter covers the following topics:

- [Specifying Design Options](#)
- [Specifying Cell Template \(Chaining Mode Only\)](#)
- [Adding Shape Settings \(Matching Mode Only\)](#)
- [Checking Errors](#)
- [About IxComponentType](#)

Specifying Design Options

The **Options > Design** command opens the **Symbolic Design Options** dialog box to specify the preferences for the Symbolic Editor.

The dialog box includes the following tabs:

- [General Tab](#)
- [Creation Tab](#)
- [Preview Tab](#)
- [Dummy Tab for MOS Devices](#)
- [Dummy Tab for Resistors](#)
- [Chaining Tab \(Chaining Mode Only\)](#)

General Tab

Figure 16 Symbolic Design Options - Chaining Mode

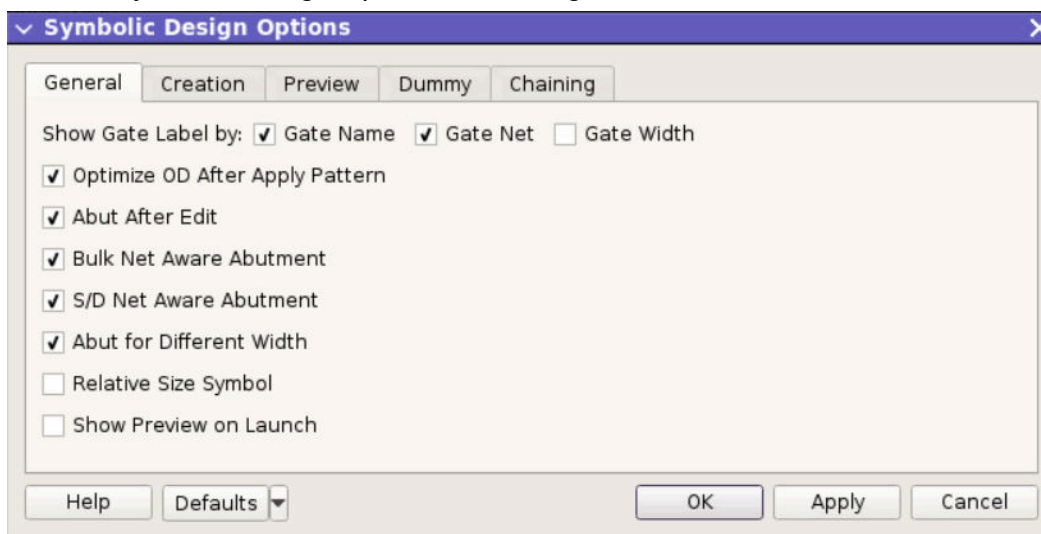
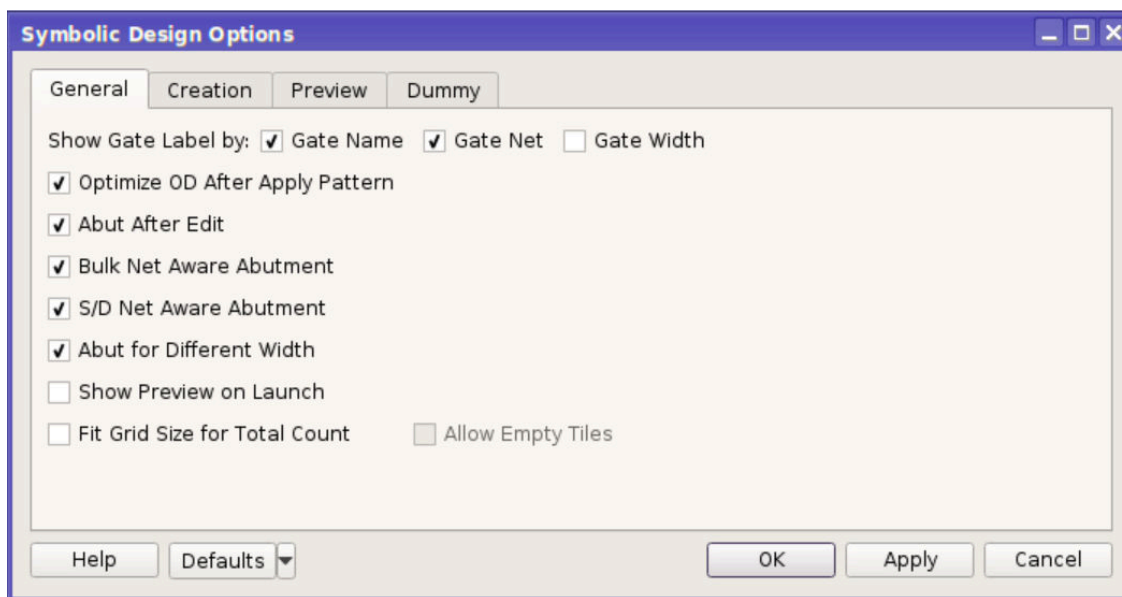


Figure 17 Symbolic Design Options - Matching Mode



This tab includes the following options:

- **Show Gate Label by:** Specify the display of gate labels by selecting one of the following options: **Gate Name**, **Gate Net**, or **Gate Width**. The default for **Gate Name** and **Gate Net** are *on*. The default for **Gate Width** is *off*.
- **Optimize OD After Apply Pattern:** When this option is turned *on*, MOS OD sharing is optimized after applying a pattern. The default is *on*.
- **Abut After Edit:** When this option is turned *on*, the devices are abutted according to the settings of abutment conditions (**Bulk Net Aware Abutment** and **S/D Net Aware Abutment** options) after you edit the devices. The default is *on*.

Two devices with different masters can be abutted only when the same abutment class (abutClass) is specified. For dummy devices, the abutClass is set based on the active device that is abutted to the dummy.



- **Bulk Net Aware Abutment:** When this option is turned *on*, the devices can be abutted only if they have the same bulk net names. When this option is turned *off*, the devices can be abutted without checking the bulk net names. The default is *on*.
- **S/D Net Aware Abutment:** When this option is turned *on*, the devices can be abutted only if they have same source/drain net names. When this option is turned *off*, a PODE (poly on OD edge) object is created when the devices are abutted with different source/drain net names. The default is *on*.
- **Abut for Different Width:** See Abut for Different Width in the **Cell Template** dialog box for details.
- **Relative Size Symbol (Chaining Mode Only):** When this option is turned *on*, the symbols are shown in three relative sizes based on gate width. The supported three relative symbol sizes represent devices with big, middle, or small gate width. The default is *off*. This option is only supported in chaining mode.
- **Show Preview on Launch:** When this option is turned *on*, the preview assistant is opened automatically and kept the previous position.
- **Fit Grid Size for Total Count (Matching Mode Only):** When this option is turned *on*, row and column would have dependency based on device total count. For example, row is 3 and column is 5. If row is changed to 5, column will be changed to 3 automatically when the **Fit Grid Size for Total Count** option is enabled. This option is only supported in matching mode.

Tcl preference: `lxFitTotalCount`

- **Allow Empty Tiles:** When this option is turned *on*, the tool allows empty tiles, when adjusting grid size for total device count. The row or column size is increased or decreased one by one and the total grid number (Row x Col) is the smallest number greater than device total count.

When this option is *off*, the tool does not allow empty tiles and the grid size must be the factor of total device count. The default is *off*.

This option is available when the **Fit Grid Size for Total Count** option is enabled.

Tcl preference: `lxAllowEmptyTiles`

Creation Tab

Use this tab to specify the settings for realized devices.

The screenshot shows the 'Symbolic Design Options' dialog box with the 'Creation' tab selected. The 'Spacing' section has 'Same Row' set to 0.5 Microns, 'Different Row' set to 1.0, and 'Empty tile size' set to Automatic. The 'Create Label' section has 'From' set to Pin, 'Height' set to 0.2, 'Font' set to stick, and 'Justification' set to a 3x3 grid. The 'Post-Process' section has 'Procedure Name' set to an empty field, 'Type' set to Post Devices Placed, and 'Group Realized Design' and 'Auto-delete Constraint' checked. The 'Keep Previous Layout Position' checkbox is also checked, and 'Keep Objects from Layout' is unchecked. The 'OK', 'Apply', and 'Cancel' buttons are at the bottom right.

This tab includes the following options:

- **Same Row Spacing:** Specify the spacing with all values (positive values, 0, negative values) between two packs in one row. The unit can be **Microns** (default) or **Polys**. The default value is 1. The `sameRowSpacing` property is added in the **Properties** list when the spacing is specified.

When spacing type is **Polys**, the same row spacing is calculated between **Poly Layer** or **Device Boundary**. When between **Device Boundary**, the same row spacing is calculated by **Active Layer** or **Instance Boundary** of the **Device Boundary** options.

- **Different Row Spacing:** Specify the spacing with all values (positive values, 0, negative values) between two rows when the unit is **Microns**. Specify the spacing with all integers (positive integers, 0, negative integers) when the unit is **Fins** (available when `wideAlignment` is defined in the Technology Library). The `diffRowSpacing` property is added in the **Properties** list when the spacing is specified.
- **Specify Boundary LPP:** When this option is enabled, the specified **Boundary LPP** is used to replace **Active Layer** and **Instance Boundary** to apply the spacing based on specified LPPs.



Spacing

Same Row: 1 Microns Boundary LPP: NWELL drawing

Different Row: 8 Fins Boundary LPP: NWELL drawing

☒ Specify Boundary LPP

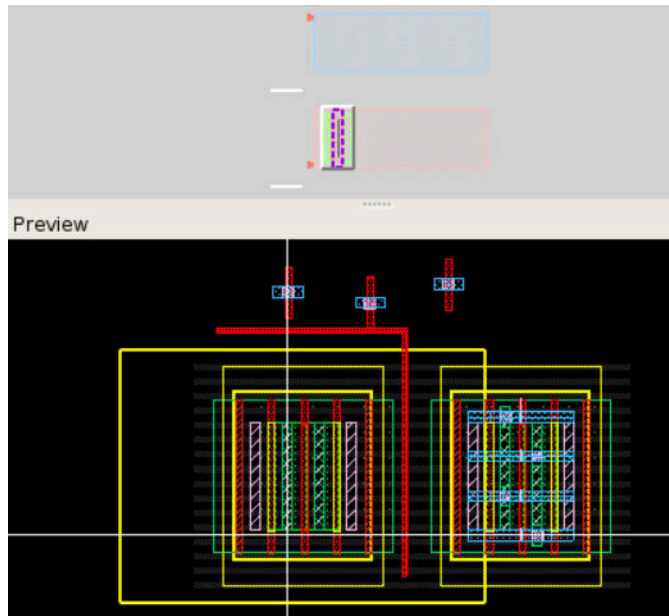
If the device doesn't have specified LPPs, the tool uses the **Instance Boundary** for spacing.

- **Device Boundary:** Specify the spacing of the instance by selecting one of the following options: **Active Layer** or **Instance Boundary**. When selecting the **Active Layer** option, the row spacing is calculated according to the active layer setting. If the active layer is not defined, the setting of `leInstBoundary` is used. When selecting the **Instance Boundary** option, the row spacing is calculated according to the setting of `leInstBoundary`. The default is **Active Layer**.
- **Empty tile size:** Specify the size of an empty tile by selecting one of the options: **Automatic** or **1 Poly Pitch**. When selecting **Automatic**, the width of an empty tile is the maximum width of the devices. The default is **Automatic**. When the value is **1 Poly Pitch**, the width of an empty tile is 1 poly pitch of the current placement. Use **Polys as Same Row Spacing** when using **1 Poly Pitch** for an empty tile size.
- **Create Label:** Enable this option to create instance labels with the following settings.
- **Post-Process:** You can specify a procedure name in the **Procedure Name** field. You can select the following types to define when to executed the procedure:
 - **Post Device Placed:** When this option is enabled, the procedure is executed after devices are placed and before routing or guard-ring is created.
 - **Pre Realized:** When this option is enabled, the procedure is executed when previewing or realizing.

If both options are enabled, the procedure is executed twice with different arguments.

- **Group Realized Design:** When this option is turned *on*, the realized devices are grouped together as one group when instances are being created in the layout. The default is *on*.

- **Auto Delete Constraint:** When this option is turned *on*, the constraint is deleted automatically after removing the group design. The default is *on*. This option is disabled when the Group Realized Design option is *off*.
- **Keep Previous Layout Position:** When this option is turned *on*, the realized devices are re-created in the layout canvas at their original positions, if you edited them in the Symbolic Editor.
- **Keep Objects From Layout:** When this option is turned on, the collected layout objects are displayed in the **Preview** assistant. The extra layout objects are shown on preview for reference. The default is off.



Post-Process Procedure

This procedure is called whenever the objects are previewed or realized in the Symbolic Editor.

Syntax:

```
procedure_name { varListOfLists }
```

Arguments:

Table 3 Post-process Arguments

Name	Type	Description
design	oaDesign	preview oaDesign

Table 3 Post-process Arguments (Continued)

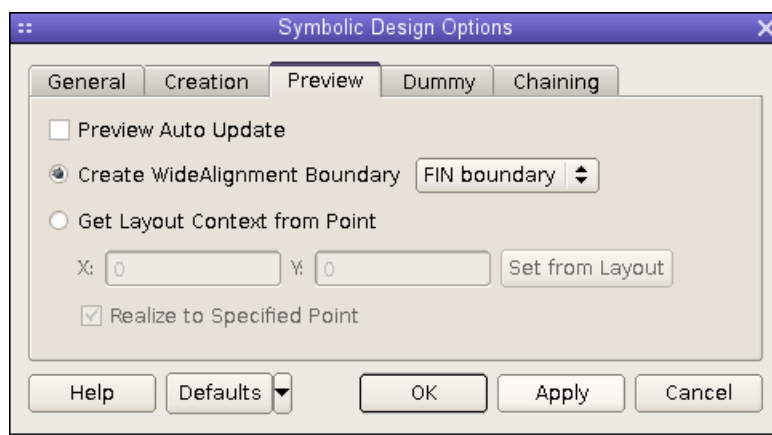
Name	Type	Description
mode	string	chain_mode or matching_mode
Type	string	MOS, Resistor, or Instance
event	string	postDevicesPlaced or preRealized

Example:

```
sedPostProc {data} {....}
```

Preview Tab

You can specify the preview settings in the tab.



This tab includes the following options:

- **Preview Auto Update:** When this option is turned *on*, the preview assistant is synchronized with edits on the symbolic canvas. The default is *off*.
- **Create WideAlignment Boundary:** Selecting this option creates the selected wide alignment boundary in the preview assistant. This option is only available for FinFET MOS devices. The default is *on*.

When the unit of **Different Row Spacing** is *Microns*, you can specify **None** to skip wideAlignment snapping in preview.

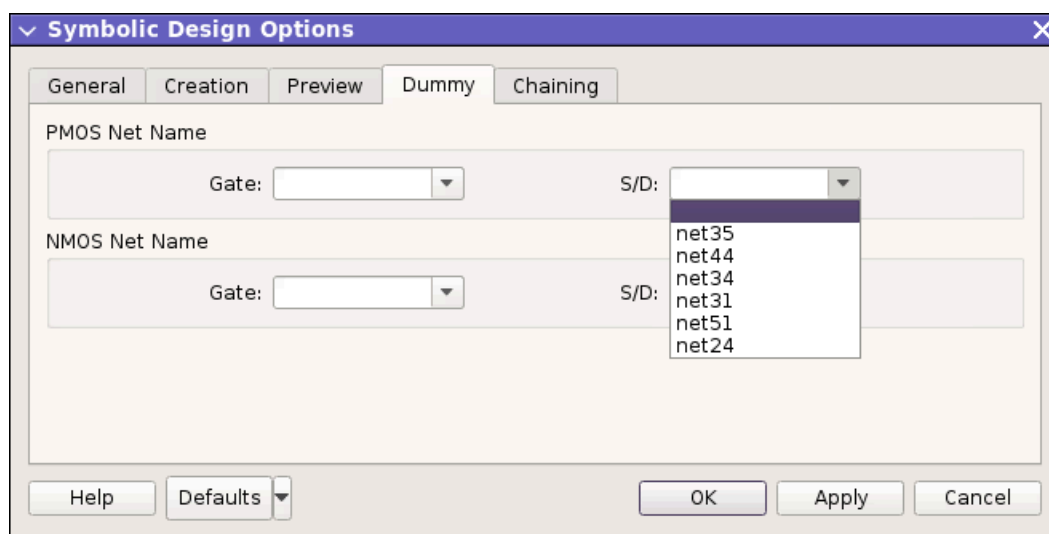
The wide alignment boundaries can be specified in the **Create > Boundary** command in the layout canvas. (see *Creating and Editing Boundaries* in the *Custom Compiler Layout Editor User Guide for Creating and Editing Objects*).

- **Get Layout Context from Point:** Selecting this option displays the layout context, including the track region, track pattern, and wide alignment boundary around the specified point (X, Y) in the preview assistant. You also can click the **Set from Layout button** to get the point (X, Y) in the layout canvas.

Selecting the **Realize to Specified Point** option creates the devices directly at the specified point (X, Y) in the layout canvas when realizing. The **Realize to Specified Point** option has higher priority than the [Keep Previous Layout Position](#) option in the **Creation** tab. When the **Realize to Specified Point** option is off, the devices are realized at their original positions if the **Keep Previous Layout Position** option is on.

Dummy Tab for MOS Devices

You can specify the dummy MOS devices settings in the tab.

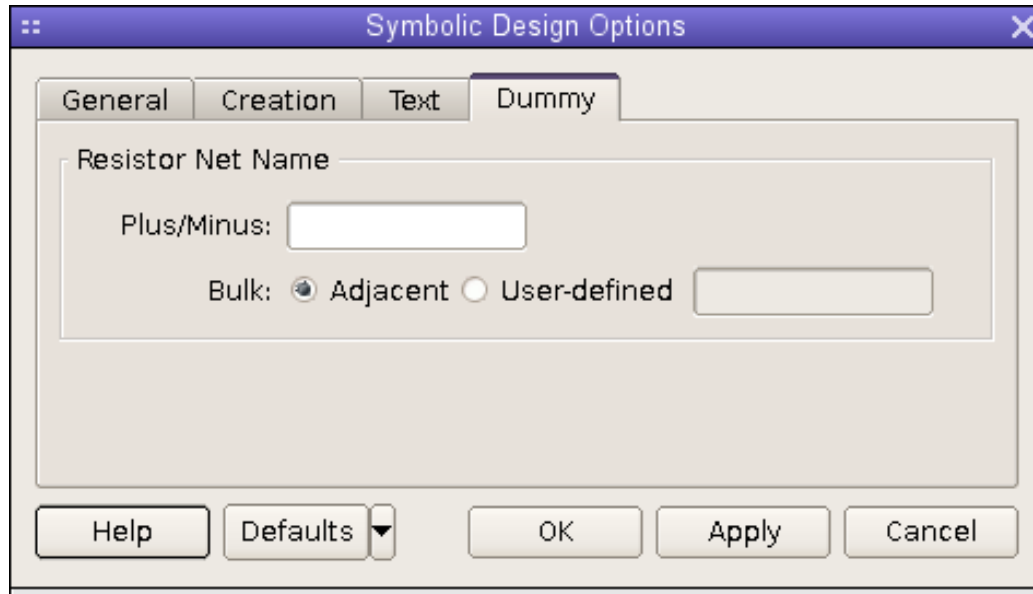


This tab includes the following options:

- **PMOS Gate Net Name:** Specify a single net name for all dummy PMOS devices. The selection field lists all connection nets.
- **PMOS S/D Net Name:** Specify a single source/drain net name for all dummy PMOS devices. The selection field lists all connection nets.
- **NMOS Gate Net Name:** Specify a single net name for all dummy NMOS devices. The selection field lists all connection nets.
- **NMOS S/D Net Name:** Specify a single source/drain net name for all dummy NMOS devices. The selection field lists all connection nets.

Dummy Tab for Resistors

This tab is used to specify settings for dummy resistors.

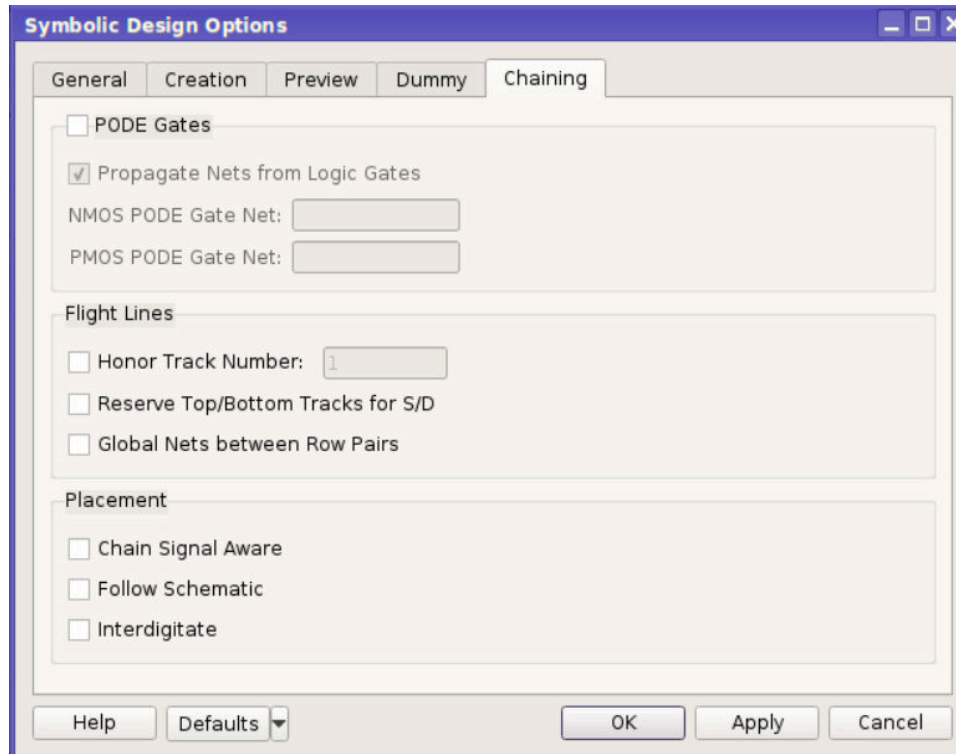


The **Resistor Net Name** field includes the following options:

- **Plus/Minus:** Specify a net name for all dummy resistors.
- **Bulk:** Specify the bulk net name for all dummy resistors by selecting one of the following options: **Adjacent** or **User-defined**.

Chaining Tab (Chaining Mode Only)

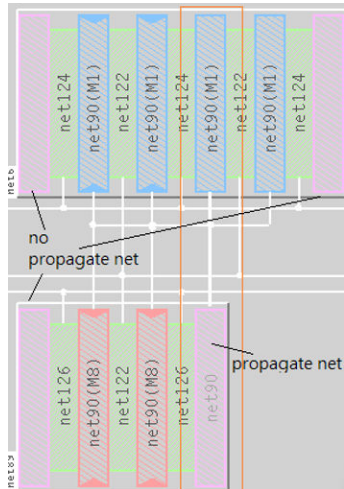
Use this tab to specify settings for PODE gates, flight lines, and placement in chaining mode.



PODE Gates

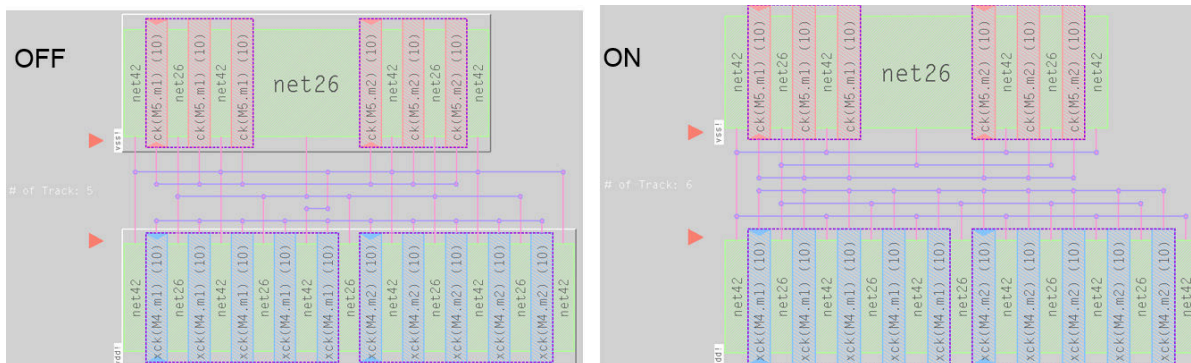
Enables this option to display the PODE gates on the left and right sides of object packs and the following option and fields. The default is off. The Maximum Poly Pitch number is displayed on the canvas.

- **Propagate Nets from Logic Gates:** Selecting this option propagates nets from logic gates.



- **NMOS PODE Gate Net:** Specify a single net name for all NMOS PODE devices.
- **PMOS PODE Gate Net:** Specify a single net name for all PMOS PODE devices.

- **Global Nets between Row Pairs:** When selected, the flight lines connected to global nets are displayed between PN pairs. The default is off.

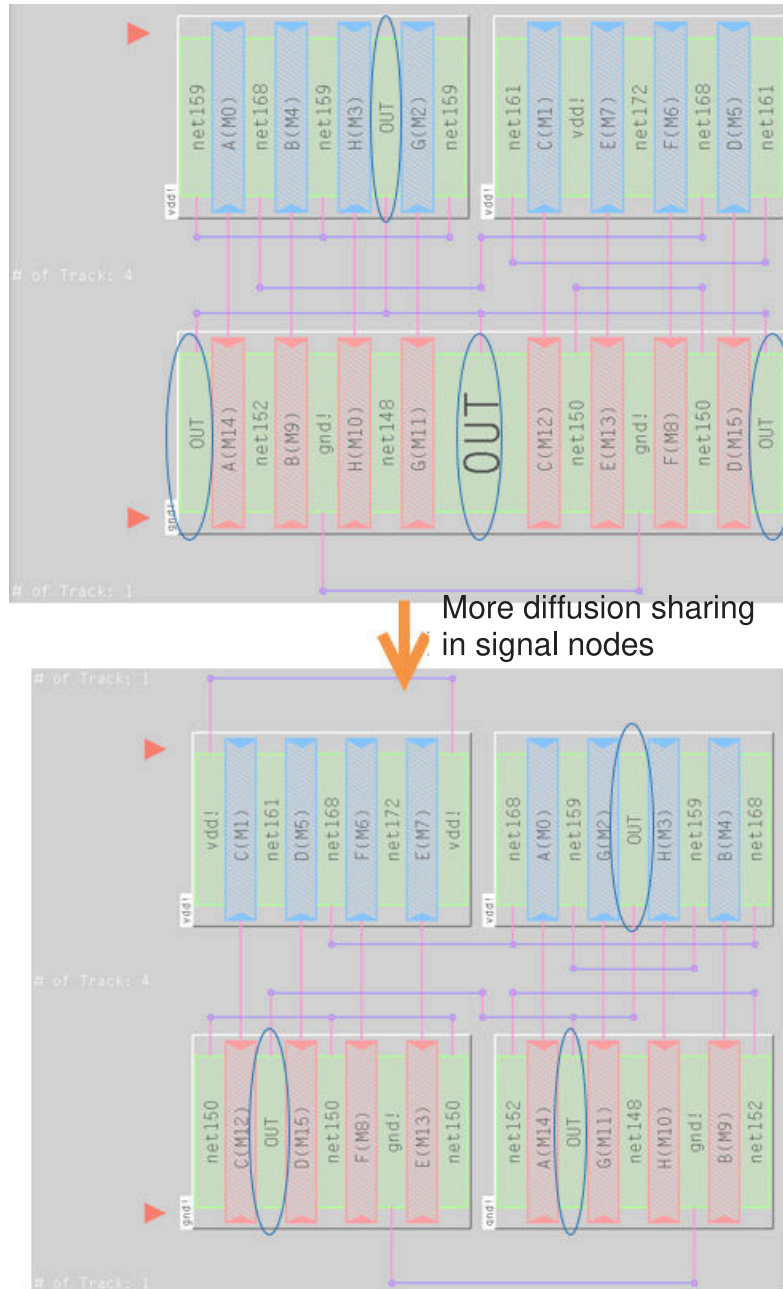


Placement

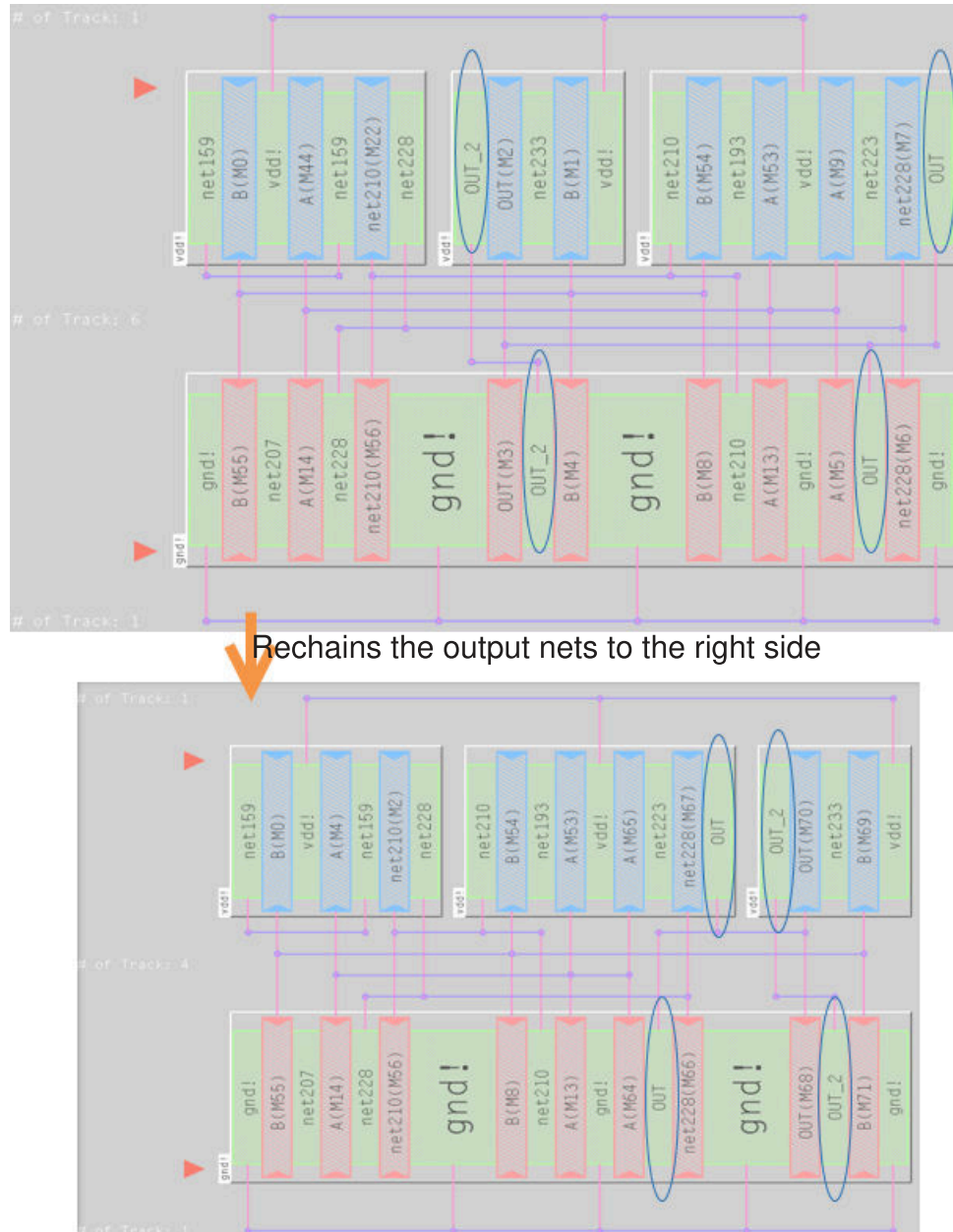
This section includes the following options for placement:

- When the **Chain Signal Aware** option is turned on, the nets are chained with the following rules:
 - Prefers diffusion sharing for signal nodes over the power or ground node.
 - Attempts to order the devices connected to input and output terminals in a left to right order.

Example1, net161 shares diffusion in the top row and OUT shares diffusion in the bottom row when turning on the **Chain Signal Aware** option.



Example 2, rechains the output nets from the middle of the design to the right side of the design when turning on the **Chain Signal Aware** option.



- When the **Follow Schematic** option is turned on, the devices are placed based on the device location in the schematic. The default is off.

In this mode, the devices are placed in left to right order as they are encountered in the schematic design. For devices in same horizontal position, the bottom to top order is followed for symbolic placement. The source drain terminals are swapped as required to optimize the diffusion sharing.

Tcl preference: `lxFollowSchematic`

- When the **Interdigitate** option is turned on, the tool performs interdigitation for multi-finger and m-factor devices. The default is off.

This option is only supported for two devices at a time and these two devices are placed in the ABBA pattern. The source drain terminals are swapped as required to optimize the diffusion sharing.

Tcl preference: `lxSEDInterdigitate`

Specifying Cell Template (Chaining Mode Only)

The **Options > Cell Template** command allows you to customize transistor placement in the **Cell Template** dialog box which also can be enabled from SDL menu (**SDL > Symbolic Editor > Cell Template**). The cell template is a set of transistor placement and realization constraints. These constraints include the key parameters for the cell layout during floor planning, such as the spacing between P/NMOS, the strip height, the PN alignment mode, and the power layer.

When you use a cell template, the symbol is created from the settings in the template, including the row pattern, chaining placement, and any constraints that apply. These settings are used to realize the result.

The placement results can be displayed in chaining mode for further manual optimization like gate swapping, moving, aligning, splitting, merging, and folding.

This dialog box includes the following tabs and dialog boxes:

- [General Tab](#)
- [Cell Height Tab](#)
- [Placement Tab](#)
- [Saving and Loading Cell Templates](#)

General Tab

Cell Template

General | Cell Height | Placement

Row Pattern: **PN** Row Number: **2**

P Row Alignment: **Top** N Row Alignment: **Bottom**

Spacing

Same Row: **1** Microns

Different Row: **8** Fins

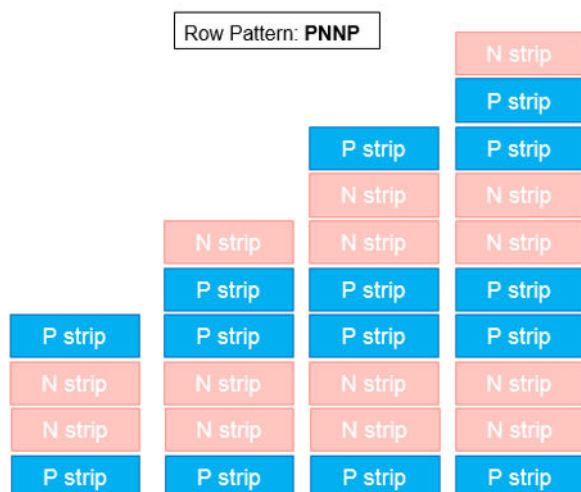
☐ Row Pair Presets

PN Preset: NP Preset:

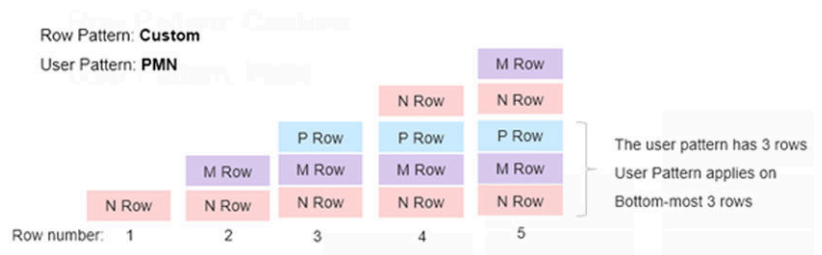
Load... Save... OK Apply Cancel

You can specify the row pattern and general settings in the tab. This tab includes the following options:

- **Row Pattern:** Specifies the row pattern from top to bottom, included **PN** (PMOS to NMOS), **NP** (NMOS to PMOS), **PNNP**, **NPPN**, or **Custom**. The default is PN.
- **Row Number:** Specifies the number of rows. The default is 2. When the **Row Pattern** is not **Custom**, the row number must be an even number.

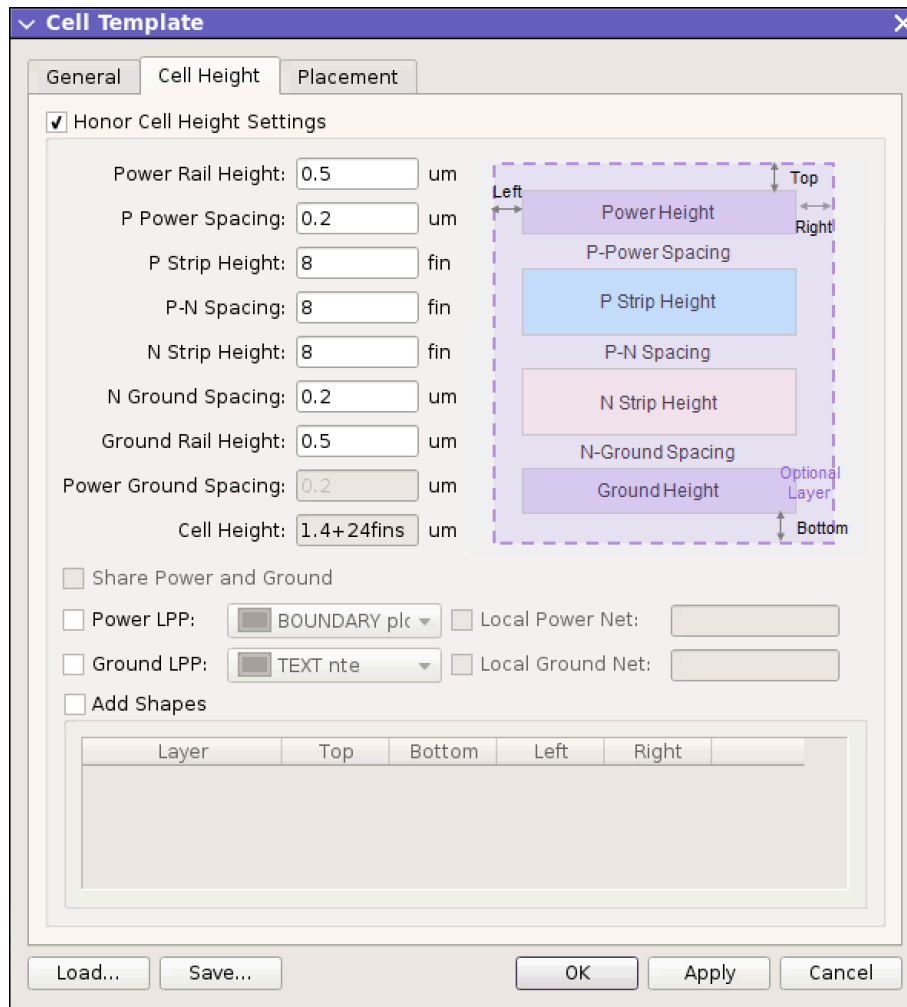


- **P Row Alignment:** Specifies the alignment mode while placing a PMOS within the P strips by selecting **Top**, **Bottom**, **Center**, **Power** or **Ground**. The default is **Bottom**. When Wide Alignment is defined, the **Center** mode is disabled.
- **N Row Alignment:** Specifies the alignment mode while placing a NMOS within the N strips by selecting **Top**, **Bottom**, **Center**, **Power** or **Ground**. The default is **Top**. When Wide Alignment is defined, the **Center** mode is disabled.
- **User Pattern:** Specifies user-defined pattern when the **Row Pattern** is **Custom**. This pattern applies on the most-bottom n rows, n is the length of this pattern. A tooltip displays the syntax of user-defined pattern.



- **Same Row Spacing:** Specifies the spacing between two packs in the same row. The value is synchronized with the value in the **Same Row Spacing** text field in the *Symbolic Design Options* form.
 - **Same Row Spacing Type:** Specifies the unit of Same Row Spacing. The value can be Microns or Polys. The default value is **Microns**.
- **Different Row Spacing:** Specifies the spacing between two rows. The value is synchronized with the value in the **Different Row Spacing** text field in the **Symbolic Design Options** dialog box. This text field is disabled when the **Honor Cell Height Settings** option is selected.
 - **Different Row Spacing Type:** Specifies the unit of the P/N Strip Height, P-N Spacing and Different Row Spacing. The default value is **Microns**. When Wide Alignment is defined, the spacing type can be **Fins**.
- **Row Pair Preset:** To specify the row pair preset parameters in the following selection fields, you must specify the preset library in the **SDL Options** dialog box. This option is disabled when the **Row Pattern** is **Custom**. For more information, see **Parameter Preset** in Main Tab.
 - **PN Preset:** The preset settings of P-N row pairs. This option is enabled when the row pattern contains PN row pairs.
 - **NP Preset:** The preset settings of N-P row pairs. This option is enabled when the row pattern contains NP row pairs.

Cell Height Tab



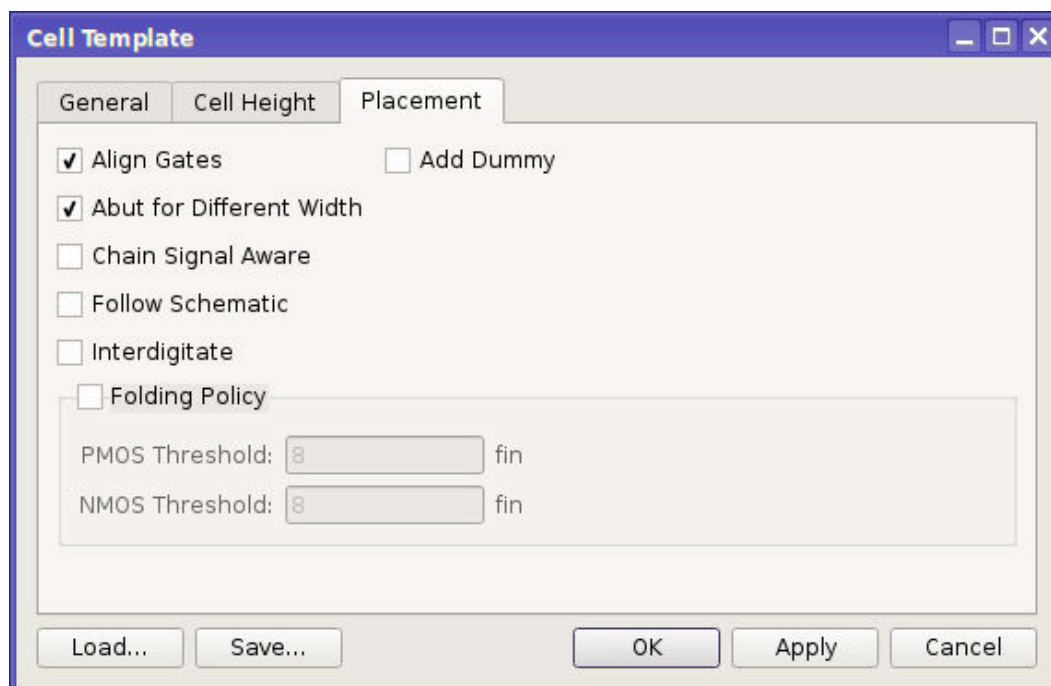
This tab includes the following options:

- **Honor Cell Height Settings:** This option is disabled when the [Row Pattern](#) is **Custom**. When selected, specify:
 - **Power Rail Height:** The height for power rails.
 - **P-Power Spacing:** The spacing between the P strip and power rail.
 - **P Strip Height:** The height for P strips.
 - **P-N Spacing:** The spacing between the P strip and N strip.
 - **N Strip Height:** The height for N strips.

- **N Ground Spacing:** The spacing between the N strip and ground rail.
- **Ground Rail Height:** The height for ground rails.
- **Power Ground Spacing:** From two different row pairs, the spacing between the nearby power and ground shapes, the nearby power and power shapes, or the nearby ground and ground shapes. The default is 0.2. This option is enabled when the **Row Pairs** is greater than 1.
- **Cell Height:** The estimated cell height is the summation of **P-N Spacing**, **P Strip Height**, **N Strip Height**, **P-Power Spacing**, **N-Ground Spacing**, **Power Rail Height**, and **Ground Rail Height** values.
- **Share Power and Ground:** Enabling this option shares the same power or ground shape for two different row pairs in the PNNP or NPPN pattern.
- **Power LPP:** The layer to be used for global power nets. The default is **off**.
 - **Local Power Net:** If a net name list is specified, the first listed net name existing in the Symbolic Editor is treated as the power net. For example, **Local Power Net** is `a1`, `a2`, `a3`. If `a1` can be found in the Symbolic Editor, `a1` is treated as the power net. If `a1` cannot be found and `a2` is found, `a2` is treated as the power net.

If the option is disabled, the power rail shape is assigned with the global power net name, such as `vdd`. For example, if the option is disabled, all nets `net1`, `net2`, `net3...netN` in the canvas are collected and sorted by net name. If `net1` is a global power net name, `net1` is treated as the power net. If `net1` is not a global power net name and `net2` is a global power net name, `net2` is treated as the power net.
- **Ground LPP:** The layer to be used for global ground nets. The default is **off**.
 - **Local Ground Net:** If a net name list is specified, the first listed net name existing in the Symbolic Editor is treated as the ground net. If the option is disabled, the ground shape is assigned with the global ground net name, such as `vss`. See [Local Power Net](#) for details.
- **Add Shapes:** If you enable **Add Shapes**, list the layers you want to use for such shapes at the bottom of the form by clicking the **Insert Layer** and **Delete Layer** buttons. You can specify different extension distances from **Top** to **Bottom** and **Left** to **Right** for each layer.

Placement Tab

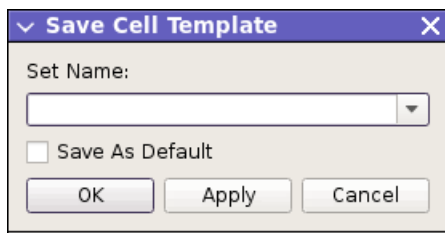


This tab includes the following options:

- **Align Gates:** When *on*, the initial placement aligns the poly gates; when *off*, initial placement does not align poly gates.
- **Add Dummy:** When this option is turned *on*, the initial placement is completed with dummy gates for perfect gate alignment.
- **Abut for Different Width:** When this option is turned *on*, MOS devices with different widths can be abutted. This option is available for the initial placement and editing commands.
- **Chain Signal Aware, Follow Schematic, and Interdigitate:** See [Placement](#).
- **Folding Policy:** When this option is turned *on*, the gate folding policy of PN transistors is determined by a threshold value. The threshold value is either a specific width at which the gate folds, or the number of fins at which the gate folds.
 - **PMOS Threshold:** Specify the threshold width or fin number for folding PMOS devices.
 - **NMOS Threshold:** Specify the threshold width or fin number for folding NMOS devices.

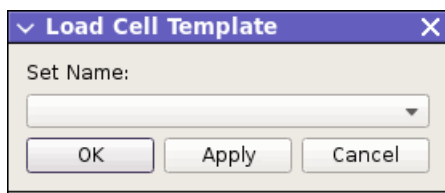
Saving and Loading Cell Templates

You can save the cell template settings in the **Save Cell Template** dialog box by clicking the **Save** button.



When saving the cell template, you can specify the settings as default by selecting the **Save As Default** option. The default settings are loaded when the Symbolic Editor is invoked.

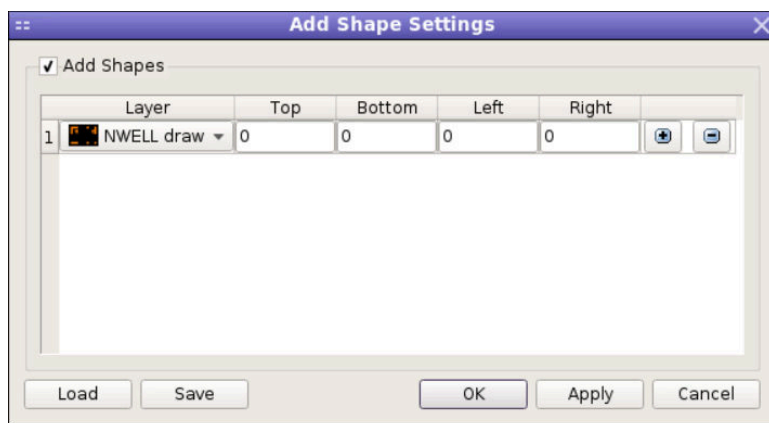
You can import the existing cell template in the **Load Cell Template** dialog box by clicking the **Load** button.



Adding Shape Settings (Matching Mode Only)

To add shapes with specifying settings using Symbolic Editor Matching Mode:

1. Choose **Options > Add Shape Settings** to open the **Add Shape Settings** dialog box.



2. Click **Add Shapes** to enable the **Add Shapes** table.

3. Specify layers with different sizes using the following options:

Option	Description
Layer	Specify the layer purpose of this additional shape.
Top	Specify the extension distance between the shape and the top of the cell.
Bottom	Specify the extension distance between the shape and the bottom of the cell.
Left	Specify the extension distance between the shape and the left of the cell.
Right	Specify the extension distance between the shape and the right of the cell.
Insert Layer	Click the button to add a new layer.
Delete Layer	Click the button to delete the layer.

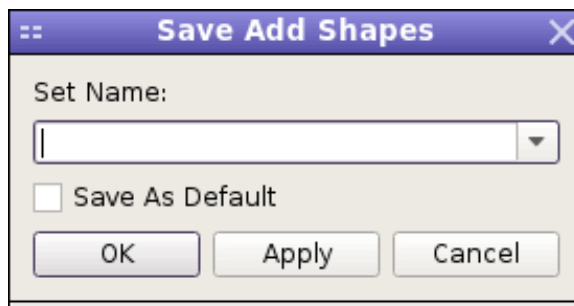
4. The specified shapes are added when using **Preview** or **Realize**.
5. Or click **OK** or **Apply** to add the specified shapes.

Saving Add Shape Settings

You must save settings permanently to a file to reuse them later.

To save add shapes settings:

1. In the **Add Shape Settings** dialog box, click the **Save** button to open the **Save Add Shapes** dialog box.



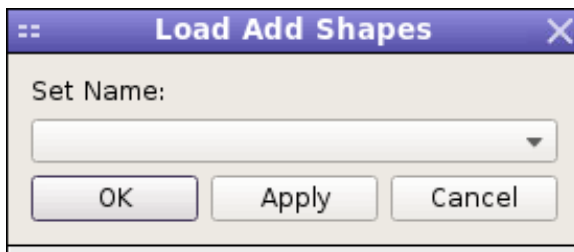
2. Type the name you want to assign to the set file in the **Set Name** text field.
3. (Optional) Click the **Save As Default** option to make the file as default layer settings.
4. Click **OK** or **Apply**. The tool saves the settings to a file.

Loading Add Shape Settings

In the **Add Shape Settings** dialog box, you must load the settings so that they appear in the **Set Name** list. You can load settings if you saved it permanently to a file. For instructions, see [Saving Add Shape Settings](#).

To load an add shapes set file:

1. In the **Add Shape Settings** dialog box, click the **Load** button to open the **Load Add Shapes** dialog box.



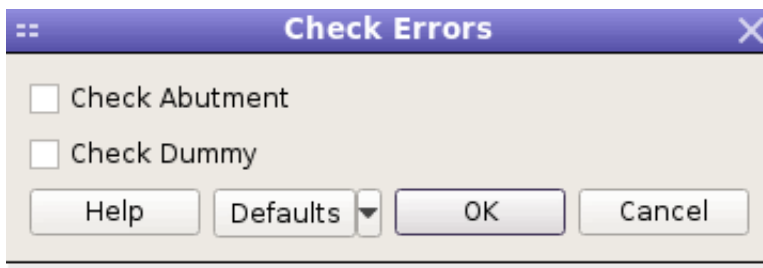
2. To load the set file, locate and select a set file in the **Set Name** selection field in the **Load Add Shapes** dialog box.

Checking Errors

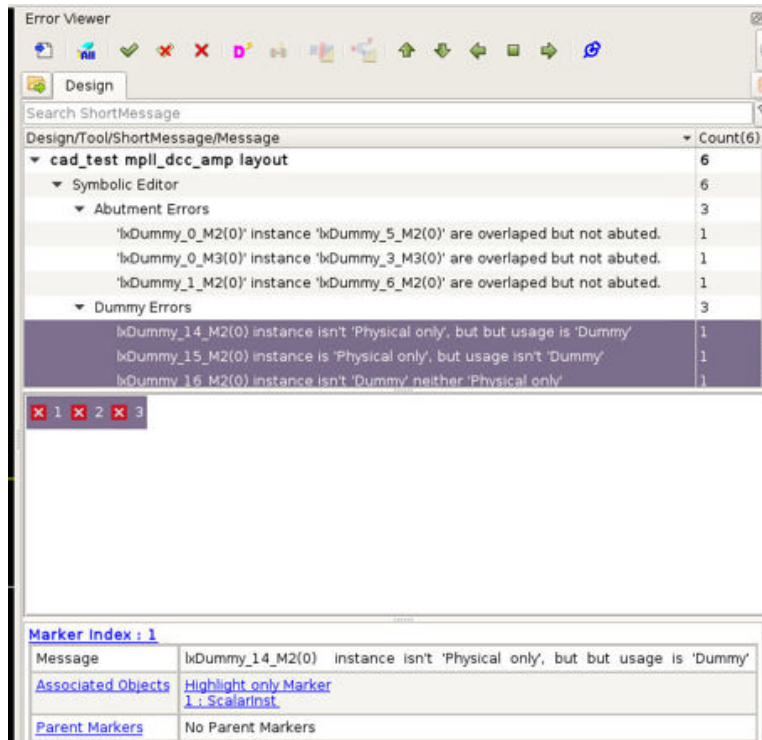
Symbolic Editor only understands active devices and dummy devices. If Symbolic Editor is enabled on a set of physical devices which do not have fig-group and constraint and devices are not correctly marked as dummies, or are not found abutting, then the result of symbolic placement may confuse. For example, to ensure that all devices that are not associated with schematic instances are brought inside Symbolic Editor correctly is to mark these devices as dummies. You can check the design before placing into Symbolic Editor using **Check Error**.

To check the errors before using Symbolic Editor:

1. Choose a set of devices selected from the layout editor.
2. Choose **SDL > Symbolic Editor > Check Error** to open the **Check Errors** dialog box.



3. Select the **Check Abutment** and **Check Dummy** options and then click **OK** to check dummy attributes or abutments on the selected devices.
4. The results of checking devices without logical counterpart, missing dummy attributes, and device pairs that should be abutted are listed in the Error Viewer with error-markers.



5. To fix the issues,
 - Make all devices associated with Dummy markers to be made dummies by setting relevant properties.
 - Or all device-pairs with Abutment markers could be triggered for abutment from their current positions.
 - Or manually update the design with resolutions.
6. Enable Symbolic Editor to optimize the design placement with correct device attributes.

About lxComponentType

This CDF attribute is used to define parameters. See About lxComponentType.

This section covers the following topics:

- [Specifying Placement Order in Chaining Mode](#)
- [Generating Instances With Equal Width in Chaining Mode](#)

Specifying Placement Order in Chaining Mode

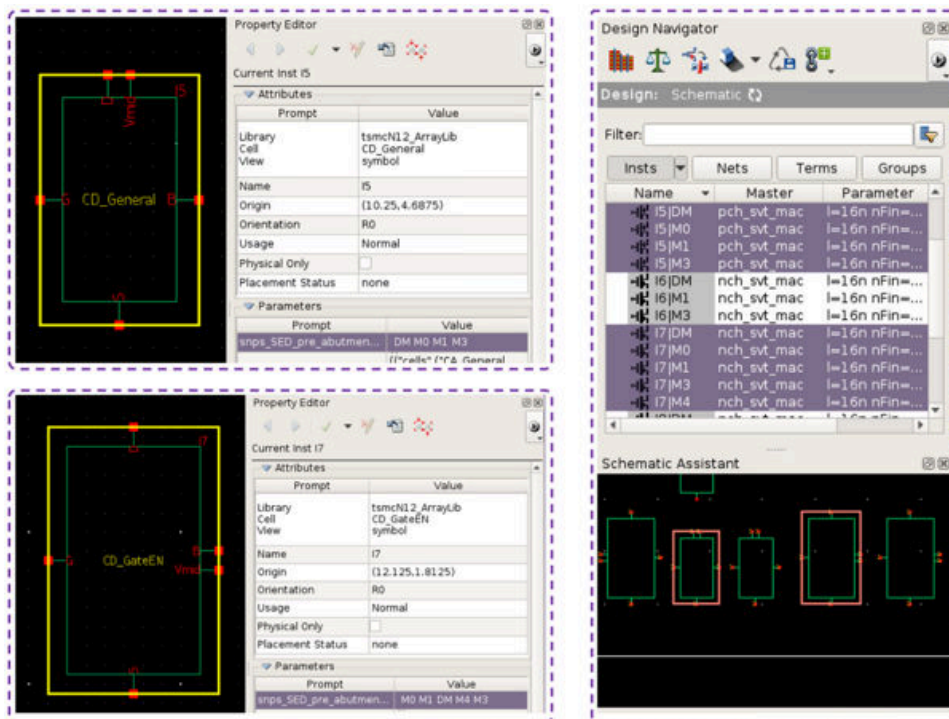
The `lxPlacementOrderParam` keyword defined in `lxComponentType` allows to specify the custom placement order in the PyCell schematic sub-master. The CDF/OA property defines a list of instance names as the placement order.

To consider the precedence, if cell-level CDF has no keyword, the SED chaining mode checks the OA property on schematic view. If there is no any specified custom placement order or the CDF/OA property is empty, SED chaining mode works as before.

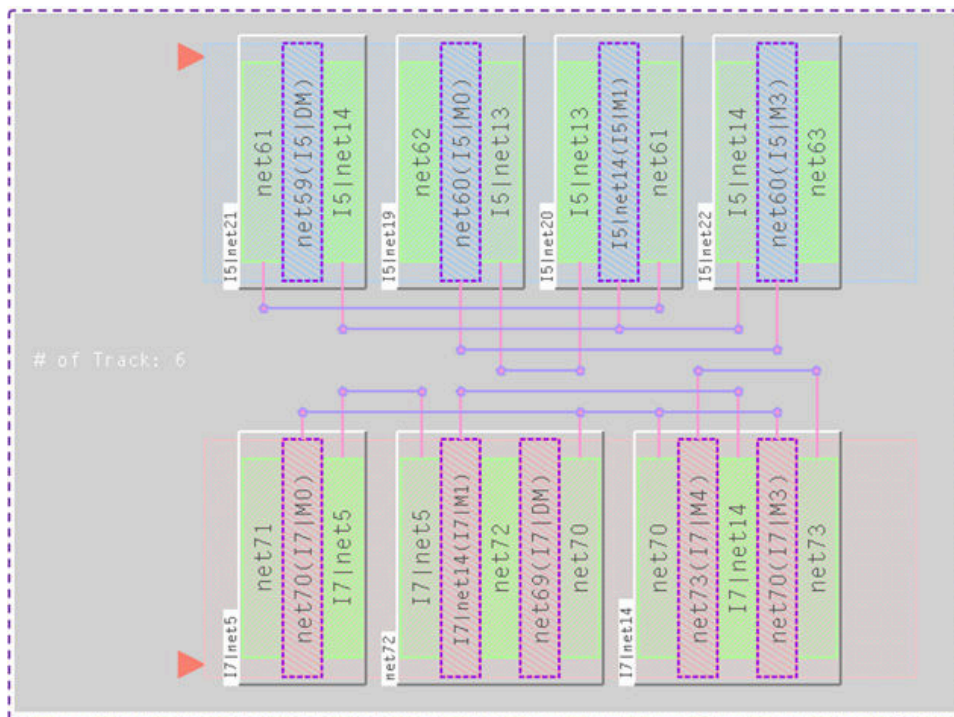
Examples

Specified the `lxPlacementOrderParam` keyword with `snps_SED_pre_abutment_order` value in `lxComponentType` and the `snps_SED_pre_abutment_order` in the CDF/OA property for different devices.

The I5 device has the DM M0 M1 M3 order and the I7 has the M0 M1 DM M4 M3 order.



After enabling SED chaining mode, the placement is shown as the following figure:



Generating Instances With Equal Width in Chaining Mode

The `lxValidWidths` keyword defined in `lxComponentType` allows to define a list of valid transistor width values when the values are discrete. The tool generates transistors with equal width with values in the `lxValidWidths` list in SED chaining mode.

4

Working With Designs

This section explains how to work with your design using the commands for Symbolic Editor.

This chapter covers the following topics:

- [Realizing and Closing Designs](#)
- [Viewing Designs](#)
- [Selecting Objects](#)
- [Highlighting Objects](#)
- [Specifying Placement and Pattern](#)
- [Optimizing Schematic ECOs](#)

Realizing and Closing Designs

After editing is completed in the Symbolic Editor window, the **Design > Realize** command realizes the devices from the Symbolic Editor window to the layout canvas. After clicking



the command , you can move the mouse cursor to the layout canvas and place the device to the canvas. You can edit the realized device with the Symbolic Editor and place to the previous layout position when the **Keep Previous Layout Position** option from the [Creation Tab](#) is selected.

The Symbolic Editor window closes after invoking the **Realize** command.

You also can use the **Design > Close** command to close the Symbolic Editor window.

Viewing Designs

This topic covers the following sections:

- [Zooming Designs](#)
- [Redrawing Designs](#)
- [Viewing the Previous Viewport or Next Viewport](#)
- [Fitting Viewing Area or Selected Objects](#)

Zooming Designs

These commands magnify the specified design view as follows:

- **Zoom Area:** To magnify a canvas area, choose **View > Zoom > Area** or click the **Set Viewport** button . Then use the mouse to set one point and rotate the mouse wheel or drag the mouse to adjust the scale.
- **Zoom In:** To show a close-up view of the data in the working window and increase magnification of the canvas area, choose **View > Zoom > In** or click the **Zoom In** button , and rotate the mouse wheel forward.
- **Zoom Out:** To show more contents in the working window at a reduced size and reduce current magnification of a canvas area, choose **View > Zoom > Out** or click the **Zoom Out** button , and rotate the mouse wheel backward.

Redrawing Designs

You can redraw a design to show recent changes. To redraw a design, choose **View >**

Redraw .

Viewing the Previous Viewport or Next Viewport

The system saves the viewport for the design. To switch to the previous viewport, choose

View > Previous Viewport or click the **Previous Viewport** button . To move

viewports between design contexts, you can switch to the next previous viewport by

choosing **View > Next Viewport** or clicking the **Next Viewport** button



Fitting Viewing Area or Selected Objects

These commands fit the viewing area as follows:

- **Fit View:** To fit the viewing area to all objects, choose **View > Fit > View** or click the **Fit to View** button .
- **Fit Selected:** To fit the viewing area to the set of selected objects, choose **View > Fit > Selected** or click the **Fit to Selected** button .

Selecting Objects

You must select devices in order to use the Symbolic Editor commands. You can invoke the command first and then select objects. In this case, the selectable objects are highlighted after you invoke the command. Or you can preselect the objects first and then invoke the command.

The tool provides the following commands to select the objects in the canvas.

- **Select By Region:** To select objects by region in the canvas, choose **Edit > Select > By Region** . (See Selecting Objects by Region in the *Custom Compiler Layout Editor User Guide for Creating and Editing Objects*.)



- **Select All:** To select all objects in the canvas, choose **Edit > Select > Select All** .
- **Deselect All:** To deselect all objects in the canvas, choose **Edit > Select > Deselect All** .
- **Select Empty** (Matching Mode Only): To select all empty grid boxes in the canvas, choose **Edit > Select > Select Empty** or click the **Select Empty** button . Nothing is selected if no empty grid box exists in the canvas.

Highlighting Objects

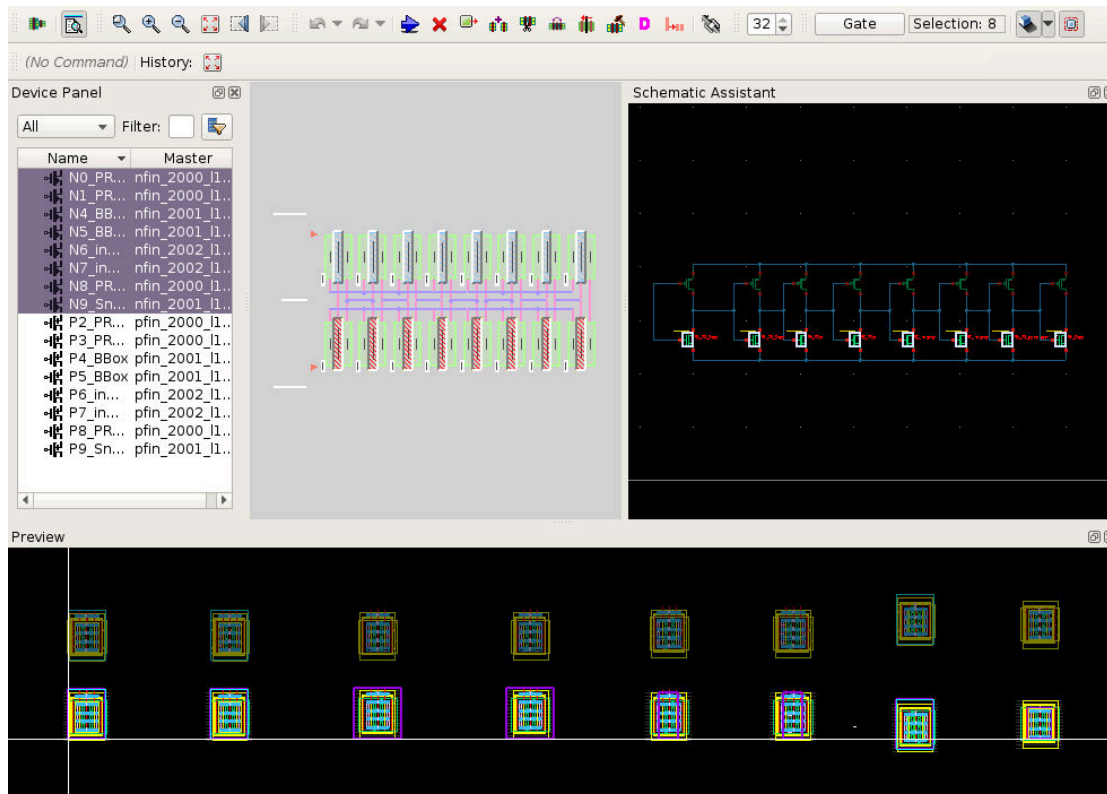
This topic covers the following sections:

- [Cross Highlighting and Selecting Objects](#)
- [Editing and Highlighting Flight Lines \(Chaining Mode Only\)](#)

Cross Highlighting and Selecting Objects

Symbolic Editor supports to cross highlight and select between different assistants.

1. Click the **Cross Shadow** icon  to dim any unselected objects.
2. Click the arrowhead in the **Cross Shadow** toolbar icon  to select the highlight color from the menu. If no color is selected, no objects are highlighted.
3. Click  in the Symbolic toolbars to enable the cross select feature to cross select the objects in the Device Panel, Symbolic Canvas, Schematic Assistant, and Preview assistant.



4. Click  in the Symbolic toolbars to enable flight lines cross highlighting in the Symbolic Canvas, Schematic Assistant, and Inspect Assistant. See [Using Symbolic Editor Assistants](#).

Editing and Highlighting Flight Lines (Chaining Mode Only)

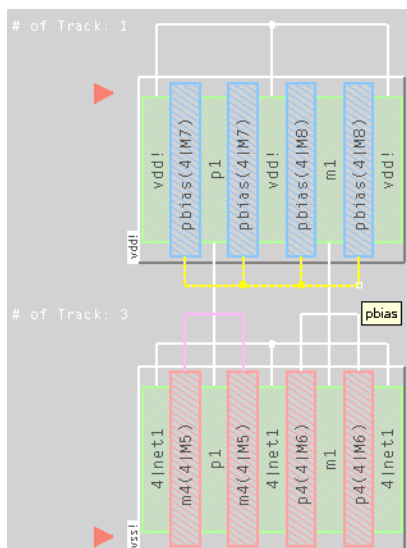
This topic covers the following sections:

- [Highlighting and Clearing Highlighting](#)
- [Editing Flight Line Modes](#)
- [Moving Flight Lines](#)
- [Inserting and Deleting Tracks \(Chaining Mode Only\)](#)
- [Modifying Flight Line Properties](#)

Highlighting and Clearing Highlighting

You can choose **View > Highlight Flight Lines** to highlight flight lines. When the mouse cursor is placed over the flight line, the line segments with same net are highlighted with

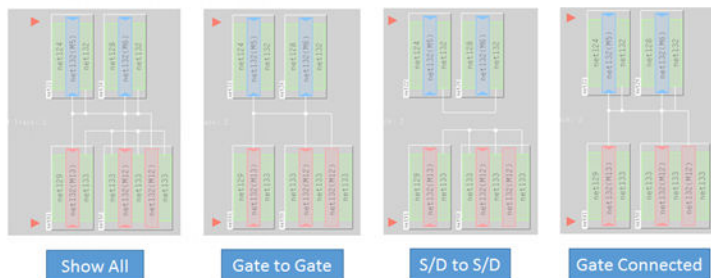
the active selection color, and the net name is shown in a yellow tip. The highlight colors cycle through the color set. To clear the highlight settings, choose **View > Clear Highlight**



Editing Flight Line Modes

Under the **View > Flight Line Mode** command, you can choose to display the flight lines by selecting one of the following commands:

- **All:** Displays all flight lines.
- **Gate to Gate:** Displays the flight lines between gate terminals.
- **S/D to S/D:** Displays the flight lines between source/drain terminals.
- **Gate to All:** Displays all flight lines connected to the gate terminal.



Moving Flight Lines

This command moves horizontal flight lines to other tracks. After executing the **Move Flight Line** command, track lines are displayed.

To move the flight lines, follow the steps below:

1. Choose **Edit > Move Flight Line** or use the shortcut key Alt+M to invoke the command. The track lines are displayed.
2. Single-clicking the horizontal flight line selects the horizontal flight line segment. Double-clicking the horizontal flight line selects the whole horizontal flight line on the same track. The layer name and net name of the selected flight line are displayed.
3. Click a different track line to move the flight line to that track. If the moved flight line conflicts with other flight lines or dots when moving, the conflicting connection is highlighted with a red dashed line.

If the net name and layer for the source flight line and target flight line are the same, these two flight lines are merged when moving.

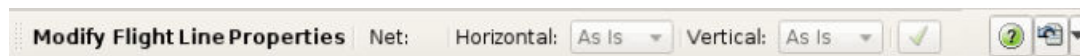
4. Press Esc to quit the command.

Inserting and Deleting Tracks (Chaining Mode Only)

The **Edit > Insert Track** command is used to insert a new empty track to the symbolic canvas. You can use the **Edit > Delete Track** command to delete a selected empty track.

Modifying Flight Line Properties

This command is used to modify the layers of flight lines.



To modify the selected flight line properties, follow the steps below:

1. Choose **Edit > Modify Flight Line Properties** or use the shortcut key Q to invoke the command.
2. The net name, horizontal layer, and vertical layer are displayed for the selected flight line.
3. You can modify the horizontal layer and vertical layer for the selected flight line in the selection fields. The conflicting connections are highlighted with a red dashed line.
4. Click the  button to apply the layer names and press Esc to quit the command.

Specifying Placement and Pattern

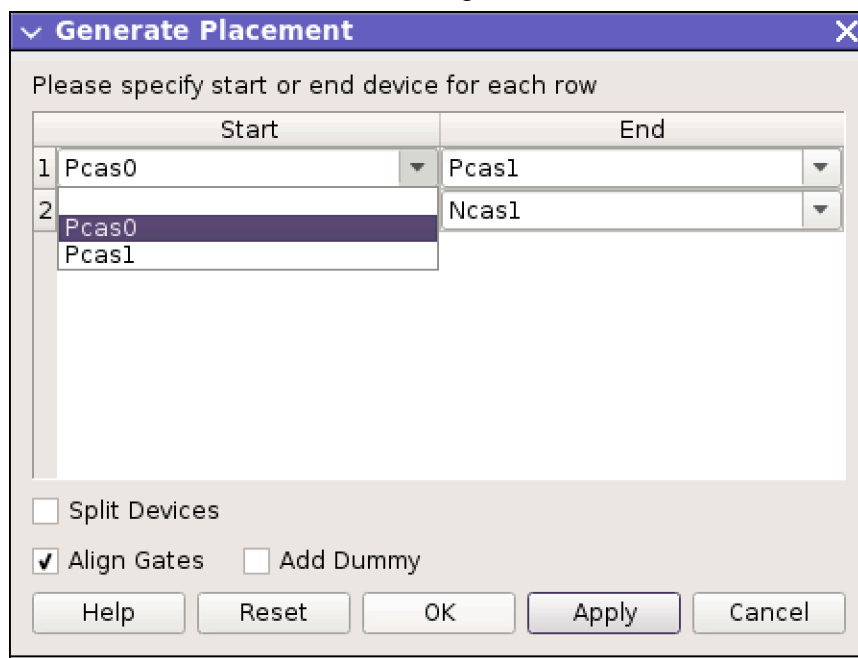
This topic covers the following sections:

- [Generating Placement \(Chaining Mode Only\)](#)
- [Generating Patterns \(Matching Mode Only\)](#)

Generating Placement (Chaining Mode Only)

This command specifies and displays the start and end devices for each row and generates the placement with the settings. This command is not available when the [Row Pattern](#) is **Custom**.

Figure 18 Generate Placement Dialog Box



To generate the placement with the settings:

1. If the devices are placed on the canvas, select **Edit > Generate Placement** to enable the **Generate Placement** dialog box.

In the dialog box, the start and end devices are shown with the current placement settings.

The Device Panel only displays the devices placed on the canvas.

2. If no device is placed on the canvas, select the devices in the Device Panel or Schematic Assistant and then enable this command.

In the dialog box, the start and end devices are empty.

3. Click  to modify the start and end devices from the list.

Or you can specify the device in the text field.

4. Select the **Split Devices**, **Align Gates**, and **Add Dummy** options to modify the placement. See **Placement** in Main Tab.
5. Click **Apply** or **OK** to generate placement with the settings.

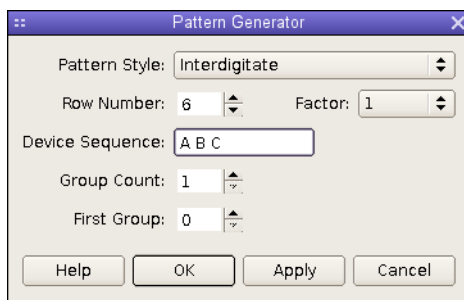
Generating Patterns (Matching Mode Only)

The **Edit > Generate Pattern** command provides the following pattern styles to regenerate the placement of all available tiles:

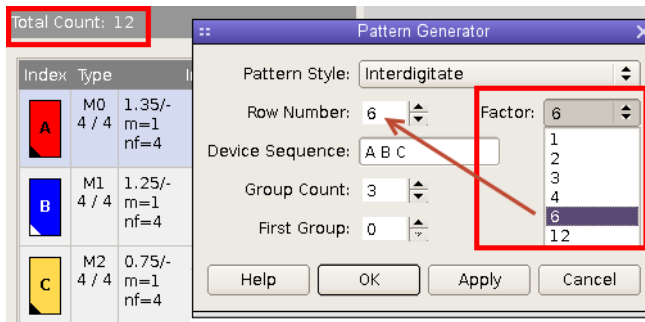
- [Interdigitate](#)
- [Common Centroid](#)
- [Central Symmetrical](#)
- [Device Group](#)
- [Text Pattern](#)
- [Series Chain](#)
- [Balanced Pattern](#)

Interdigitate

This pattern places all device tiles from the device panel one by one.



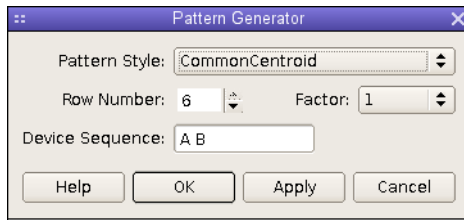
- **Row Number:** Specify the number of rows. The default is 1.
- **Factor:** Lists the factors of the total device number. You can specify a row number by choosing a factor. For example, if the total count of devices is 12, the factors are 1, 2, 3, 4, 6, and 12.



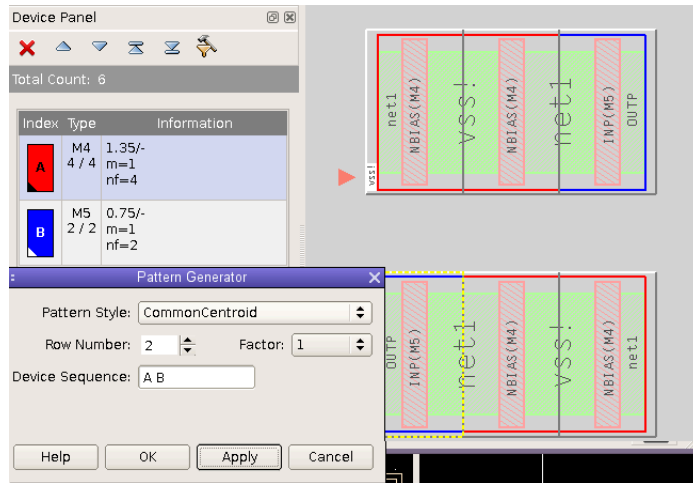
- **Device Sequence:** Specify the order in which devices are placed in the generated pattern. The default is the order of device indexes (A, B, ...) in the device panel.
- **Group Count:** Specify the number of the same type of tiles in a single group. For example, if **Group Count** is 3, the pattern is AAABBBCCCC.
- **First Group:** Specify the number of tiles to use in the first group. The default is 0. This option is available when First Group is larger than zero. For example, if **First Group** is 0 and **Group Count** is 3, the pattern is AAABBBCCCC . . . ; if **First Group** is 2 and **Group Count** is 3, the pattern is AABBBCCCCAAABBBCCC.

Common Centroid

This pattern places half of the tiles for each device type.

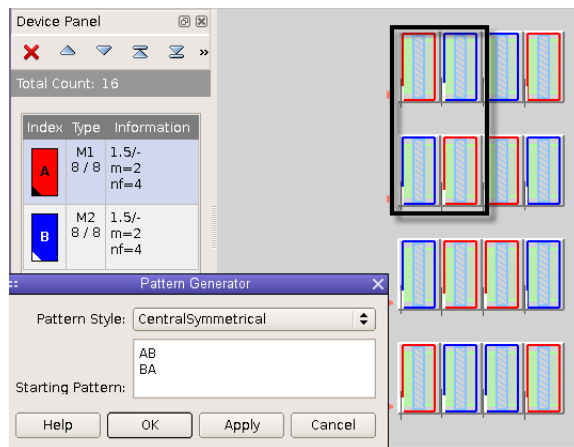
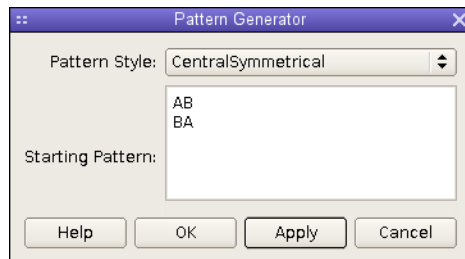


For example, index A has 4 tiles, index B has 2 tiles, and the row number is 2. Half of the A tiles (2 tiles) and half of the B tiles (1 tiles) are placed in the first row.



Central Symmetrical

This pattern lets you define an optimal, symmetrical pattern to place two device types. It can only be applied to two device types. This placement result (AB_BA) is symmetrical in any perspective.



- **Starting Pattern:** To define the pattern, enter an integer showing the number of times the pattern is repeated, and then alphabetic characters (A and B) representing the indexes from the Device Panel.
 - The repeat count is an integer that is greater than 0.
 - (underline) is supported to place devices in separate rows. For example, AB BA places AB in one row, and BA in the next row.
 - { } (braces) is supported to group indexes. For example, 3{AB} = ABABAB.

Device Group

This pattern places contiguous rows or columns for each group of devices.

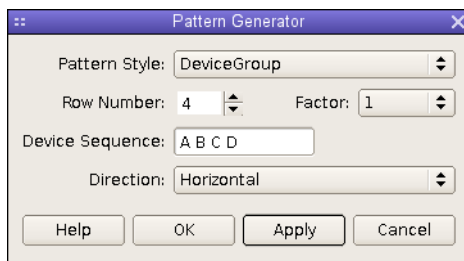


Figure 19 Device Group in Vertical Placement

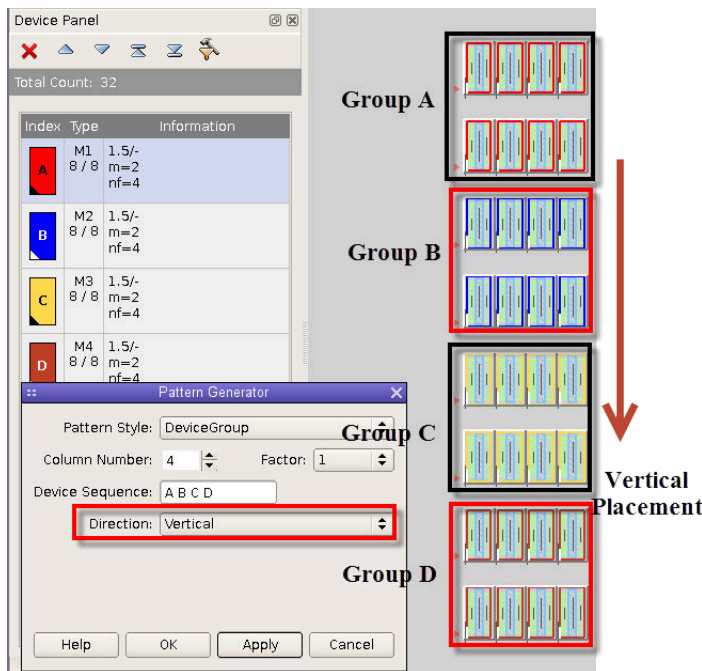
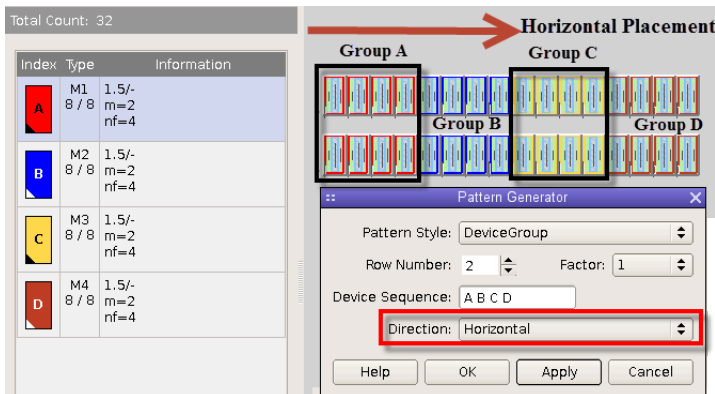
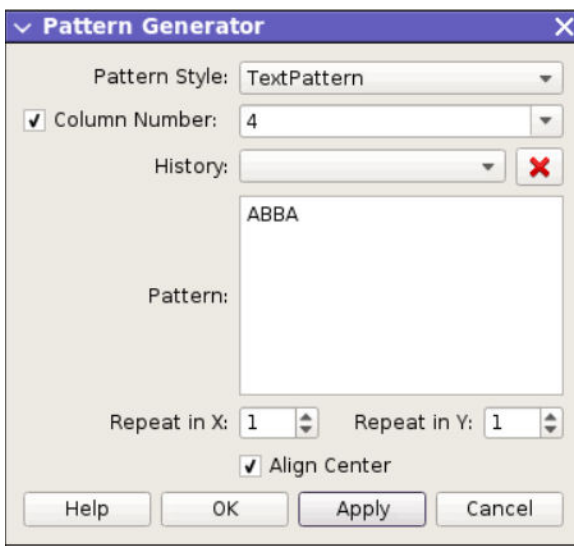


Figure 20 Device Group in Horizontal Placement



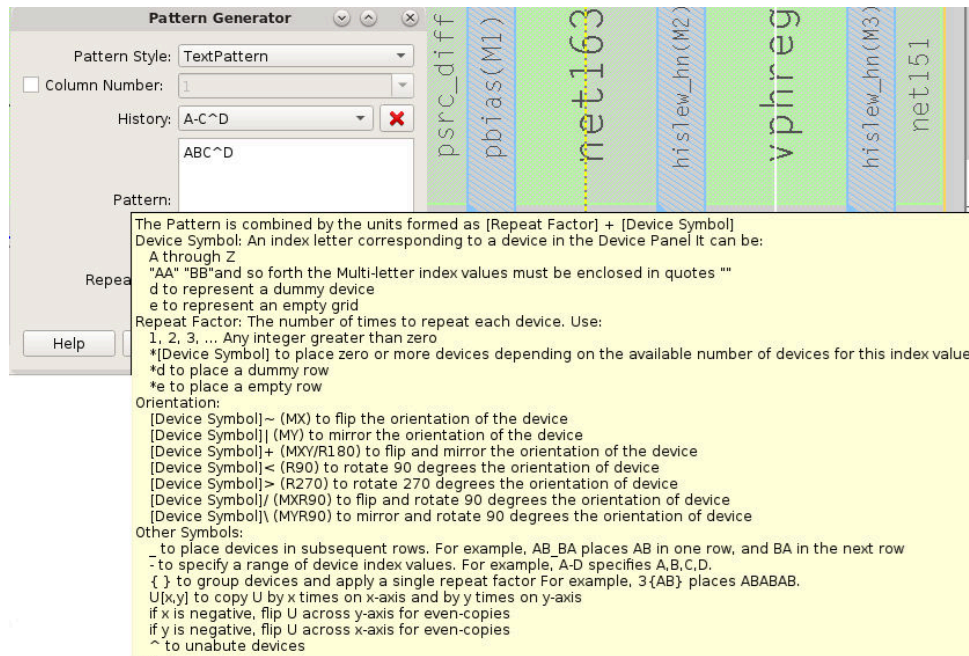
Text Pattern

This pattern places all tiles using an expression you define.



- **Column Number:** Specify the number of columns in the pattern. The default is *off*.
- **History:** Lists the last 10 patterns you used. Click the  button to delete all patterns in the **History** selection field.
- **Pattern:** Specify the user-defined pattern in the format [Repeat Factor] [Device Index(s)]. You can add special characters to group a series or separate devices by

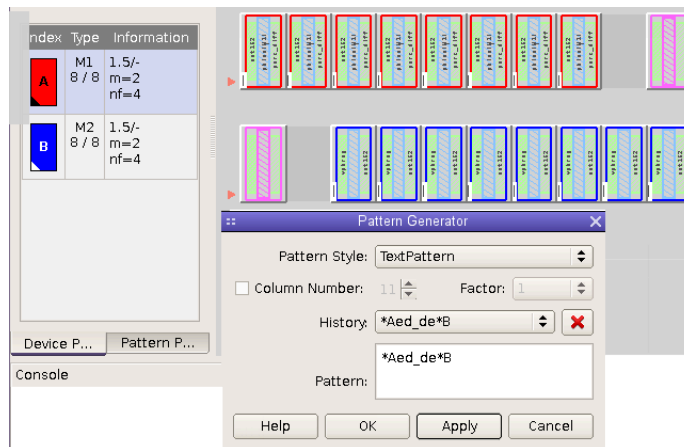
rows, as described below. You can also place the cursor over **Pattern** to display a pop-up containing more information about the following pattern syntax:



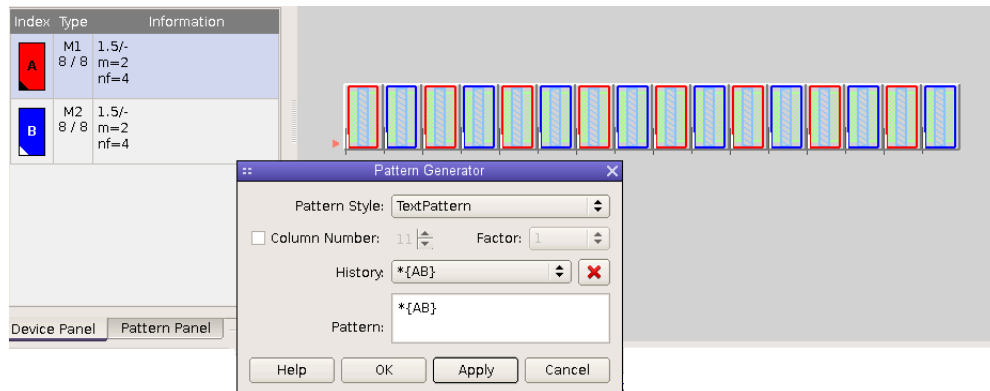
The pattern includes a Repeat Factor, one or more Device Index values, and optional Other Symbols.

- Device Index: An index letter corresponding to a device in the Device Panel. It can be:
 - A through Z.
 - "AA" "BB" and so forth. Multi-letter index values must be enclosed in quotes "".
 - d to place a dummy device.

- e to place an empty grid box.



- Repeat Factor: The number of times to repeat each device. Use:
 - 1, 2, 3 ... Any integer greater than zero.
 - *[Device Index] to place zero or more devices depending on the available number of devices for this index value. For example, *{AB} results in a placement as shown in the next figure.



- *d to place a dummy row.
- *e to place an empty grid row.
- Extend Symbol: Specify the repeat pattern in the following format: [Device Index(s)] [x-extend, y-extend]. You also can specify the x-extend and y-extend in the **Repeat in X/ Repeat in Y** command. If the repeat number is negative, even units on x-axis or y-axis will be mirrored. For examples:

Pattern = AB_BA[-2,2], the result in the symbolic canvas will be
ABBA_BAAB_ABBA_BAAB.

Pattern = AB_BA[2,-2], the result in the symbolic canvas will be
ABAB_BABA_BABA_ABAB.
Pattern = AB_BA[-2,-2], the result in the symbolic canvas will be
ABBA_BAAB_BAAB_ABBA

◦ Orientation Symbols:

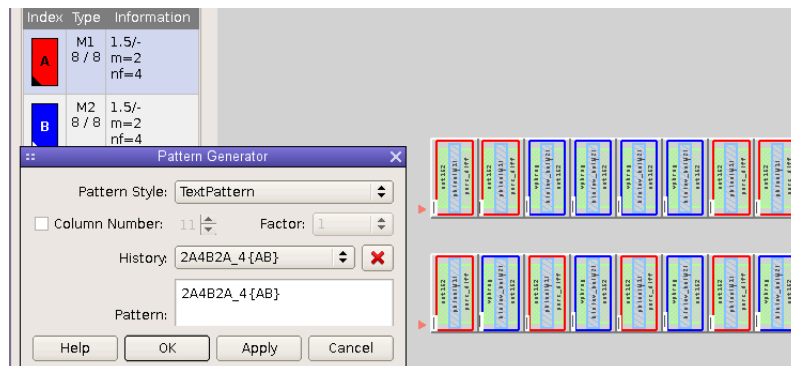
Orientation Symbols	Description	For Example
[Device Index]~ (MX)	Flips the device around the X axis.	A~
[Device Index] (MY)	Mirrors the device around the Y axis.	A
[Device Index]+ (MXY/R180)	Flips and mirrors the orientation of the device.	A+
[Device Index]< (R90)	Rotates the device in 90-degree.	A<
[Device Index]> (R270)	Rotates the device in 270-degree.	A>
[Device Index]/ (MXR90)	Flips the device and then rotates in 90- degree.	A/
[Device Index]\ (MYR90)	Mirrors the device and then rotates in 90-degree.	A\

Note:

For MOS devices, if the **Optimize OD After Apply Pattern** option in the **General Tab** is *on*, the placement result doesn't honor the mirror operation.

◦ Other Symbols:

- _ to place devices in subsequent rows. For example, 2A4B2A_4{AB}.



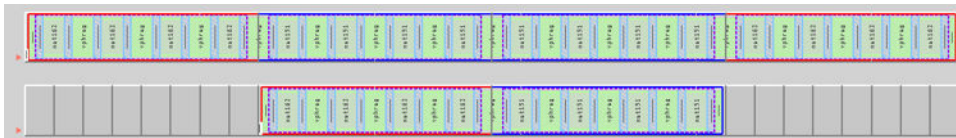
- - to specify a range of device index values. For example, A-D specifies ABCD.

- `{ }` to group devices and apply a single repeat factor. For example, `3{AB}` places ABABAB.
- `^` (caret) to unabut devices. Indicates that the device for which the symbol is applied is unabuted from the next one.

For example:

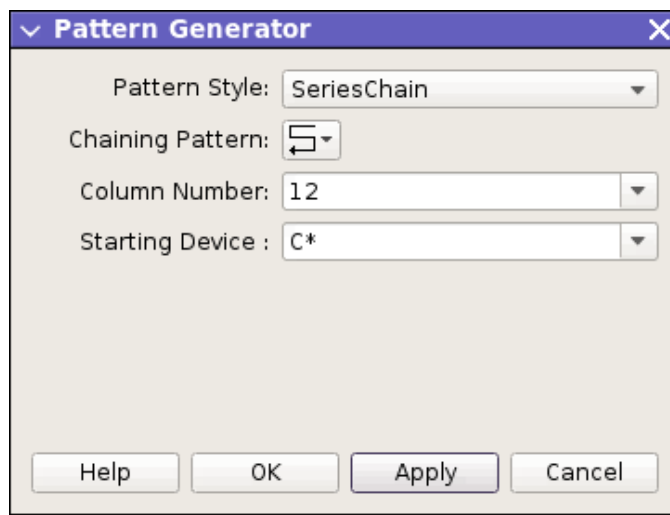
- If `TextPattern = AB^AB`, the result is `AB^AB_AB^AB`, where A device count = 4, B device count = 4.
 - If `TextPattern = 2A^2B`, the result is `AA^BB`, where A device count = 2, B device count = 2. Unabutment symbol applies only for the second 'A' device and not for the all 'A' devices.
- **Repeat in X/ Repeat in Y:** Specify the repeat number for extension on x-axis and y-axis. See [Extend Symbol](#).
 - **Align Center:** When enabling this option, center all devices on the canvas. When the preference `lxSEDTextpatternCenterAlign` is false, the placement is aligned left.

For example, center the pattern `ABBA_BA` on the canvas.



Series Chain

This pattern places devices in groups of horizontal or vertical chaining patterns.

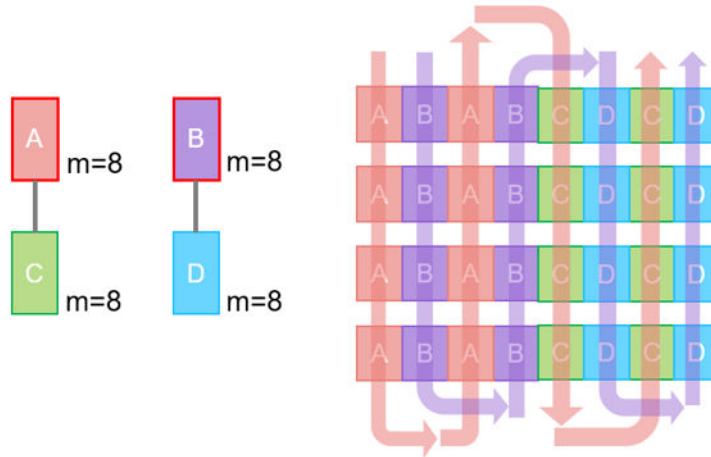


- **Chaining Pattern:** Specifies the chaining pattern. The W1 is only available for resistor.

	S1	Upper Left Horizontal Snake
	S2	Lower Left Horizontal Snake
	S3	Upper Right Horizontal Snake
	S4	Lower Right Horizontal Snake
	S5	Upper Left Vertical Snake
	S6	Lower Left Vertical Snake
	S7	Upper Right Vertical Snake
	S8	Lower Right Vertical Snake
	Z1	Upper Left Horizontal Zigzag
	Z2	Lower Left Horizontal Zigzag
	Z3	Upper Right Horizontal Zigzag
	Z4	Lower Right Horizontal Zigzag
	Z5	Upper Left Vertical Zigzag
	Z6	Lower Left Vertical Zigzag
	Z7	Upper Right Vertical Zigzag
	Z8	Lower Right Vertical Zigzag
	W1	W Pattern

- **Column Number:** Specifies the maximum number of placed columns in a horizontal chaining pattern. If you select a vertical chaining pattern, this option becomes **Row Number** and specifies the maximum number of placed rows.
- **Starting Device:** Specifies the starting devices to chain all devices. Default (marked "**") is the terminal device of the series chain in the schematic design. If no terminal device exists, the default is A.

You also can input multiple starting devices by using the same chaining pattern. The placement is created by multiple series chains. For example, A B. The first chain starts at device A and the second chain starts at device B.



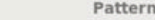
The chain direction of the starting device is prioritized by the following:

- Resistors: minus is higher than plus
- MOS: source is higher than drain

The W pattern is used for special case and only available for resistors. The **Row Number** of W pattern must be even number. The default is 2. The number of starting devices must be 1 or 2.

1	7	3	5
8	2	6	4
9	15	11	13
16	10	14	12

1	b	3	D
a	2	c	4
8	g	6	e
h	7	f	5



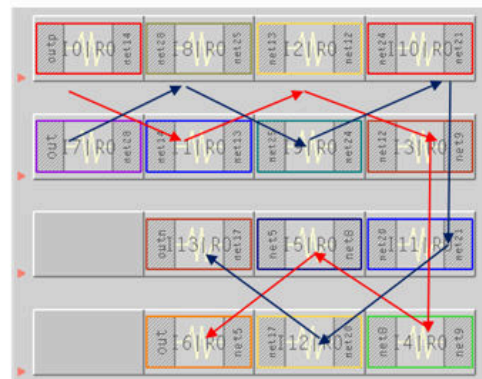
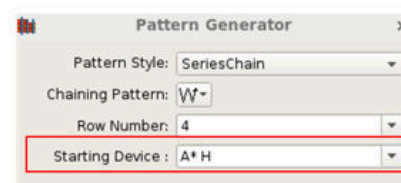
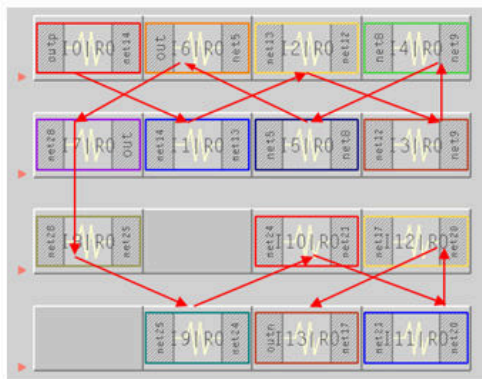
Pattern Generator

Pattern Style: SeriesChain

Chaining Pattern: W-

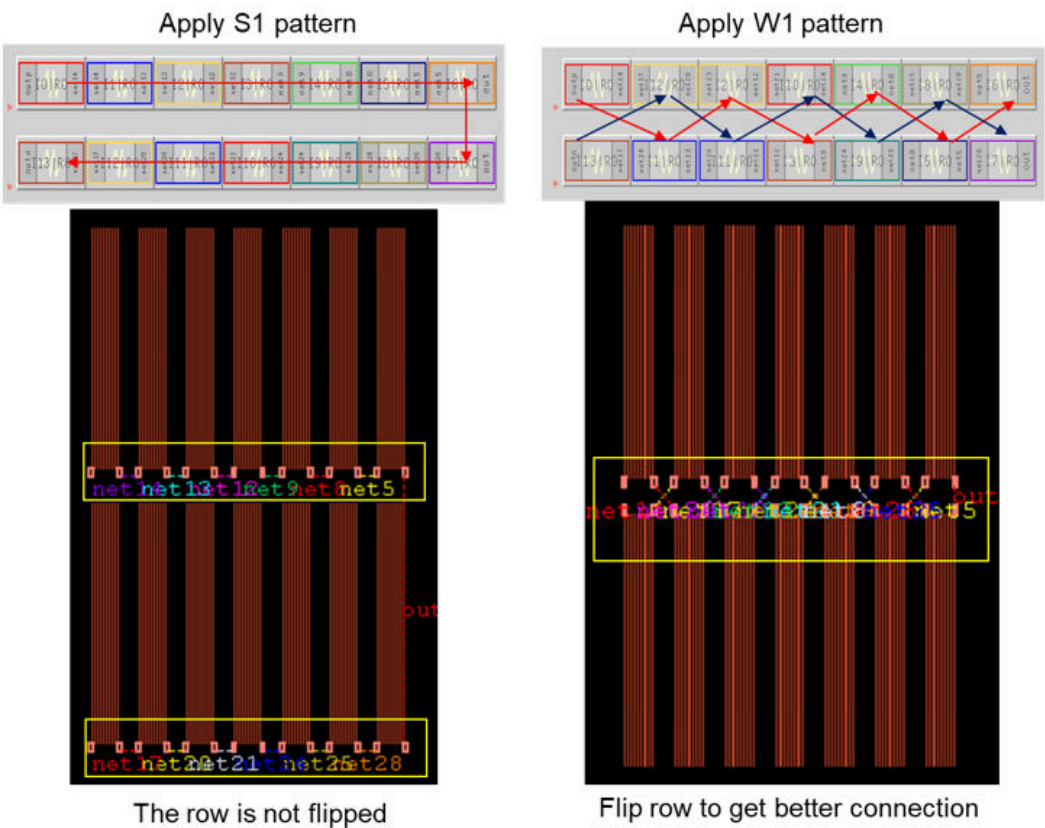
Row Number: 4

Starting Device: A*



- The W pattern is applied.
- The plus/minus terms are at the same side of corner.
- The symbol is horizontal.

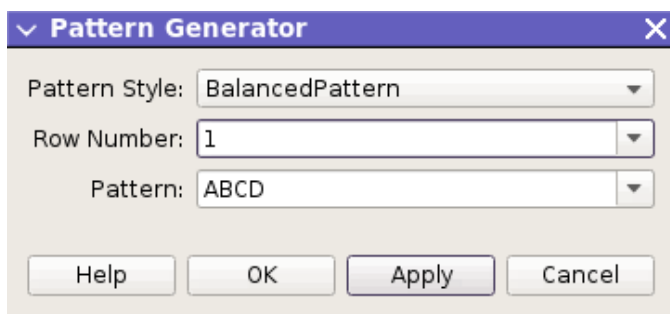
For examples:



Balanced Pattern

This pattern places all tiles using one of the following styles according to the device number and device count.

Pattern Style	Description
Differential Pair	Only for two devices with the same m-factor and finger number.
Cascode Pair	Only for two devices with different m-factor or finger number.
Current Mirror	More than two devices and determined by connection.
Cascode Mirror	More than two devices and determined by connection.



You can select the customized pattern in the **Pattern** selection field. The candidates are shown according to the device number and device counts.

Optimizing Schematic ECOs

When changing the device total widths, the schematics need to be updated, which results into device counts to change in the Symbolic Editor, and modifications are needed in layout fig-groups.

If the total width for a device has been increased in the schematic, you can add more devices into that device's group in the Symbolic Editor. For example, when the device count changes from 4 to 8, you can reapply the pattern with counts being 8, or place extra 4 devices manually into the symbolic canvas.

If the total width for a device has been decreased, you can delete some devices, or replace existing extra devices to dummies into that device's group using Symbolic Editor. For example, when the device count changes from 4 to 2, extra two devices are deleted and dummy devices are used to replace the deleted devices in the in symbolic canvas.



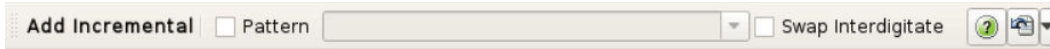
5

Creating and Editing Basic Objects

This chapter explains how to work with basic objects.

This chapter covers the following topics:

- [Adding Objects](#)
- [Creating and Editing Dummy Devices](#)
- [Deleting Objects](#)
- [Moving Objects](#)
- [Copying Objects \(Matching Mode Only\)](#)
- [Swapping Devices](#)
- [Flipping Objects](#)
- [Aligning Rows](#)
- [Flipping Rows \(Matching Mode Only\)](#)
- [Modifying Preset](#)
- [About Guard Rings \(Matching Mode Only\)](#)
- [Calculating Grid Size \(Matching Mode Only\)](#)
- [Transposing Devices \(Matching Mode Only\)](#)
- [Adding Penalty Tiles \(Matching Mode Only\)](#)
- [Removing Layout Parameters](#)
- [Undo and Redo](#)

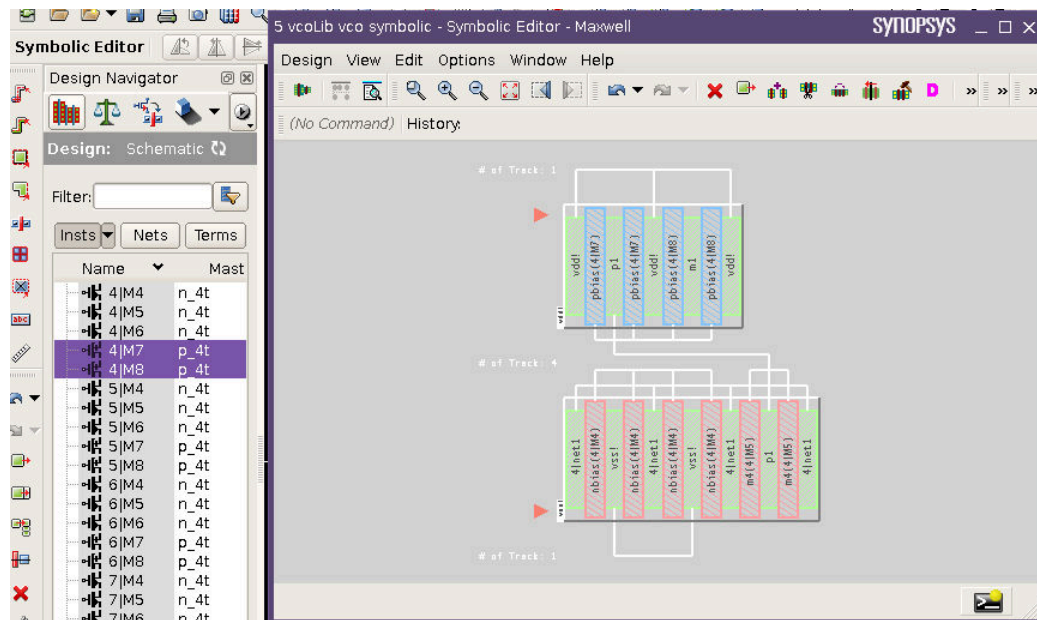
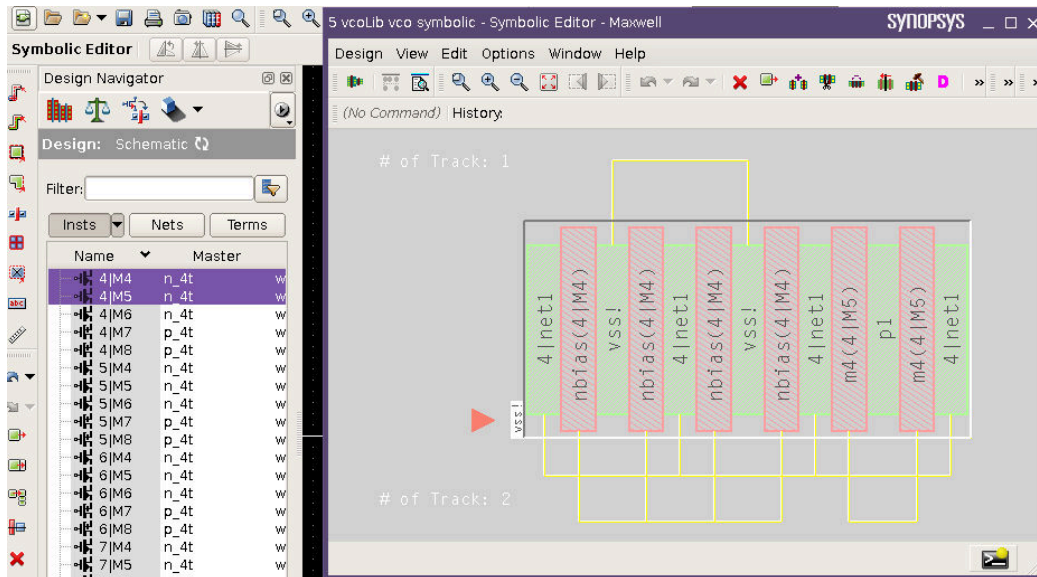


Option	Description
Pattern	Enables this option to specify the text patten or choose the existed text pattern. The default is off.
Swap Interdigitate	Enables this option to swap the order of A and B for the placement pattern ABBA or BAAB. When the pattern is ABBA, after applying this option, the pattern changes to BAAB. This option is available in the following conditions: • The Interdigitate option in the Placement Tab from Cell Placement is enabled. • The Pattern option is disabled.

Adding More Objects From Schematic

After you invoke the Symbolic Editor, you can still add more devices from the Design Navigator in Schematic Mode or Schematic Assistant to the Symbolic Editor window by

invoking the Symbolic Editor with Chaining Mode again. Or you can use **Edit > Add** in the [Working With Chaining Mode](#) section to add the devices.



Working With Matching Mode

To manually add new device objects to the Symbolic Editor canvas:

1. Choose **Edit > Add** or click the **Add** button .
2. Select the devices that you want to added in the Device Panel. You can also select multiple devices.
3. (Optional) Choose the **Pattern** option, specify the pattern type in the text field, and specify the number of specified patten to be repeated in the **Repeat in X** or **Repeat in Y** selection fields.

Note:

The text pattern with dummy or empty is not supported.

4. Select the grid boxes and then the selected pattern is assigned to the grid boxes. The patterns can be assigned continuously when selecting multiple grid boxes.

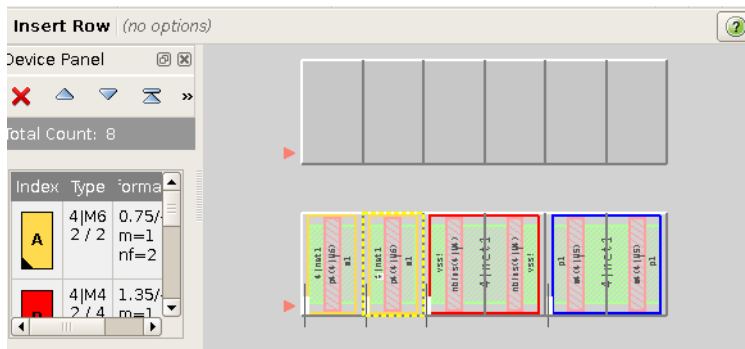
The following figure illustrates the **Add Device** toolbar. You can specify the text pattern in the text field and the number to be repeated when enabling the **Pattern** option.



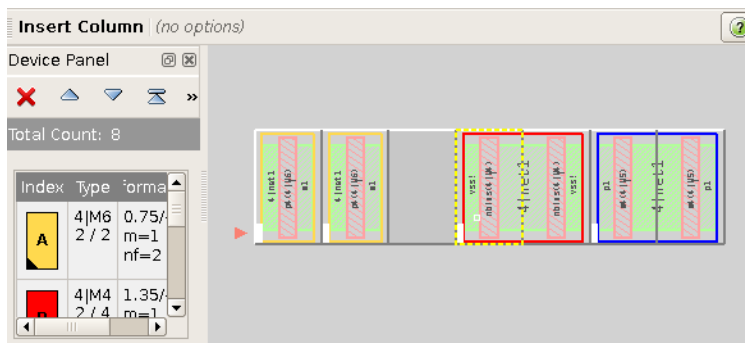
Option	Description
Pattern	Click this option to specify the text patten or choose the existed text pattern. The default is off.
Repeat in X	Specify the number of specified patten to be repeated in x direction.
Repeat in Y	Specify the number of specified patten to be repeated in y direction.
Align Center	Click this option to center all devices on the canvas. When the preference <code>lxsEDTextpatternCenterAlign</code> is false, the placement is aligned left.

Inserting Rows and Columns

To insert empty rows into the canvas, place your cursor where you want the row and choose **Edit > Insert > Insert Row** or click the **Insert Row** button .



To insert empty columns into the canvas, place your cursor where you want the column and choose **Edit > Insert > Insert Column** or click the **Insert Column** button .



Inserting Tiles and Shifting Devices

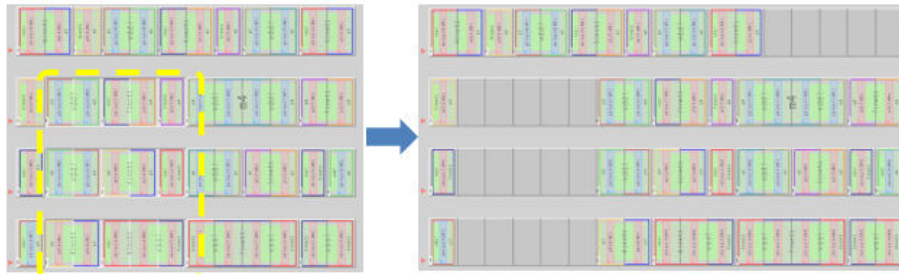
To move down the devices and insert empty tiles based on the selection:

1. Choose **Edit > Insert > Insert Shift Down** or click the **Insert Shift Down** button .
2. Area-select the devices. Moves selected devices down and all devices located on the down side of the selection. Empty tiles are inserted in the place of selected tiles and added to fill the new rows.



To move the selected devices to right side and insert empty tiles based on the selection:

1. Choose **Edit > Insert > Insert Shift Right** or click the **Insert Shift Right** button .
2. Area-select the devices. Shifts selected devices to right and all devices located on the right side of the selection. Empty tiles are inserted in the place of selected tiles. Newly added columns count depends on selected tiles count on a row.

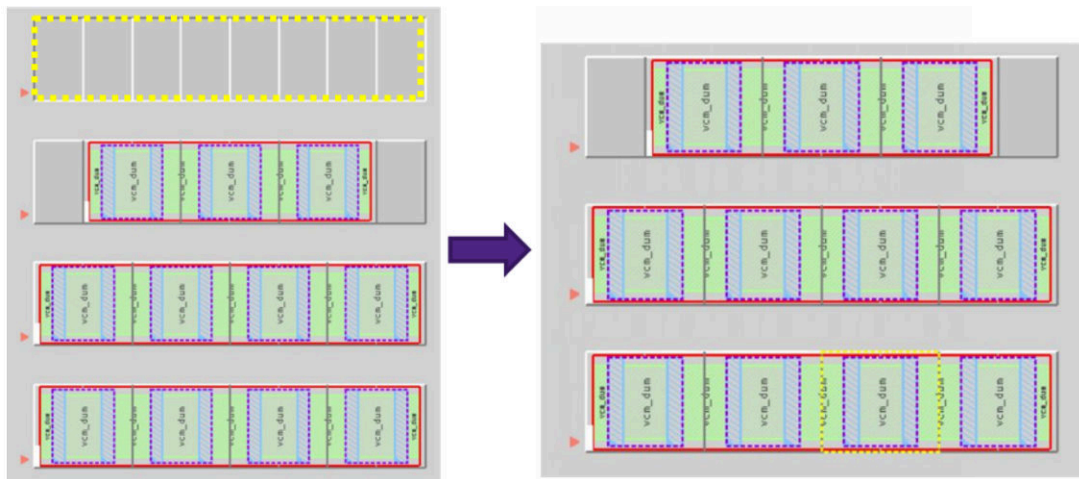


Removing Rows and Columns

Remove rows or columns when the selected rows or columns are empty tiles. If there are other active devices on the rows or columns, the command is disabled and the error messages are shown.

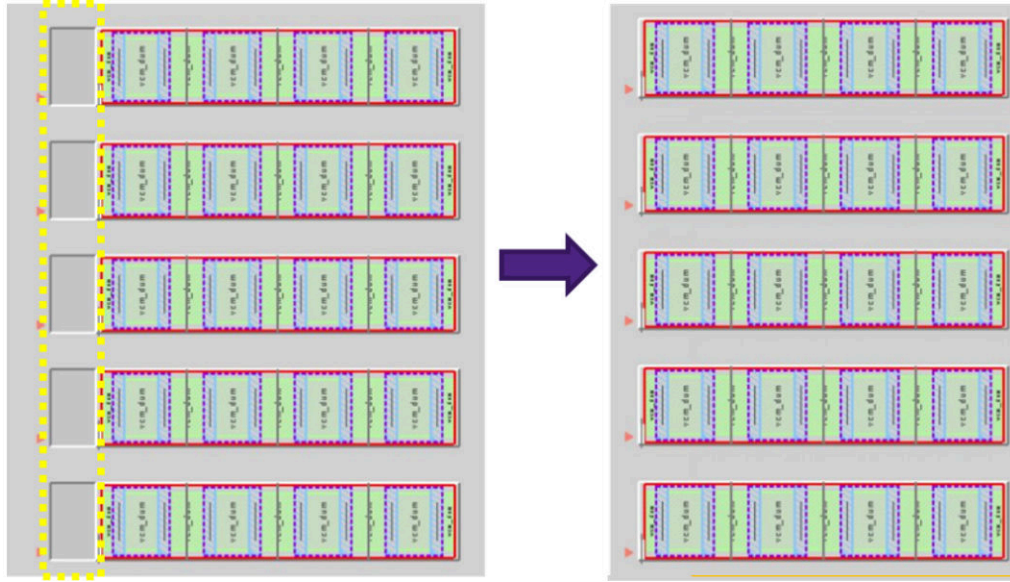
- **Remove Row:** To remove the empty row, place your cursor where you want the row and choose **Edit > Remove > Remove Row** or click the **Remove Row** button .

Figure 21 Remove an Empty Row



- **Remove Column:** To remove empty columns into the canvas, place your cursor where you want the column and choose **Edit > Remove > Remove Column** or click the **Remove Column** button .

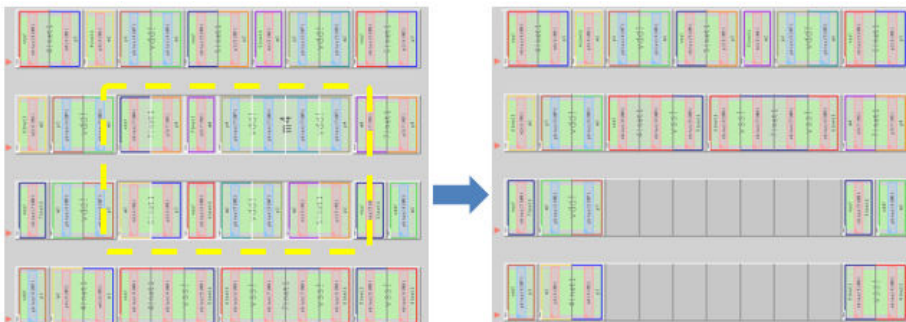
Figure 22 Remove an Empty Column



Removing and Shifting Devices

To remove the selected devices and moves up all devices located on the bottom of selected devices:

1. Choose **Edit > Remove > Remove Shift Up** or click the **Remove Shift Up** button .
2. Removes the devices by area-selecting devices. All devices located on the bottom of selected devices are moved up and empty tiles are added to fill the removed area.

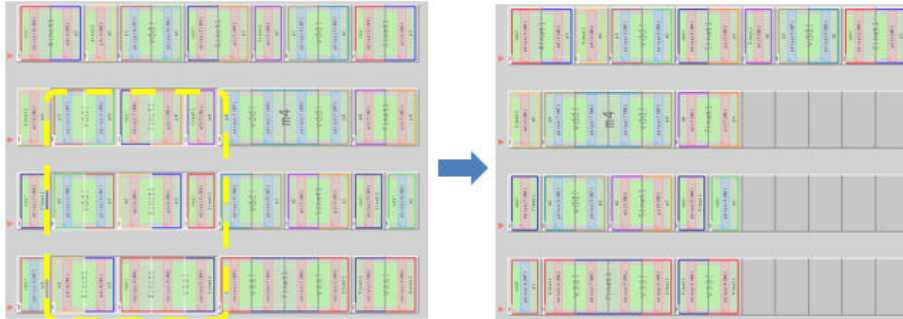


To remove selected devices and shift left all devices located at the left side of selected devices:

1. Choose **Edit > Remove > Remove Shift Left** or click the **Remove Shift Left** button

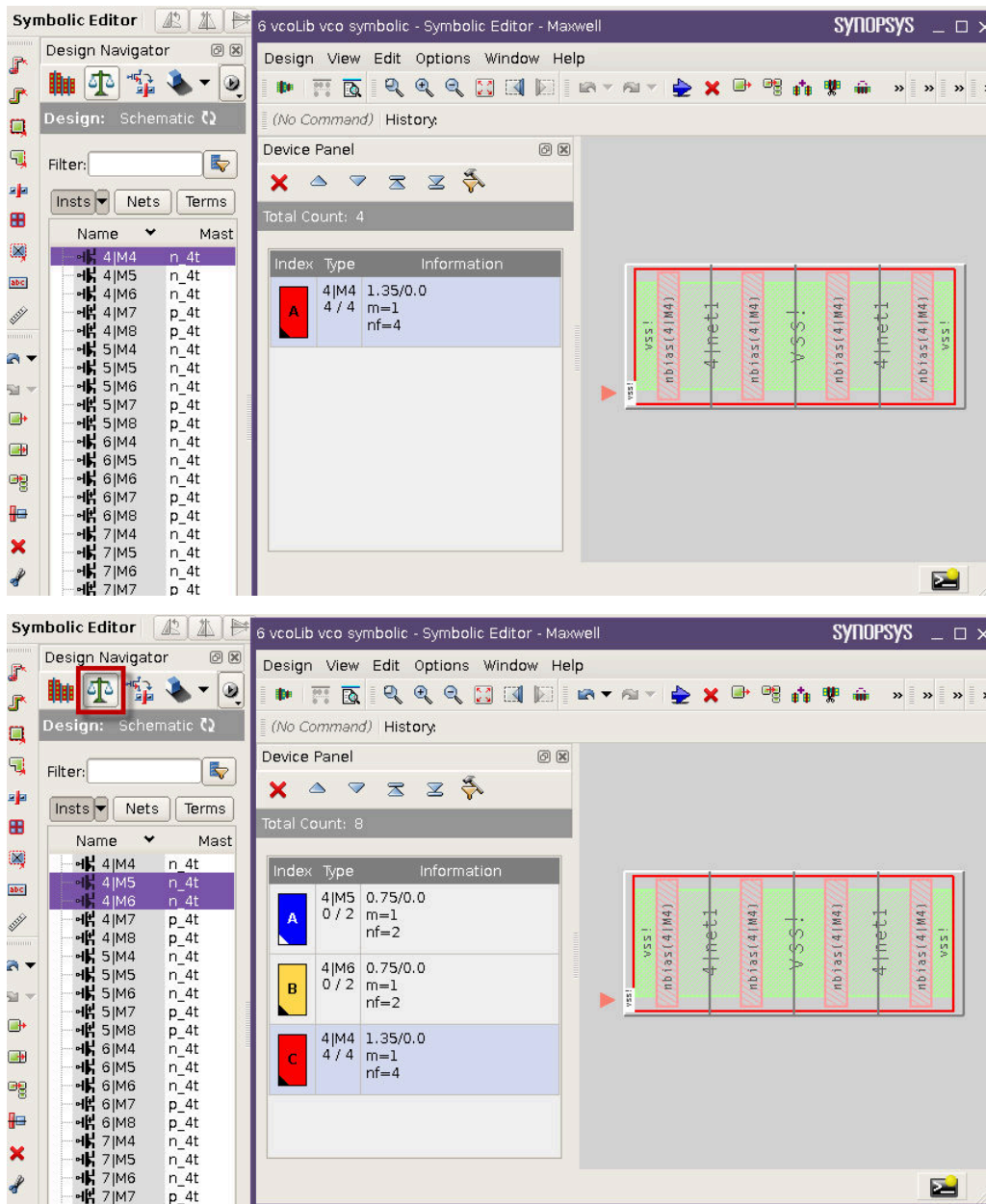


2. Removes the devices by area-selecting devices. All devices located at the left side of selected devices are moved left and empty tiles are added to fill the removed area.



Adding More Objects From the Schematic

After you invoke the Symbolic Editor, you can add more devices from the Design Navigator or Schematic Assistant to the **Device Panel** in the Symbolic Editor window by invoking the Symbolic Editor with Matching Mode again.



Creating and Editing Dummy Devices

You can create and modify the dummy devices using the following commands:

- [Adding Dummy Devices](#)
- [Adding Peripheral Dummies](#)
- [Modifying Dummy Properties](#)

Adding Dummy Devices

You can add dummy gates on the selected OD (oxide definition) area.

This topic describes the following:

- [Adding Dummy Toolbars](#)
- [Adding Dummy Dialog Box Options](#)

Adding Dummy Toolbars

To see more options of **Add Dummy** toolbar, choose **Edit > Add Dummy** or click the **Add Dummy** button .

Figure 23 Add Dummy for MOS Devices

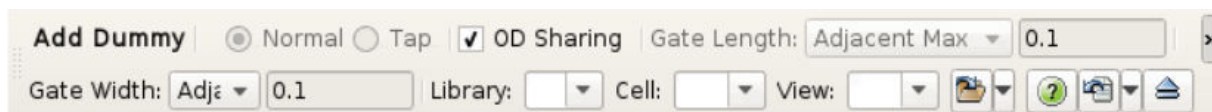


Figure 24 Add Dummy for Instances (Matching Mode Only)



Figure 25 Add Dummy for Resistors (Matching Mode Only)

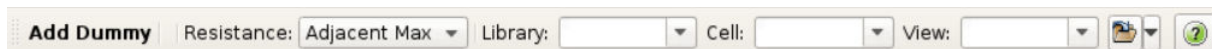


Figure 26 Add Dummy for Tap Cell



The toolbar contains the following options:

- **Normal/Tap:** When the **Normal** option is selected, the command is used to add dummy. When the **Tap** option is selected, the command is used to add the tap row. See `1xTap` in `About IxComponentType`.

In chaining mode, the **Tap** option is available when the **Row Pattern** in **Cell Template** is **Custom**. The tap row only can be added to mixed row.

- **OD Sharing:** When this option is turned *on*, the newly added dummy gates share the OD area with the selected OD. The default is *on*.
- **Gate Length:** Specify the length of dummy gates for MOS devices to be added by selecting one of the options: **Adjacent Max**, **Adjacent Min**, or **User-defined**.
- **Gate Width:** Specify the width of dummy gates for MOS devices to be added by selecting one of the options: **Adjacent Max**, **Adjacent Min**, or **User-defined**.
- **Resistance** (Matching Mode only): Specify the width or length for dummy resistors by selecting one of the options: **Adjacent Max** or **Adjacent Min**. The default is **Adjacent Max**.
- **Reference** (Tap only): Specifies the reference row of tap row by selecting the following directions: **Top** and **Bottom**. The default is **Top**.
- **Library/Cell/View:** Click  to select the cellview.

Adding Dummy Dialog Box Options

When the **Edit > Add Dummy** command is active, pressing F3 or clicking the **Eject** button on the toolbar shows the **Add Dummy** dialog box.

Figure 27 Add Dummy for MOS Devices in Chaining Mode

The 'Add Dummy' dialog box is shown with a purple title bar. It contains several sections: 'Mode' with radio buttons for 'Normal' (selected) and 'Tap'; 'OD Sharing' with a checked checkbox; 'Library', 'Cell', and 'View' dropdown menus; 'Net Name' with 'Left S/D', 'Right S/D', 'Gate', and 'Bulk' dropdown menus; 'Parameters' with a 'Finger Number' spinner and 'Gate Length' and 'Gate Width' settings (each with radio buttons for 'Adjacent Max', 'Adjacent Min', and 'User Defined', and a text input field set to '0.1'). At the bottom are 'Help', 'Hide', 'Defaults' (with a dropdown arrow), and 'Cancel' buttons.

Figure 28 Add Dummy for MOS Devices in Matching Mode

The 'Add Dummy' dialog box features a title bar with a collapse icon and a close button. The 'Mode' section has radio buttons for 'Normal' (selected) and 'Tap'. The 'Library' section includes a checked 'OD Sharing' checkbox, followed by dropdown menus for 'Library', 'Cell', and 'View' (which includes a folder icon). The 'Net Name' section contains dropdowns for 'Left S/D', 'Right S/D', 'Gate', and 'Bulk'. The 'Parameters' section, indicated by a blue arrow icon, includes 'Finger Number' with 'Auto' and 'User Defined' (selected) options, and a numeric input set to '4'. It also has 'Gate Length' and 'Gate Width' sections, each with 'Adjacent Max' (selected), 'Adjacent Min', and 'User Defined' options, with numeric inputs set to '0.1'. At the bottom are 'Help', 'Hide', 'Defaults' (with a dropdown arrow), and 'Cancel' buttons.

Figure 29 Add Dummy for Instances (Matching Mode Only)

This version of the 'Add Dummy' dialog box is simplified. It has the same title bar and 'Mode' section with 'Normal' selected. The 'Library' section contains the 'Library', 'Cell', and 'View' dropdowns. The 'Net Name' and 'Parameters' sections are absent. The bottom buttons are 'Help', 'Hide', 'Defaults' (with a dropdown arrow), and 'Cancel'.

Figure 30 Add Dummy for Resistors (Matching Mode Only)

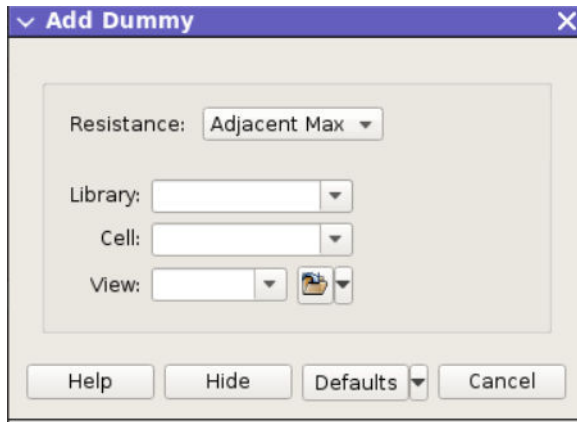
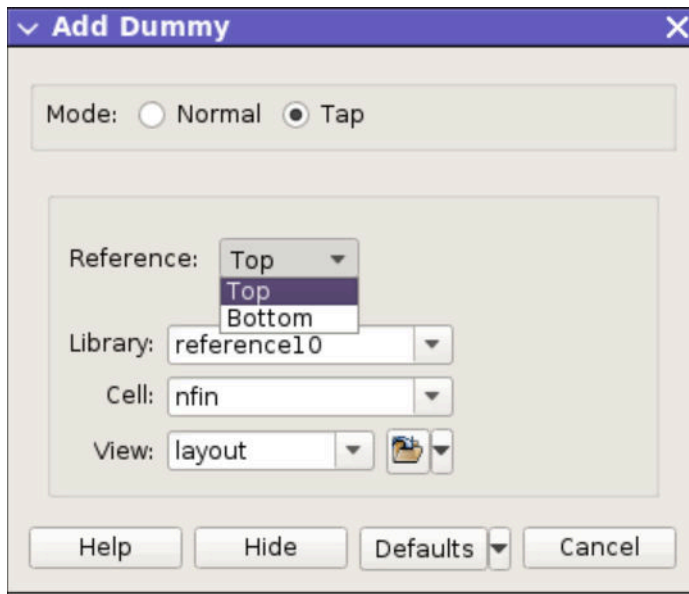


Figure 31 Add Dummy for Tap Cell



The dialog box contains the following options:

Option	Description
Normal/Tap	Normal: When this option is selected, the command is used to add dummy. Tap: When this option is selected, the command is used to add tap row. See <code>1xTap</code> in <code>About IxComponentType</code>. In chaining mode, the Tap option is available when the Row Pattern in Cell Template is Custom. The tap row only can be added to mixed row.

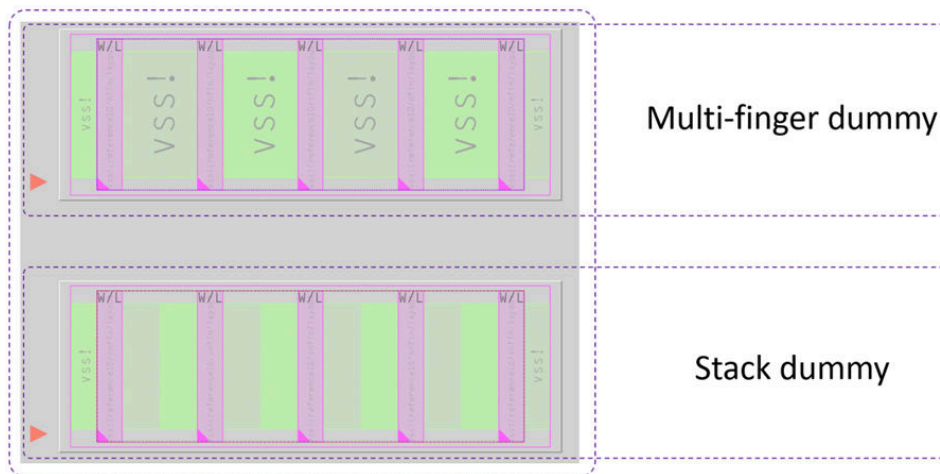
Option	Description
OD Sharing	When this option is turned on, the newly added dummy gates share the OD area with the selected OD. The default is on.
Reference (Tap only)	Specifies the reference row of tap row by selecting the following directions:: Top and Bottom . The default is Top .
Resistance (Matching Mode only)	Specify the width or length for dummy resistors by selecting one of the options: Adjacent Max or Adjacent Min . The default is Adjacent Max .
Library/Cell/View	Click  to select the cellview.
Parameters	
Finger Number (Chaining Mode only)	Specifies the fingers count. If the finger number is larger than one, the tool creates multi-finger dummy (finger group). Valid Values: 1 to 999. The default value is 1.
Finger Number (Matching Mode only)A	Specifies the fingers count. If the finger number is larger than one, the tool creates multi-finger dummy (finger group). Valid Values: 1 to 999. The default creation mode is User Defined with value 1. • Auto: In this mode, adjacent dummies created in one command merge to one multi-fingered dummy. • User Defined: In this mode, specifies the finger count. The fingers count cannot be higher than the available tiles count.
Net Name	For MOS device type. There are four options to specify the dummy: Left S/D , Right S/D , Gate and Bulk net names. These user net names have higher priority over default values from Design Options .
Gate Length	Specify the length of dummy gates for MOS devices to be added by selecting one of the options: Adjacent Max , Adjacent Min , or User-defined .
Gate Width	Specify the width of dummy gates for MOS devices to be added by selecting one of the options: Adjacent Max , Adjacent Min , or User-defined .

Stack dummy and Multifinger dummies

The **Add Dummy** option allows Stack dummy and Multifinger dummies.

If user specified **Finger Number** is set to **User Defined** value '1', then a single finger dummy is created for each dummy tile. The LCV for the dummy is selected from the reference logic device. If the **Finger Number** is set to **Auto**, then a multi-tile dummy is created from the reference logic device. If the logic device is multi-fingered, then a corresponding n-tile multi-finger device is created if there are 'n' contiguous adjacent

dummy tiles present. If the logic device is a stack device, then **n-series** connected stack device is created in **n** adjacent dummy tiles.



The LCV (and parameters) of the dummy devices can also be specified through `lxComponentType` setup on a cell using `lxDummyMappingLCVList`.

Adding Peripheral Dummies

This topic covers the following sections:

- [Adding Peripheral Dummies in Chaining Mode](#)
- [Adding Peripheral Dummies in Matching Mode](#)

Adding Peripheral Dummies in Chaining Mode

You can add dummies around the selected object.

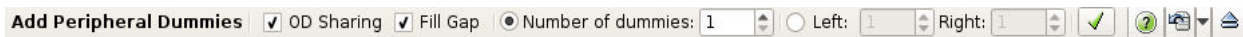


Figure 32 Add Peripheral Dummies Dialog Box in Chaining Mode

✓ OD Sharing

✓ Fill Gap

Create Peripheral Dummies By:

• Number of dummies: 1

• Left: 1 Right: 1

Parameters:

Finger Number: • One • Refer to Adjacent

Gate Length: • Adjacent Max • Adjacent Min

Gate Width: • Adjacent Max • Adjacent Min

Help Hide Defaults OK Apply Cancel

To see more options, choose **Edit > Add Peripheral Dummies**  to open the dialog box.

- **OD Sharing:** When this option is turned *on*, the OD area of MOS devices is shared with adjacent gates when adding dummies unless the OD area of the adjacent gate below or above the MOS device is not shared.
- **Fill Gap:** When this option is turned *on* (default), the gates of the original active devices are aligned before the peripheral dummies are added, and the dummies are filled to a rectangular placement.
- **Create Peripheral Dummies By:** Select one of the following methods to create peripheral dummy devices:
 - **Number of dummies:** Specifies the number of dummies to be added on the left and right sides of the selected devices. The value must be a positive integer from 1 to 256. The default is 1.
 - **Left, Right:** Specify the number of dummies by entering the **Left** and **Right** values for the number of columns. The valid value is from 1 to 256. The default value for these fields is 1.

- **Parameters:** This section specifies the following options:
 - **Finger Number:** Specify the number of fingers by selecting one of the following options:
 - **One:** Always use 1 as the number of fingers. The default is **One**.
 - **Refer to Adjacent:** Refer to the number of fingers of adjacent MOS devices. Referenced logic MOS devices are located by columns first, and then by rows.
 - **Gate Length:** Specify the channel length of dummy gates by selecting one of the following options:
 - **Adjacent Max:** When selected, the channel length is obtained from the maximum of adjacent MOS device gate lengths. The default is **Adjacent Max**.
 - **Adjacent Min:** When selected, the channel length is obtained from the minimum of adjacent MOS device gate lengths.
- Adjacent MOS devices are located by columns first, and then by rows.
- **Gate Width:** Specify the channel width of dummy gates by selecting one of the following options: **Adjacent Max** or **Adjacent Min**. The default is **Adjacent Max**.

Adding Peripheral Dummies in Matching Mode

You can add dummies around the selected object.



Figure 33 Add Peripheral Dummies Dialog Box in Matching Mode



To see more options, choose **Edit > Add Peripheral Dummies**  to open the dialog box.

- **OD Sharing:** When this option is turned *on*, the OD area of MOS devices is shared with adjacent gates when adding dummy unless the OD area of the adjacent below or above MOS is not shared.
- **Create Peripheral Dummies By:** Select one of the following methods to create peripheral dummy devices:
 - **Ring Number:** Specify the number of rings to be added around the selected grid boxes. The default is **Ring Number**. The value must be a positive integer from 1 to 256. The default is 1.
 - **Left, Right, Top, Bottom:** Specify the amount of dummy devices by entering the **Left** and **Right** values for the number of columns, and the **Top** and **Bottom** values for the number of rows. The valid value is from 1 to 256. The default value for these fields is 1.
- **Parameters:** This section specifies the following options:
 - **Finger Number:** Specify the number of fingers by selecting one of the following options:
 - **One:** Always use 1 as the number of fingers. The default is **One**.
 - **Refer to Adjacent:** Refer to the number of fingers of adjacent MOS devices. Referenced logic MOS devices are located by columns first, and then by rows.
 - **Max:** Create dummy devices by automatically detecting the highest possible number of fingers for adjacent dummy tiles.

Note:

This option is supported only in **Matching** mode of the **Symbolic Editor** and not supported in **Chaining** mode.

- **Gate Length:** Specify the channel length of dummy gates by selecting one of the following options:
 - **Adjacent Max:** When selected, the channel length is obtained from the maximum of adjacent MOS device gate lengths. The default is **Adjacent Max**.
 - **Adjacent Min:** When selected, the channel length is obtained from the minimum of adjacent MOS device gate lengths.
 - **User Defined:** When selected, specifies one or two values in the text field to be the channel length for the left or right dummy gates. If only one value is specified, the value is the channel length for both left and right dummy gates. If two values are specified, the first value is the channel length of the left dummy gates, and the second value is the channel length of the right dummy gates.

Adjacent MOS devices are located by columns first, and then by rows.

- **Gate Width:** Specify the channel width of dummy gates by selecting one of the following options: **Adjacent Max**, **Adjacent Min**, or **User Defined**. The default is **Adjacent Max**.

When **User Defined** is selected, specifies one or two values in the text field to be the channel width for top or bottom dummy gates. If only one value is specified, the value is the channel width for both top and bottom dummy gates. If two values are specified, the first value is the channel width of the top dummy gates, and the second value is the channel width of the bottom dummy gates.

Modifying Dummy Properties

You can modify dummy gates in the selected OD area. To see more options, choose **Edit** > **Modify Dummy Properties**  to open the dialog box.



Press **F3** to display the **Modify Dummy Properties** dialog box.

Figure 34 Modify Dummy Properties Dialog Box for MOS Devices

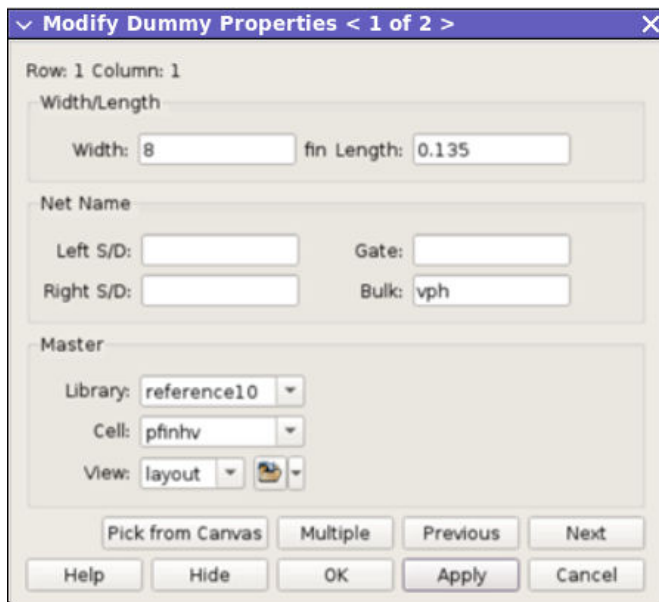


Figure 35 *Modify Dummy Properties Dialog Box for Instances (Matching Mode Only)*

▼ Modify Dummy Properties < 1 of 1 > X

Row: 1 Column: 4

Instance Terminal

Term Name	Net Name
in	net19
out	net30
vdd!	vdd!
vss!	vss!

Master

Library: workshop_test

Cell: inv

View: layout

Multiple Previous Next

Help Hide OK Apply Cancel

Figure 36 *Modify Dummy Properties Dialog Box for Resistors (Matching Mode Only)*

▼ Modify Dummy Properties < 1 of 1 > X

Row: 1 Column: 3

Net Name

Plus: Minus:

Bulk:

Master

Library:

Cell:

View:

Multiple Previous Next

Help Hide OK Apply Cancel

- **Width/Length:** You can modify the width and length of the dummy gates.
- **Net Name:** For MOS devices, you can modify the **Left S/D** (Left source/drain), **Right S/D**, **Gate**, and **Bulk** for the dummy gates. Bulk is enabled when the selected dummy gate is inside the pack composed of dummy. If the dummy gate is abutted with the logic device, the bulk net of dummy cannot be modified. See [Figure 34](#).

For resistors, you can modify the **Plus**, **Minus**, and **Bulk**. See [Figure 36](#).

- **Master:** You can specify the cellview in the selection fields. For matching mode, this option only supports for MOS devices.
- **Pick from Canvas:** After selecting the dummy devices, you can select any active dummy device or any of the properties assigned to the device from the SED canvas. All the selected dummy active devices now get their LCV and parameters assigned from the selected device.

Note:

If you modify the properties of a dummy device using the **Pick from Canvas** option, then the original LCV assigned by the **Symbolic Editor** (using `lxDummyMapping`) is overwritten.

- **Multiple/Single:** You can click the **Multiple** or **Single** button to switch to modify multiple dummy gates or a single dummy gate.
- **Instance Terminal (Matching Mode Only):** For instance, you can modify the net names for dummy instances.
- **Previous/Next:** You can click the **Previous** and **Next** buttons to switch to the previous or next dummy gate when multiple dummy gates are selected.

Deleting Objects

The **Edit > Delete** command deletes selected gates or all gates in the selected transistor packs. Area select is supported.

To see more options, choose **Edit > Delete** or click the **Delete** button  to open the dialog box.



- **Dummy Only:** You can select this option to delete dummy gates only.

You can preselect the gates and then invoke the **Delete** command. The command ends when the gates are deleted.

You can invoke the **Delete** command first, and then select the objects to delete. The command is still activated so you can continue to delete objects.

Moving Objects

You can move the objects using the following commands:

- [Working With Chaining Mode](#)
- [Working With Matching Mode](#)

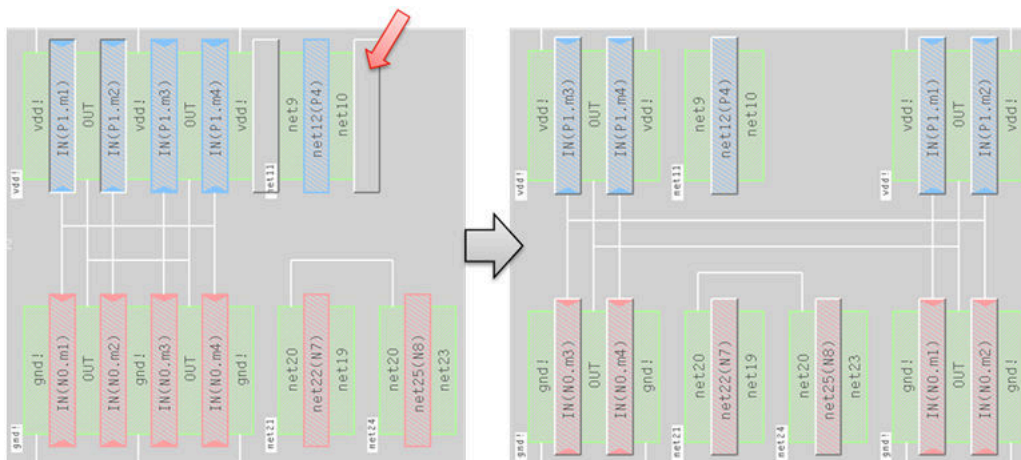
Working With Chaining Mode

The **Edit > Move** command moves the packs of MOS devices to another position.

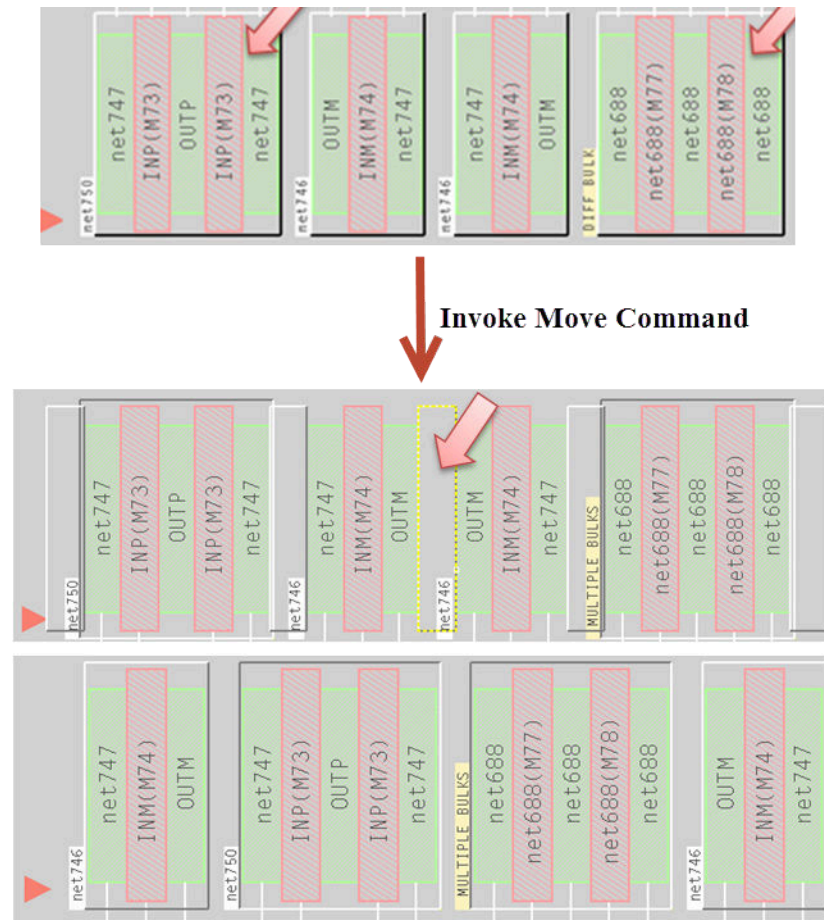
To move selected objects to the specified position, choose **Edit > Move** or click the **Move** button .



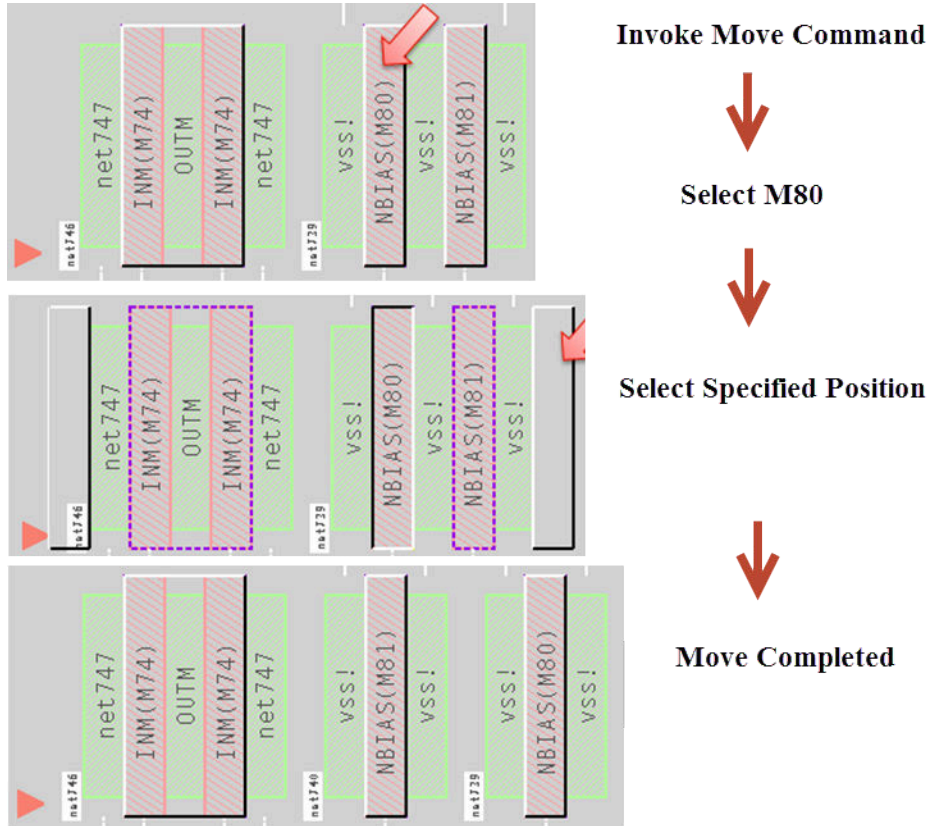
- **Keep Alignment:** When selected, any gates aligned to the selected gates are moved with the selected gates. The **Keep Alignment** setting works when moving objects in multiple row pairs. This option is disabled when the [Row Pattern](#) is **Custom**.



You can preselect the objects and then invoke the **Move** command. After invoking the command, you can select the available position to move the objects.



You can invoke the **Move** command first and then select a gate or area select multiple gates. After selecting the objects you want to move, click where you want to move the objects.



Working With Matching Mode

The **Edit > Move** command moves selected objects to the specific destination.

You can disable the **Infix Mode** option in the **General Options** dialog box in the layout canvas to specify the first point for the **Move** command. If the infix mode is enabled, this command use the left-top point of the selection set as anchor point.

When executing the **Move** command using drag and drop the selected objects, this command use the point that starts dragging as anchor point. If the start drag point is a finger group or stack MOS, then the left-most tile will be use.

When the select count is one, use the selected tile as anchor point. If the selected object is a finger group or stack MOS, then the left-most tile will be used.

For more information about **Infix Mode**, see Infix Mode in the *Custom Compiler™ Layout Editor User Guide for Creating and Editing Objects*.

Copying Objects (Matching Mode Only)

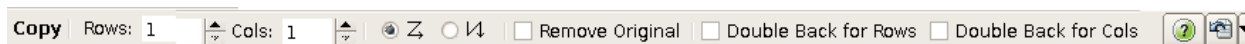
The command copies selected objects to paste to a specified destination. You can disable the **Infix Mode** option in the **General Options** dialog box in the layout canvas to specify the first point for the **Copy** command. If the infix mode is enabled, this command use the left-top point of the selection set as anchor point.

When executing the **Copy** command using drag and drop the selected objects, this command use the point that starts dragging as anchor point. If the start drag point is a finger group or stack MOS, then the left-most tile will be used.

When the select count is one, use the selected tile as anchor point. If the selected object is a finger group or stack MOS, then the left-most tile will be used.

For more information about **Infix Mode**, see Infix Mode in the *Custom Compiler™ Layout Editor User Guide for Creating and Editing Objects*.

The **Copy** toolbar options are a subset of the dialog box options.



To see more options, choose **Edit > Copy** or click the **Copy** button  to open the dialog box.

- **Rows/Columns:** Specify the number of repeated rows/columns to be copied. The default is 1.
- **Sequence:** Specify the placement sequence by selecting one of the following options:
 -  (Left to right first): Place the matching patterns from left to right first, and then from top to bottom. This is the default.
 -  (Top to bottom first): Place the matching patterns from top to bottom first, and then from left to right.
- **Remove Original:** When this option is on, the original selected objects are removed.
- **Double Back for Rows:** When this option is *on*, even rows are flipped. This option is disabled when the row orientation is flipped by using [Flipping Rows \(Matching Mode Only\)](#).
- **Double Back for Cols:** When this option is turned *on*, even columns are flipped.

Swapping Devices

This topic covers the following sections:

- [Working With Chaining Mode](#)
- [Working With Matching Mode](#)

Working With Chaining Mode

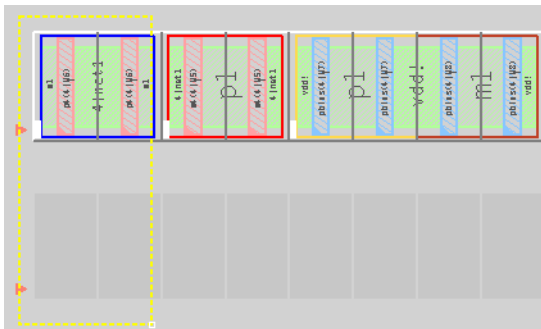
The **Edit > Swap Device** command swaps the position of the selected devices. This command breaks the alignment after swapping.

Working With Matching Mode

The **Edit > Swap Device** command swaps the position of the selected devices when choosing **Edit > Swap Device**. The selected objects can be swapped when the size of the two selection sets are the same.

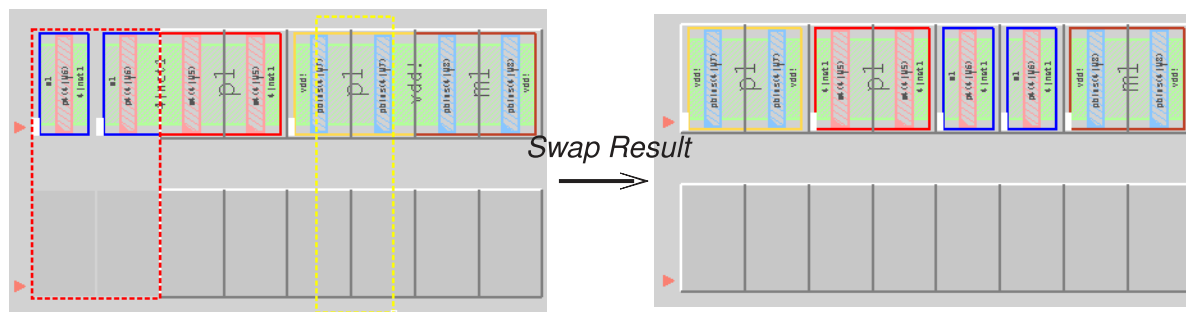
Use the following procedure to swap devices:

1. Choose **Edit > Swap Device** to invoke the **Swap Devices** command in matching mode.
2. Select two objects in the Symbolic Editor canvas.



3. Select another two objects in the Symbolic Editor canvas to swap.

The selected objects and target objects must use the same pattern.



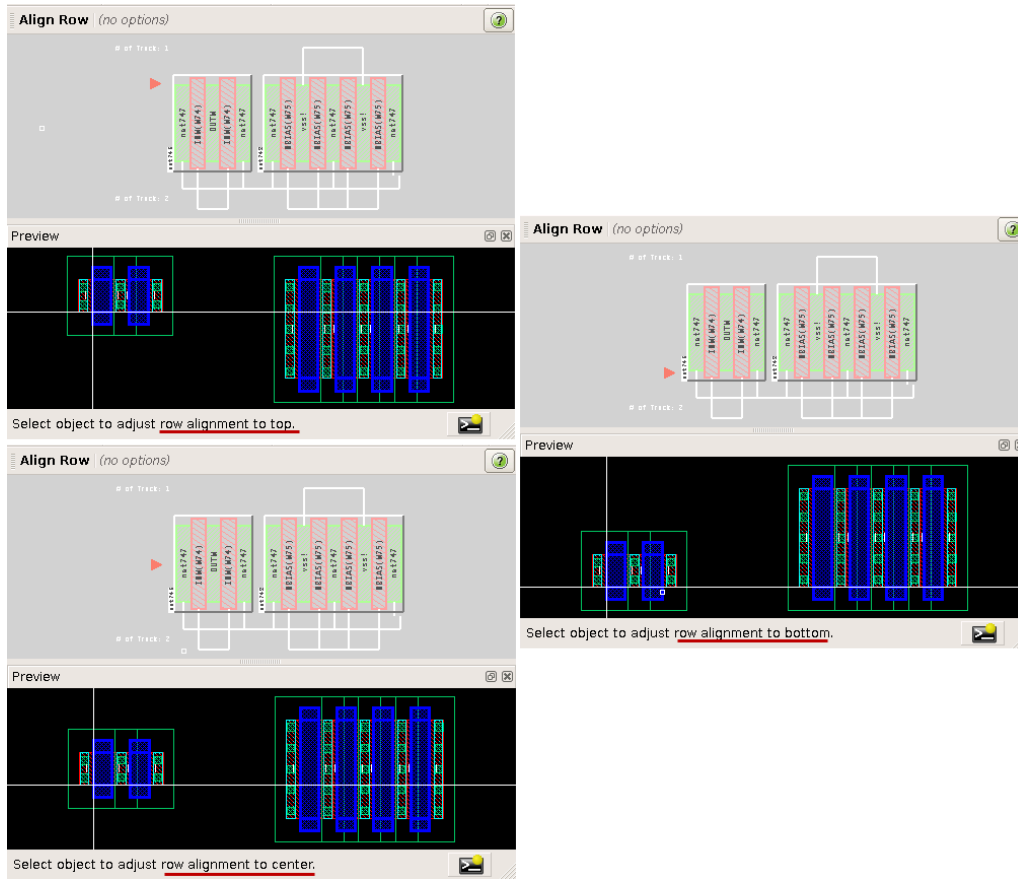
Flipping Objects

The **Edit > Flip** command flips the selected patterns in the same row (if they are vertically placed) into a different position. You can manually arrange the device placement order when choosing **Edit > Flip** or click the **Flip** button .

Aligning Rows

These command align OD edges of selected gates when gate widths are different in the same row. To rearrange the alignment of MOS devices in the selected row, choose one of the following sub-commands:

- To re-align all MOS devices in the selected row to the top of the row, choose **Edit > Align Row > Top** .
- To re-align all MOS devices in the selected row to the center of the row, choose **Edit > Align Row > Center** .
- To re-align all MOS devices in the selected row to the bottom of the row, choose **Edit > Align Row > Bottom** .



Flipping Rows (Matching Mode Only)

This command sets all devices orientation regardless of their previous orientation. Four types of row orientations are supported for all MOS devices or instances.

The following figure illustrates the **Flip Row** toolbar. After clicking , the specified row orientation is applied to set the orientation for all placed devices.



Option	Description
None	Set orientations of all devices to be R0 / MY if the original orientation is MX / R180.
Flipped	Set orientations of all devices to be MX / R180 if the original orientation is R0 / MY.

Option	Description
Even Flipped	Set orientations of all devices on even rows (from bottom to top and included dummy rows) to be MX / R180 if the original orientation is R0 / MY.
Odd Flipped	Set orientations of all devices on odd rows (from bottom to top and included dummy rows) to be MX / R180 if the original orientation is R0 / MY.

If the current row orientation is **Flipped**, **Even Flipped**, or **Odd Flipped**, fail to invoke the **Mirror X** command and display the error message.

Modifying Preset

You can use **Edit > Modify Preset** to specify and modify the preset parameters individually using the **Preset** selection field for the selected objects.

The following figure illustrates the **Modify Preset** toolbar for changing the preset parameter in the selection field:



[Preset Toolbar](#) is also supported to specify the preset parameters.

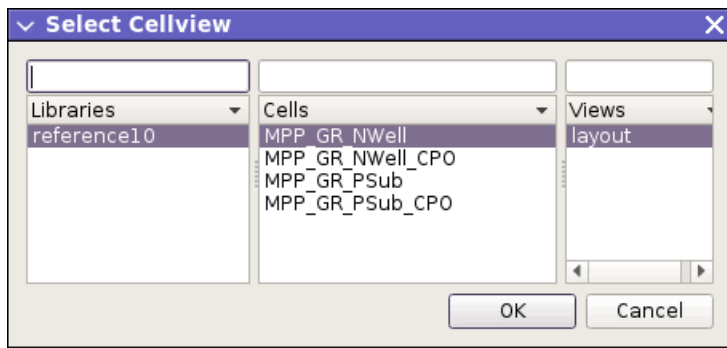
The preset parameters in the selection field are specified in the preset library, which is specified in the **SDL Options** dialog box. For more information, see **Parameter Preset** in Main Tab.

About Guard Rings (Matching Mode Only)

The **Edit > Generate Guard Rings** command generates single or multiple guard rings with the default settings for all instances.

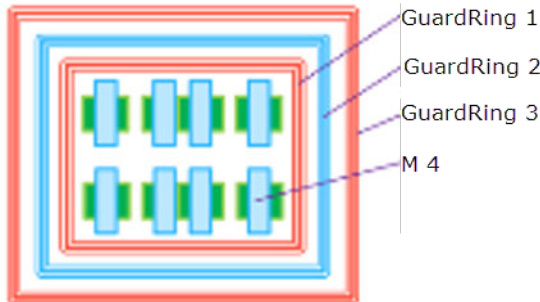


Click  to browse the available cellviews.



After completing the placement in matching mode, you can use **Edit > Generate Guard Rings** to create the guard rings. After committing this command, one or more guard rings

are created in the symbolic canvas. In the display, only the ring number corresponds to the actual guard ring properties; the displayed colors and widths are for illustration only.



In the **Generate Guard Rings** dialog box, you can specify the following:

Option	Description
Guard Rings	Click the Add Guard Ring and Delete Guard Ring buttons to add a new row or delete the selected row in the Guard Rings table. Click the browser icon to browse the available cellviews. Alternatively, click Library, Cell, View, and Net cells to load the library name, cell name, view name and net name from the guard ring configuration file. Click the Move Top Guard Ring, Move Up Guard Ring, Move Down Guard Ring, and Move Bottom Guard Ring buttons to move the selected guard ring.
Parameters of Selected Guard Ring	Specifies the following parameters for guard rings. Style: Style of the guard ring Rectangle, Conformal, Horizontal, Vertical, and Edge. The Edge style has Top, Bottom, Left, and Right options. Individual Wrap: Enables this option to generate guard ring to surround each individual element with the merge distance specified in the Merge Distance field. Note: This option is enabled only for Rectangle and Conformal styles and disabled when Horizontal, Vertical, or Edge styles are selected. Notch: The notch of the guard ring. Note: The Notch option is enabled only if the Conformal style is selected.
Spacing	Boundary: Defines the minimum spacing between the guard ring boundary and the boundary of enclosed objects.

Option	Description
	Layers: Defines the minimum spacings between guard ring layer boundaries and the layer boundaries of enclosed objects. The Layers table includes of the following fields and buttons: • Guard Ring: Contains a default boundary and also obtains LPPs from the guard ring LPPs procedure. • Reference: Shows LPPs from the selected valid objects and preference. If a bound instance or array instance is selected, the LPP across the hierarchy is considered. • Spacing: The spacing can be a positive or negative value. Clicking the Plus or Minus button adds or deletes a layer in the Layers table. Tcl preferences: <code>leGRSpacingMethod</code>, <code>leGRSpacing</code>, and <code>leGRLayerSpacing</code>.
Body Width	You can specify the reference body width by selecting Cut Rows or Specify for MCell guard rings. Parameters: Show all parameters in the table for editable PCells. You can specify the parameters in the table to control the guard ring.

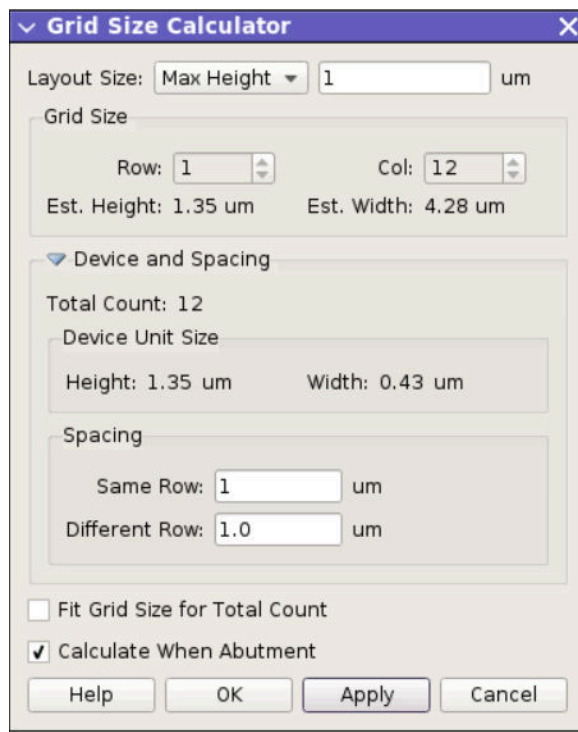
Calculating Grid Size (Matching Mode Only)

The **Edit > Calculate Grid Size** command calculates a number of tiles required in the Row and Col in Symbolic Editor matching mode.

The tool calculates the grid size based on the input size and the known information (device total count and spacing). To calculate the grid size, the tool calculates the grid size based on max height or max width.

Note:

This command assumes the multiple devices with same size and chooses the first device to calculate the device boundary.



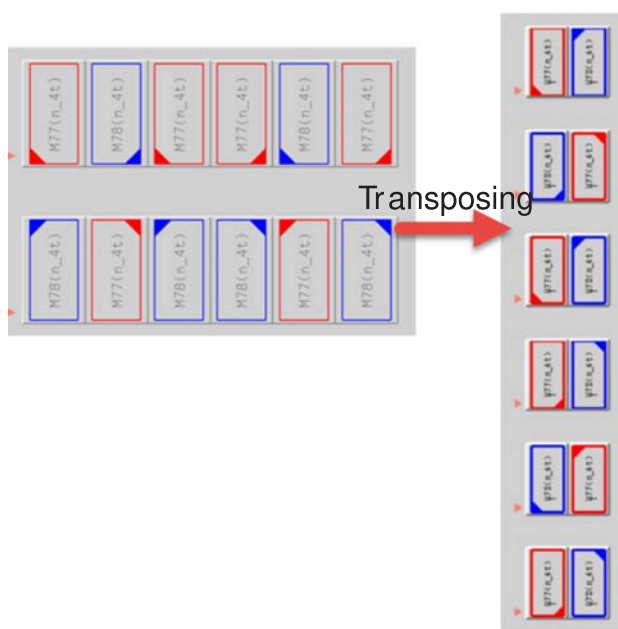
In the **Grid Size Calculator** dialog box, you can specify the following:

Option	Description
Layout Size	If the input value is Max Height, the tool calculates the required row from the value first and then gets the column number based on the row number and total device count. If the input value is Max Width, the tool calculates the required column from the value first and then gets the row number based on the column number and total device count.

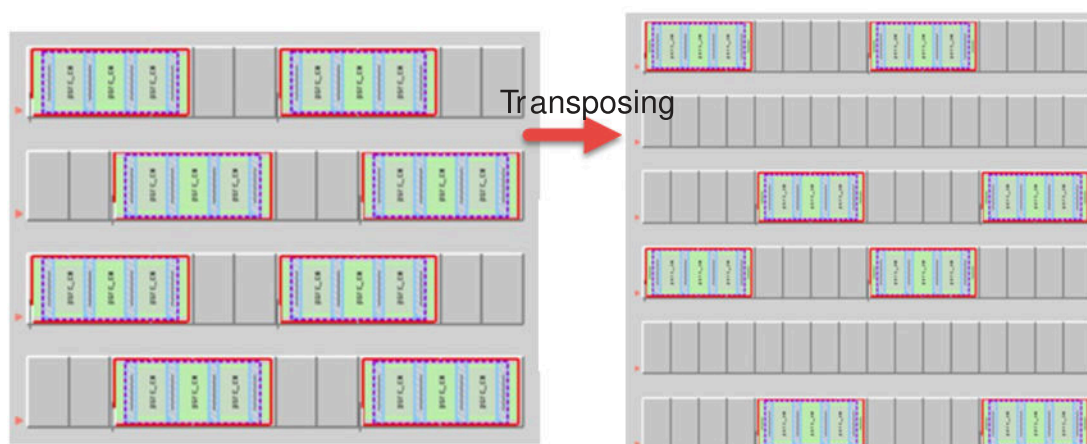
Option	Description
Grid Size	Displays the calculation result.
Device and Spacing	Displays the known information for calculating.
Fit Grid Size for Total Count	When this option is turned on, row and column would have dependency based on device total count. For example, row is 3 and column is 5. If row is changed to 5, column will be changed to 3 automatically when the Fit Grid Size for Total Count option is enabled.
Calculate When Abutment (MOS Only)	When this option is turned on, the tool considers all MOS devices are abutted for calculation.

Transposing Devices (Matching Mode Only)

The **Edit > Transpose** command switches the row and column indexes of all devices placement. For example, when transposing, the text pattern `ABAABA_BABBAB` is transposed to the equivalent text pattern `AB_BA_AB_AB_BA_AB`.



When transposing devices with empty tiles, this command keeps the largest finger size. For example, when transposing the text pattern `A2eA2e_2eA2eA_A2eA2e_2eA2eA` is transposed to the equivalent text pattern `A4eA4e_16e_4eA4eA_A4eA4e_16e_4eA4eA`.



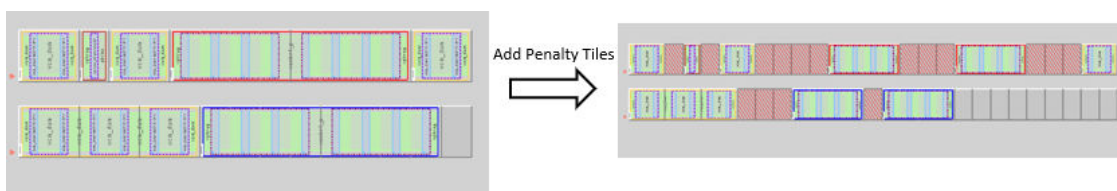
Adding Penalty Tiles (Matching Mode Only)

The **Edit > Add Penalty**  command iterates to add penalty tiles between each pair of devices from top left to bottom right on the canvas.

To add penalty tiles:

1. Specify the penalty procedure using the preference `lxSEDPenaltyProcName`. For example:

```
db::setPrefValue lxSEDPenaltyProcName -value penalty_proc
```
2. Choose the **Edit > Add Penalty** command to add penalty tiles. The command adds empty tiles between pair of devices. The number of added penalty tiles is specified by the penalty procedure. Penalty tiles are highlighted with the `lxPenaltyTiles` packet.



The preview design leaves one poly-pitch spacing per empty or penalty tile. The spacing value is computed by number of penalty tiles * poly-pitch. The penalty spacing correction is needed; otherwise it should specify to 0. The procedure returns in format `[list $numPenaltyTiles $spacingCorrectionValue]`.

Removing Layout Parameters

This **Edit > Remove Layout Parameters** command is available only when you launch the Symbolic Editor from layout. Layout parameters are copied with devices when the Symbolic Editor is launched from layout. Click this command to remove layout parameters from the selected devices.

Undo and Redo

To undo the last performed command, choose **Edit > Undo** or click the **Undo** button .

To repeat the last performed command, choose **Edit > Redo** or click the **Redo** button .

6

Editing MOS Devices

This chapter explains how to edit MOS devices in the symbolic editor.

This section covers the following topics:

- [Editing Gates \(Chaining Mode Only\)](#)
- [Mirroring Objects](#)
- [Editing Source/Drain](#)
- [Modifying PODE Properties \(Chaining Mode Only\)](#)
- [Swapping Interdigitate](#)
- [Rechaining MOS Devices \(Chaining Mode Only\)](#)
- [Optimizing OD \(Matching Mode Only\)](#)
- [Unsetting Finger Group](#)
- [Customizing Spacing \(Matching Mode Only\)](#)
- [Editing Individual Alignment \(Matching Mode Only\)](#)

Editing Gates (Chaining Mode Only)

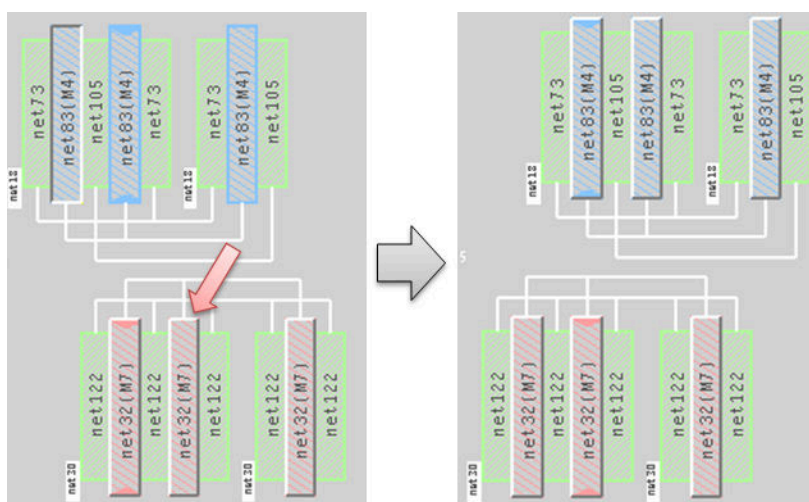
This topic covers the following sections:

- [Aligning Gates](#)
- [Unaligning Gates](#)
- [Folding Gates](#)
- [Merging Gates](#)

Aligning Gates

The **Edit > Align Gates** command fine-tunes the placement of MOS devices by aligning the centerline of the device gates onto the same vertical line. This alignment relationship is kept until the gates are unaligned or a new operation is applied to these gates.

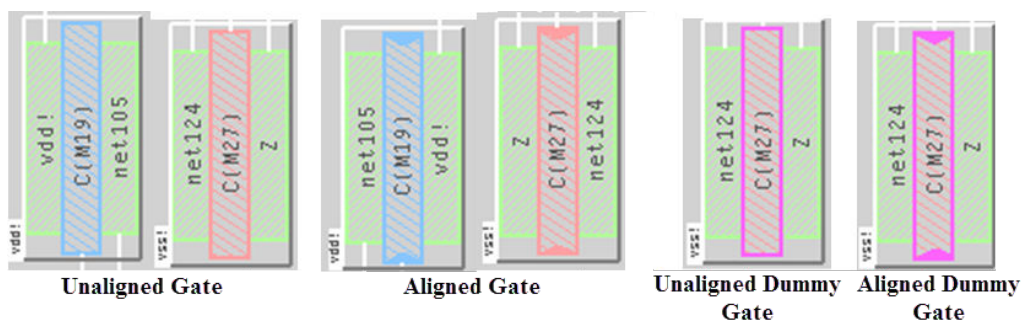
1. Select the **Edit > Align Gates**  or use the shortcut key **A** to invoke the **Align Gates** command.
2. Left-click to select a gate.
3. Next, left-click the reference gate to complete the aligning task.



Only one gate can be aligned and the OD (oxide definition) shapes cannot be stretched for single packs, finger groups, or stack units when the PODE gates are displayed. The gate alignments on the same pack are broken if the OD shape must be stretched. See [Chaining Tab \(Chaining Mode Only\)](#) in the **Symbolic Design Options** dialog box for details.

Unaligning Gates

The **Edit > Unalign Gates** command breaks the alignment relationship in the selected gates of MOS devices. To remove the gate alignments on the selected objects, choose **Edit > Unalign Gates** .



Folding Gates

You can expand a wide MOS device into several small MOS devices using the **Edit > Fold Gates** command.

Note:

If the gate width cannot be evenly divided by the manufacturing grid resolution, the **Fold** operation fails.

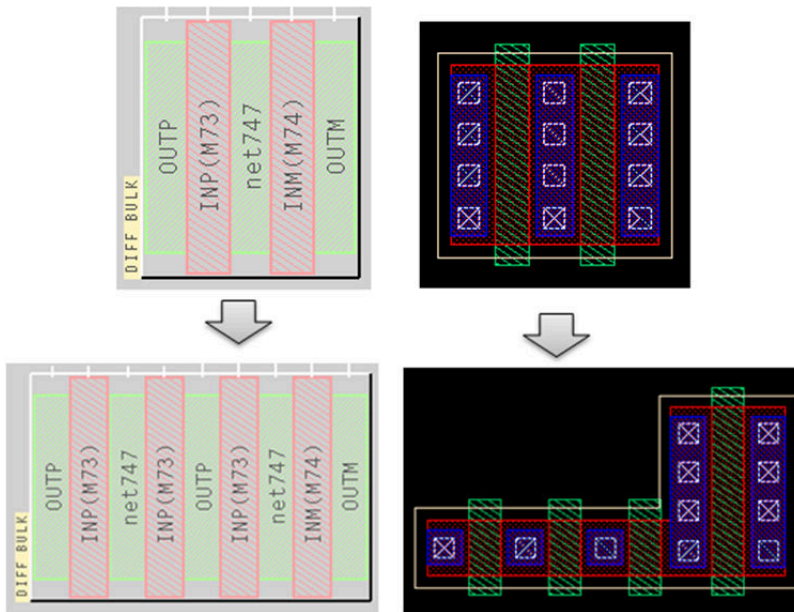
To see more options, choose **Edit > Fold Gates**  to open the dialog box.

Fold Gates ☒ Segment ☐ Same Gate Width ☐ Allow Discrepancy ☐ Threshold

- **Segment:** Specify the number of segments into which you want to fold the gate. The fin number is divided by this segment number. If the number of fins for the resulting gate is smaller than the `lxMinFin` defined in `lxComponentType`, the gate cannot be folded.

For example, after selecting M73, the gate width 3.45 is divided by 3 (segment number = 3) and become 1.15. The resulting gate width is 1.15 and the gate is folded to 3 gates. The additional gates are added to the right side.

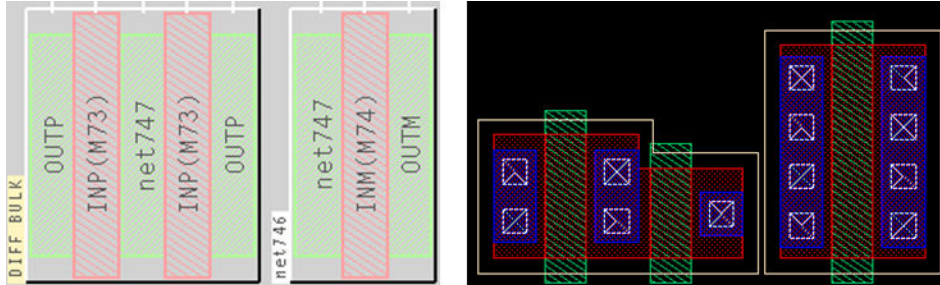
Figure 37 Fold Gate Example (Segment = 3)



Support the following options in the **Segment** mode:

- **Same Gate Width:** Enabling this option folds gates with the same gate width. If the gate width is not a multiple of the segment number, the selected gate cannot be folded (unless **Allow Discrepancy** is checked; see below). The default is *off*.
- **Allow Discrepancy:** Enabling this option allows a discrepancy for folding gates with the same gate width. If the gate width is not a multiple of the segment number, the selected gate is folded with the same gate width. Select the **Same Gate Width** option to enable this option. The default is *off*.
- **Threshold:** Specify the threshold value of the gate width or fin number. You can fold the selected gates into the width indicated in the text widget. If the threshold value or the fin number of the remaining gate is smaller than the `lxMinWidth` or `lxMinFin` defined in `lxComponentType`, the gate cannot be folded.

For example, after selecting M73, the gate width 3.45um is folded by 2um (Threshold = 2um). The first resulting gate width becomes 2um and the second gate width becomes 1.45um. The additional gates are added to the right side of the original gate.

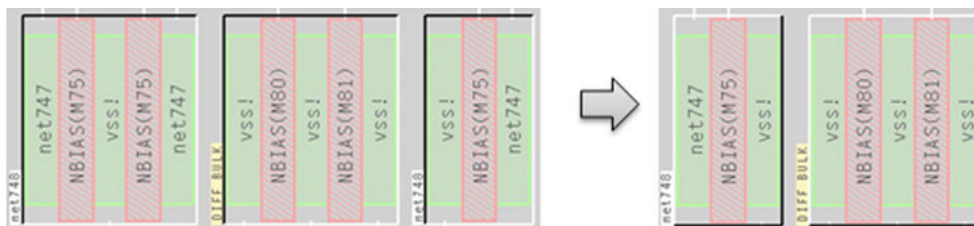


Merging Gates

By sharing the connected source/drain portions of any MOS device on the same net, the **Merge Gates** command combines the selected gates in the same MOS device. If the resulting gate width is larger than `1xMinWidth` or `1xMinFin` defined in `1xComponentType`, the gates can not be merged.

- You can preselect the merged gates or you can invoke the command first.
- When you choose **Edit > Merge Gates**  to invoke the command, you can select a single gate first. The gates with the same gate name as the selected gate become selectable. You can select a second gate, and then the first selected gate is merged to the second selected gate.

Area select is supported. You can select multiple gates. The gates with same gate name are merged.



Mirroring Objects

The **Edit > Mirror X**  command makes a vertical mirrored reflection of the selected pattern. This command is available only for instances, selected gates, finger groups, or stack groups.

The **Edit > Mirror Y**  and **Rotate 90**  commands are available only for instances. See [Mirroring and Rotating Instances](#).

Editing Source/Drain

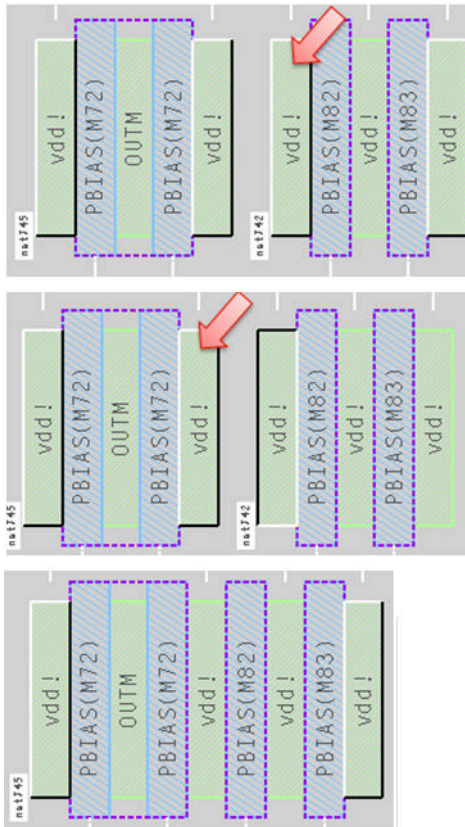
This section covers the following topics:

- [Merging Source and Drain](#)
- [Splitting Source and Drain](#)
- [Swapping Source and Drain](#)

Merging Source and Drain

This command merges SD (source/drain) in the selected packs or gates when choosing **Edit > MergeSD** or click the **Merge SD** button .

After you invoke the **Merge SD** command and then click the first source/drain, all the selectable source/drains are highlighted. When the second source/drain is selected, the two packs are merged.



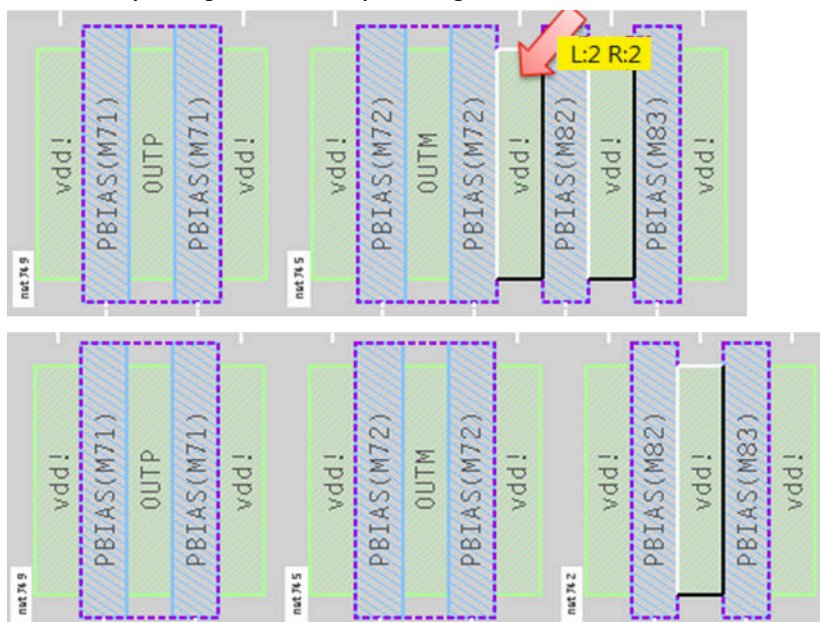
Splitting Source and Drain

This command separates the pack of MOS devices into two parts by simply clicking on the pack. To split the pack of MOS devices, choose **Edit > SplitSD** or click the **Split SD** button



. When moving the cursor to the selectable pack, the result of the gate count for left of pack and right of pack is shown in a yellow tip.

Figure 38 Splitting the Corresponding Pack Into Two Packs



Swapping Source and Drain

The **Edit > SwapSD** command swaps the source and drain of all of the selected MOS devices.

To switch the nodes between MOS device packs, follow the steps below:

1. Select the **Edit > SwapSD** or click the **Swap SD** button  or use the shortcut key S to invoke the **SwapSD** command.
2. Once the MOS devices are selected, this command swaps device sources and device drains immediately without changing the gate order.

Modifying PODE Properties (Chaining Mode Only)

You can specify and modify net names for PODE gates. To assign net for PODE gates, you need to define the terminal name of any PODE gates in the `lxTermMap` Tcl preference. See About `lxComponentType` for details.

To specify net names for PODE gates:



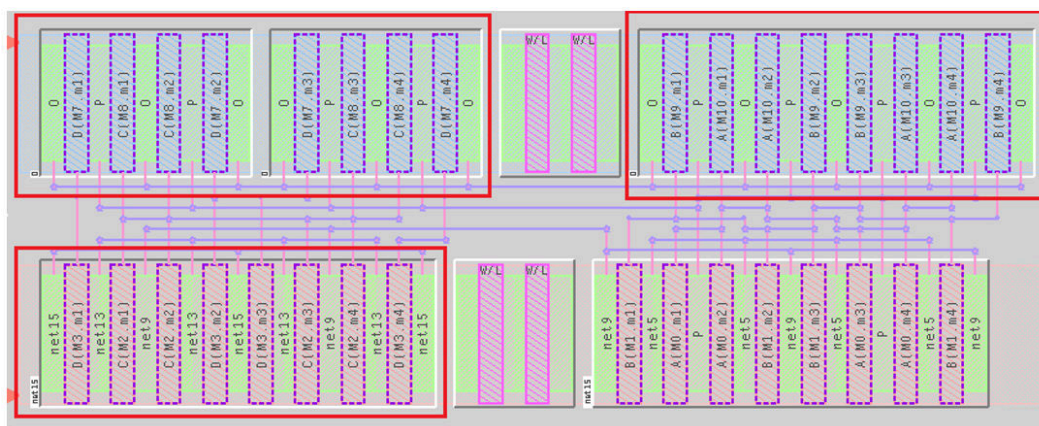
1. Select the **PODE Gates** option in the **Chaining Tab (Chaining Mode Only)** tab from the **Symbolic Design Options** dialog box (**Options > Design**) to display the PODE gates.
2. Select the **Edit > Modify PODE Properties** command and specify a net name in the **Net** text field.
3. Next, select the PODE gates and then click the button to apply the net names.

Swapping Interdigitate

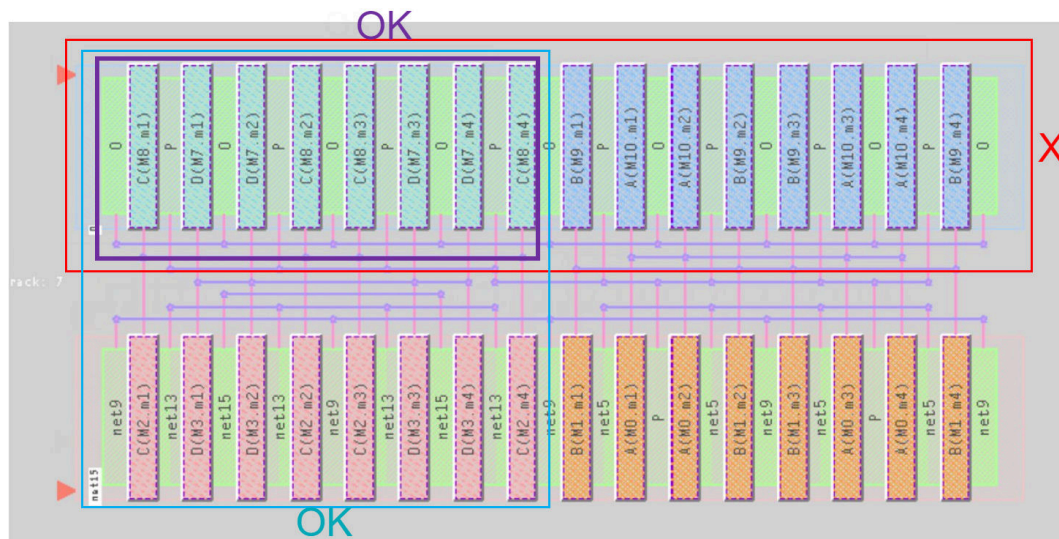
The **Edit > Swap Interdigitate** command swaps the selected pair devices. This command allows users to quickly modify the interdigitated placement to get better connection. When the devices placement is ABBA, the placement is changed to BAAB after applying this command.

To use this command:

- This command is only available for MOS devices.
- This command works in each continuous selection set. It validates continuous selection sets and swaps interdigitated devices for each set separately. If the selected objects are in the same row and not separated by other objects, the devices are in the same continuous selection set. The following figure shows three different continuous selection sets:

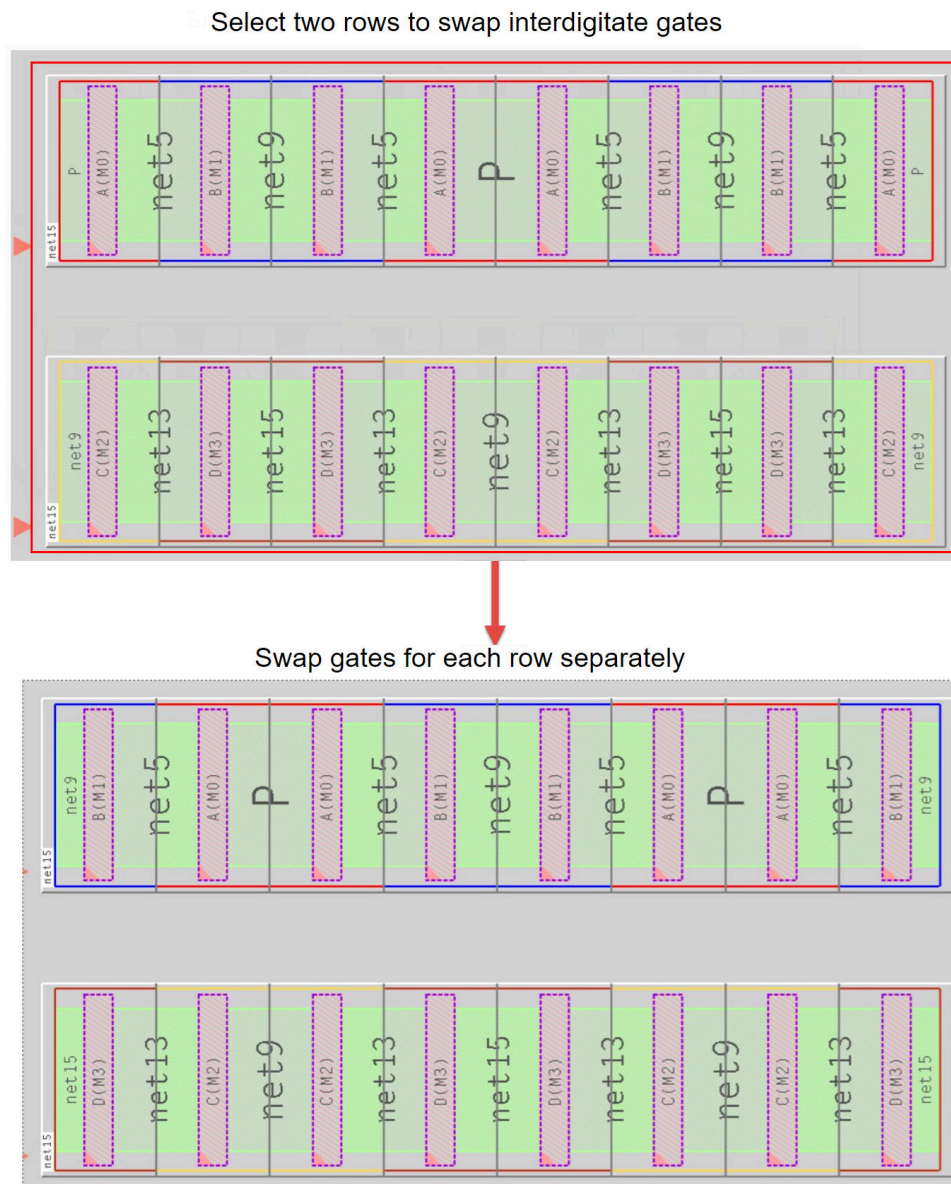


The following figure shows the validation for each continuous selection set.



- In each continuous selection set, the selected tiles should contain just two devices. For example, {M0 M1} is valid and {M0 M1 M2} is invalid.
- In each continuous selection set, the count of selected two devices should be the same. For example, {M0 M1 M0 M1} is valid and {M0 M1 M0} is invalid.

- The process breaks gate alignment when swapping pair devices.

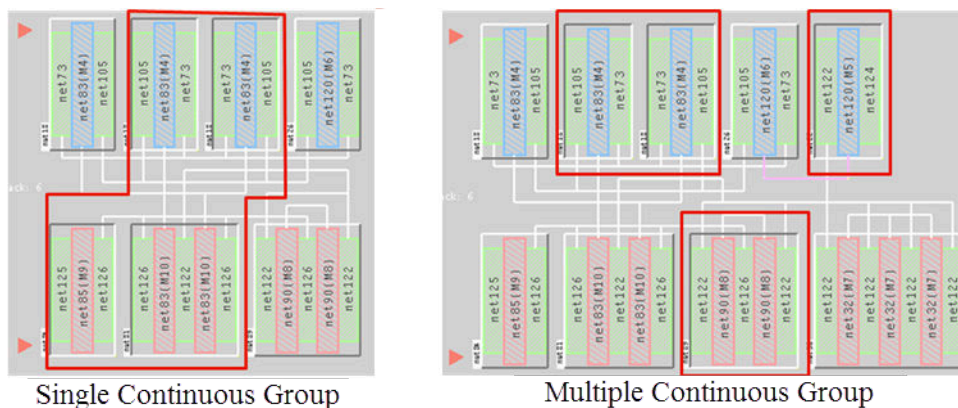


Rechaining MOS Devices (Chaining Mode Only)

The **Edit > Rechain** command rearranges selected MOS devices to return them to their optimal chaining status. This command only supports a single continuous group.

Chapter 6: Editing MOS Devices

Rechaining MOS Devices (Chaining Mode Only)



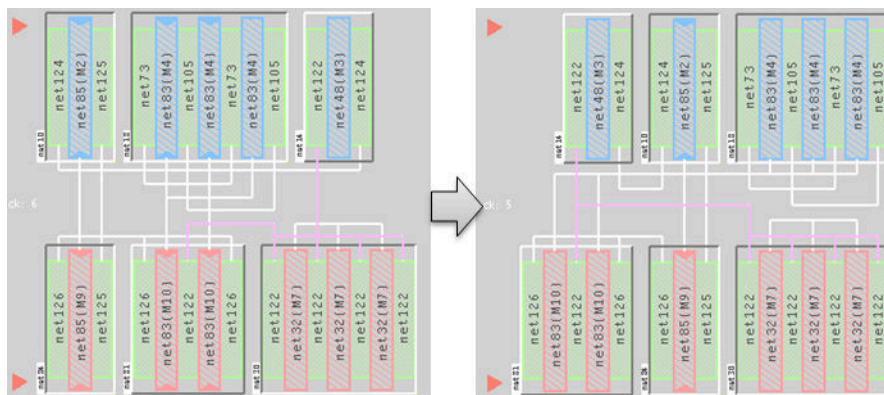
To see more options, choose **Edit > Rechain** or click the **Rechain** button  to open the dialog box.

Rechain Append: ☒ right ☐ left 

- **Append:** You can select **right** or **left** option to place the rechain devices.

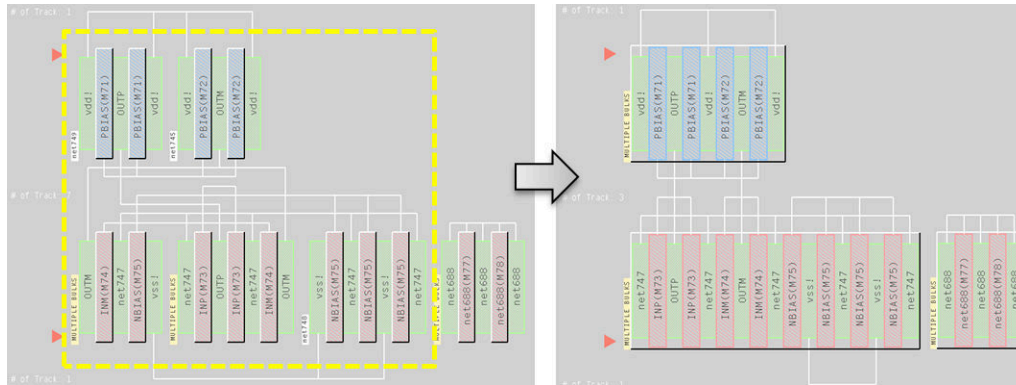
You can preselect the objects before invoking the **Rechain** command. Rechain devices immediately after you invoke **Rechain** command.

Figure 39 Rechain After Selection (Append to right)



You can invoke the **Rechain** command and then specify the **Append** option to **left**. The command rechains the objects after selecting the objects by area selection.

Figure 40 Rechain Before Selection (Append to left)



Optimizing OD (Matching Mode Only)

The **Edit > Optimize OD** command optimizes OD (oxide definition) sharing for selected gates, finger groups, or stack groups to quickly arrange device abutment without manually swapping the source and drain.

Unsetting Finger Group

This command is available only when the selected devices have a finger-factor. It unsets the finger group for the selected set. If the selected set does not have any finger groups, no action is taken.

You can select the objects and then invoke the command by choosing **Edit > Unset Finger Group** or click the **Unset Finger Group** button  to unset the selected packs.



Customizing Spacing (Matching Mode Only)

You can use **Edit > Spacing** to specify and clear the customized spacing between MOS devices.

The **Edit > Spacing** command covers the following sections:

- [Customize Spacing Command](#)
- [Reset Customized Spacing Command](#)

These command are only available for matching mode and MOS devices.

Customize Spacing Command

This command specifies customized horizontal spacing between two instances. Different spacings can be specified for the devices on particular columns. The global intra-row spacing (See [Figure 8](#)) is overridden if the customized spacing is specified.

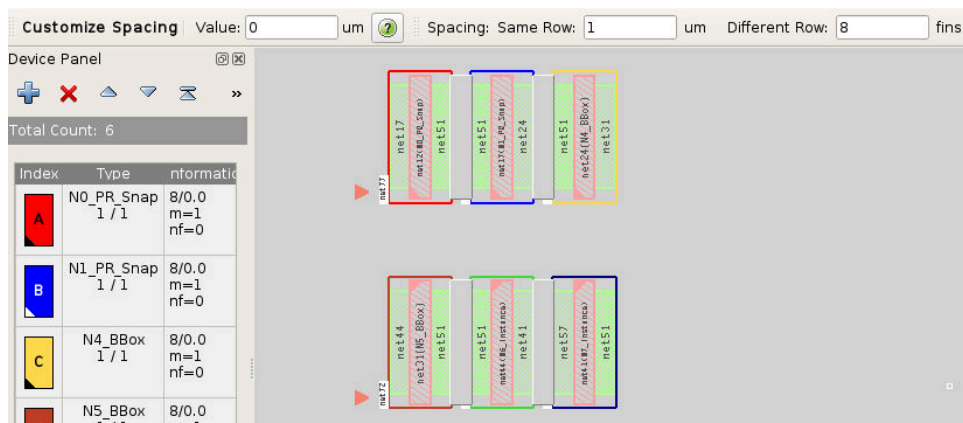
To customizing the spacing between two MOS tiles:

1. Click the **Edit > Spacing > Customize Spacing**  command to enable the toolbar.

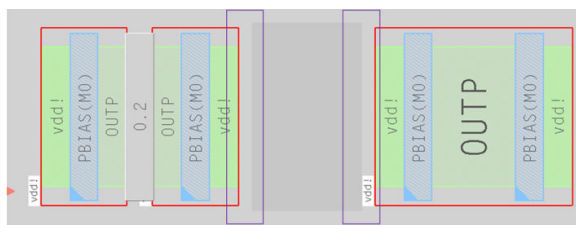


2. The separator relation is selectable if it is between two splitter MOS tiles. If the separator relation has customized spacing, the customized spacing is displayed on the separator relation when the command is activated.

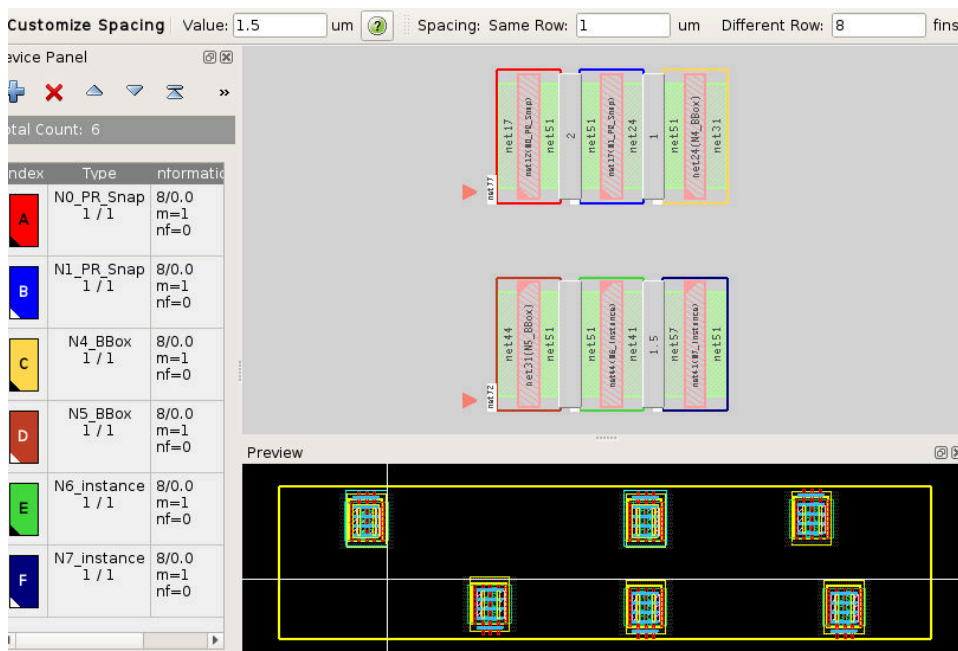
Chapter 6: Editing MOS Devices Customizing Spacing (Matching Mode Only)



The separator relation is not selectable if it is nearby the empty tile.



- Specify the custom spacing on the toolbar, and select the separator relation to commit it.



Reset Customized Spacing Command

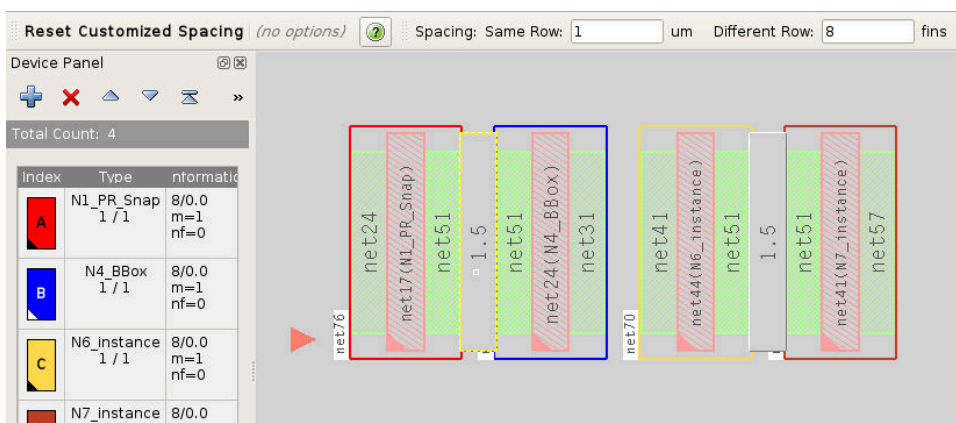
This command clears customized spacing from the selected separator relation. The customized spacing can also be cleared from all separators if the global intra-row spacing is updated.

To clear the customized spacing between two MOS tiles:

1. Click the **Edit > Spacing > Reset Customized Spacing**  command to enable the toolbar.



2. Select the separator relation to clear the customized spacing.



To clear all customized spacing, you can update the global intra-row spacing.

Editing Individual Alignment (Matching Mode Only)

You can use **Edit > Individual Alignment** to add and clear the individual alignment for the MOS tile, finger group, or stack group.

This **Edit > Individual Alignment** command covers the following sections:

- [Align Individual Devices Command](#)
- [Reset Individual Alignment Command](#)

These command are only available for matching mode and MOS devices.

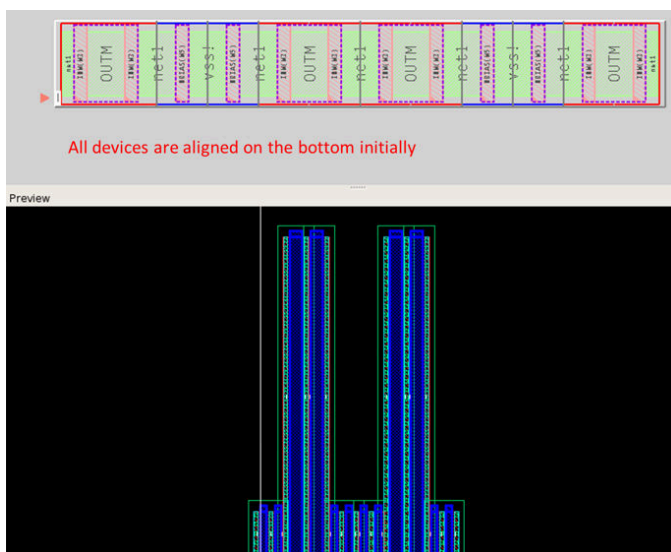
Align Individual Devices Command

This command sets the individual alignment for the MOS tile, finger group, or stack group. If the individual alignment is specified, the row alignment is overridden.

The row alignment is a common setting for all devices in this row, and the individual alignment is a local setting for the specific device. Both of them are applied based on the biggest device in this row. When applying individual alignment and row alignment, the biggest device is found in the row to decide the row height. The device can be aligned on the top or bottom base on the setting.

To set the individual alignment:

1. All devices are aligned on the bottom initially. See [Aligning Rows](#) for row alignment.



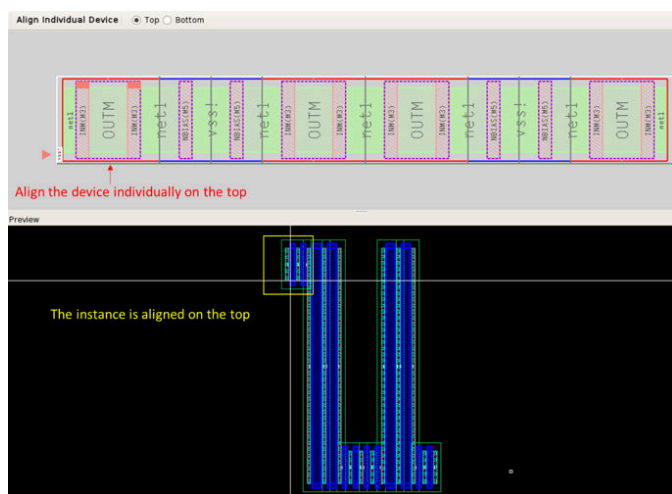
2. Click the **Edit > Individual Alignment > Align Individual Device**  command to enable the toolbar.



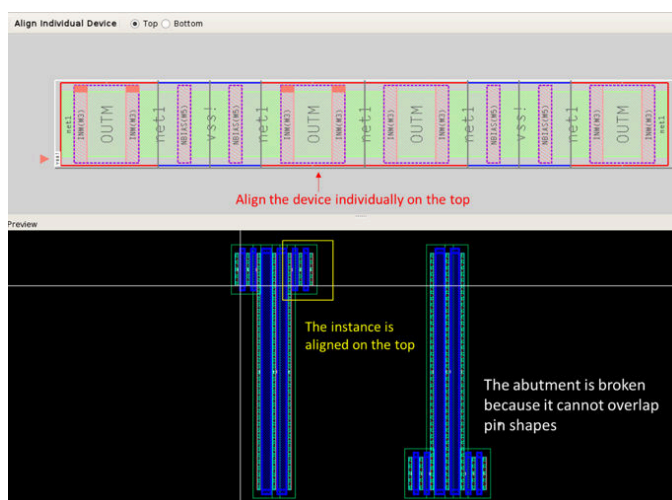
3. Select **Top** and click the left-most device to align individually on the top. When the individual alignment is specified, the row alignment is overridden.

Chapter 6: Editing MOS Devices

Editing Individual Alignment (Matching Mode Only)



4. Click the middle device to align individually on the top. The abutment is broken because the pin shapes are not overlapped.



Reset Individual Alignment Command

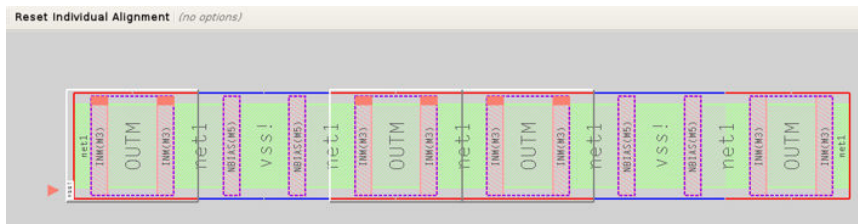
This command clears the individual alignment for the selected MOS tile, finger group, or stack group. The individual alignment also can be cleared if the row alignment is updated.

To clear the individual alignment:

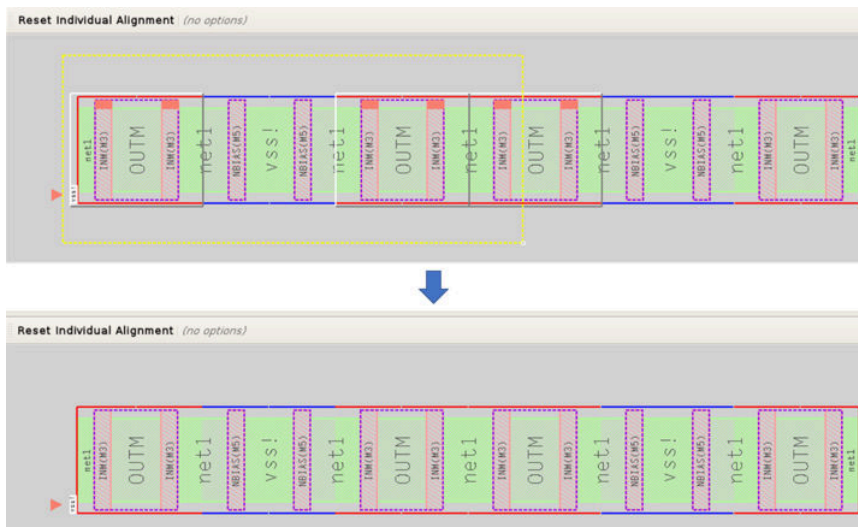
1. Click the **Edit > Individual Alignment > Reset Individual Alignment**  command to enable the toolbar.



The devices with individual alignment are selectable.



2. Area-select the devices to clear the individual alignment.



7

Creating and Editing Instances and Resistors

This chapter explains how to edit instances and resistors in the symbolic editor.

This chapter covers the following topics:

- [Mirroring and Rotating Instances](#)
- [Editing Resistors](#)

Mirroring and Rotating Instances

The **Edit > Mirror Y**  command makes a horizontal mirrored reflection of the selected pattern. This command is available only for instances.

The **Edit > Rotate 90**  command rotates the selected instance-tiles in 90-degree orientation. This command is available only for instances.

For Mirror X, See [Mirroring Objects](#).

Editing Resistors

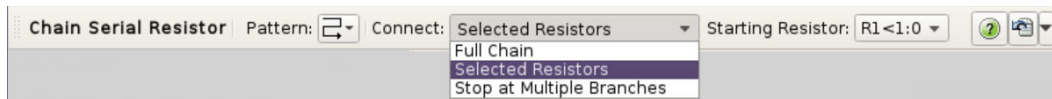
This topic covers the following sections:

- [Chaining Serial Resistor Devices](#)
- [Reassigning Nets](#)
- [Swapping Terms](#)

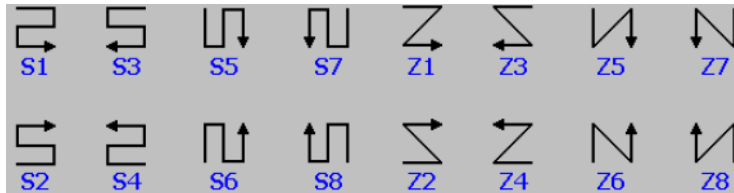
Chaining Serial Resistor Devices

The **Edit > Chain Serial Resistor** command can automatically place serial connected resistor devices in a certain sequence and make the resistor chain use the minimum routing resource. The start point can be specified by selecting a grid box in the matching canvas pane before placement. This command is only available for resistor devices.

To see more options, choose **Edit > Chain Serial Resistor** or click the  button.



- **Pattern:** There are 16 placement patterns that control the direction of placement. The default is the first pattern S1.



- **Connection:** When this option is turned *on*, this command is stopped when multiple connected devices are found. When this option is turned *off*, the resistors connected to the minus net have higher priority than the plus net.
 - **Full Chain:** When this option is selected, this command chains resistors from the selected resistor until the chain is end. If there is a branch, choose one resistor to continue to chain.
 - **Selected Resistor:** When this option is selected, this command place selected multi-segment resistors in a serial chain pattern. The selected resistors need from one chain. The **Starting Resistor** is the head or tail of the selected resistors.
 - **Stop at Multiple Branches:** When this option is selected, this command is stopped when multiple connected devices are found. When this option is turned off, the resistors connected to the minus net have higher priority than the plus net.

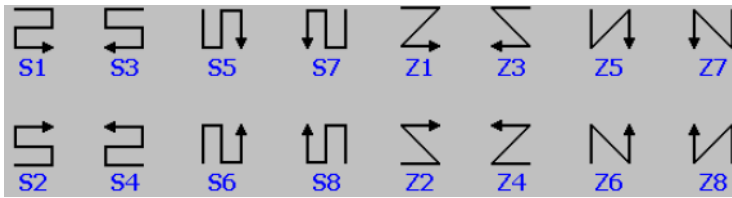
Reassigning Nets

The **Edit > Reassign Net** command reassigns the nets for the placed resistor devices in the canvas. This command is only available for resistor devices.

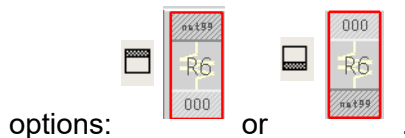
To see more options, choose **Edit > Reassign Net** or click the  button.



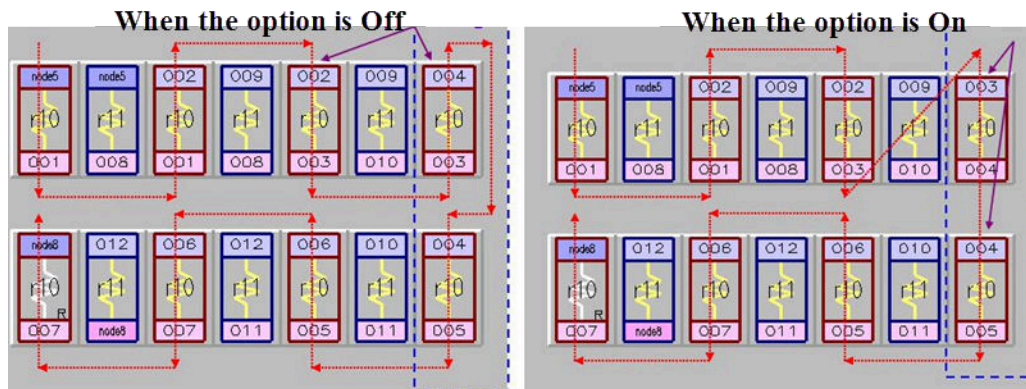
- **Pattern:** There are 16 placement patterns that control the direction of placement. The default is the first pattern S1.



- **Starting Net:** Specify the starting net by selecting one of the following options: **Plus** or **Minus**.
- **Net Terminal:** Specify the starting net pattern by selecting one of the following



- **Shortest Connection Between Rows:** When selected, the last resistor in the row changes its orientation according to the shortest connection between rows.



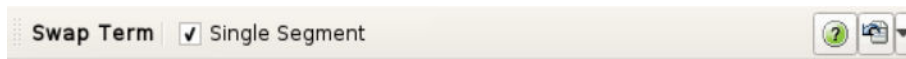
- **Extend Connected Device:** When this option is selected, the following connected resistors are reassigned:

- Column-based pattern:
- Row-based pattern:

Swapping Terms

The **Edit > Swap Terms** command swaps the top and bottom nets of the selected resistor. This command is only available for resistor devices.

To see more options, choose **Edit > Swap Terms** to open the dialog box.



- **Single Segment:** When one resistor is separated into multiple segments (serial connected with internal nets), this option can be used to decide whether to swap terms on one segment or all segments. **Swap Terms** on all segments swaps the nets on the two external terms (PLUS and MINUS) and keeps all internal nets serially connected. The default is TRUE.

8

Design Centering

This chapter explains Design Centering functionality and capabilities.

This section covers the following topics:

- [Overview](#)
- [Associating Spare Dummy](#)
- [Generating Group Placements](#)

Overview

Design Centering is a process of design refinement to meet various specifications using different methodologies. Symbolic Editor provides capabilities that enables you to plan for the device size changes.

Typically, due to parasitic effects, the design specifications are not met. This requires updates to layout placement and routing for improved parasitics and / or change the device sizes (width / length) to mitigate the parasitic effect and try to meet the specification. After the device sizes are changed in schematic, the corresponding layout also needs to be updated. For example, if a device width is increased by increasing its multiplier factor, then in the layout, additional physical instances need to be created and placed alongside the original physical instances for the device. This change can require a lot of manual edits, some with Symbolic Editor for the placement of the device, and more with Layout Editor to push and shove other layout objects in the neighborhood of the changed device. Furthermore, this needs to be repeated multiple times in a design optimization loop, in which a single iteration involves device size change, layout update, post-layout layout simulation, analyses and decision on next device size change. If you anticipate sizes (width / length) of some devices to change during Design Centering, then you can create extra physical space upfront for these devices.

Symbolic Editor lets you create extra physical instances as dummy devices, and associate these with one or more devices. Furthermore, using Symbolic Editor, you can create a list of possible placements for a given set of devices by varying their sizes. These can be stored and used later in the design centering flow. Symbolic Editor also provides commands to replace the current placement of devices with the one corresponding

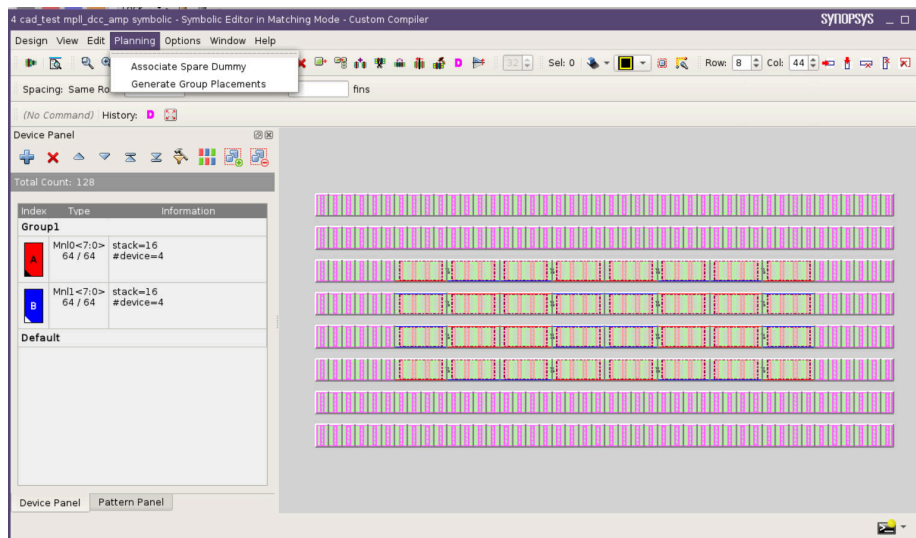
to updated device values in the loop. Design centering capabilities are available under **Planning** menu in the SED main window and with the environment variable `SYNOPTSYS_FEATURE_DESIGN_MIGRATION`.

Associating Spare Dummy

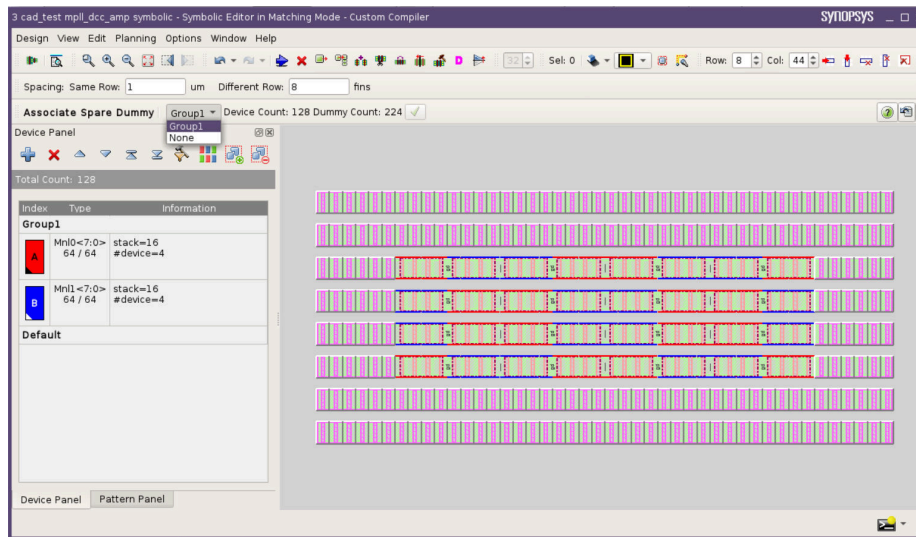
The device changes in the schematic after being introduced into the layout can cause the overall device footprint to change. Therefore, layout update requires a lot of manual effort, and consequently the Design Centering Optimization loop is also expected to have manual intervention. One solution to reduce the manual effort is to add dummies for a device and associate these dummies to the active device. After this is done, whenever the device size (count) increases, the available associated dummies can be converted to active devices; and whenever the device count decreases, some of the active devices can be converted to dummies. This ensures that overall footprint of devices plus the associated dummies is not changed. The act of converting dummy to active devices (or vice versa) is essentially a routing change, which changes connections on a dummy associated to VDD/VSS to nets incident on active devices.

To associate a spare dummy to device groups:

1. In the **Symbolic Editor**, choose **Planning > Associate Spare Dummy**.



2. Choose **Group1** from the **Associate Spare Dummy** box and select **Dummy Count**. Click **Apply**.

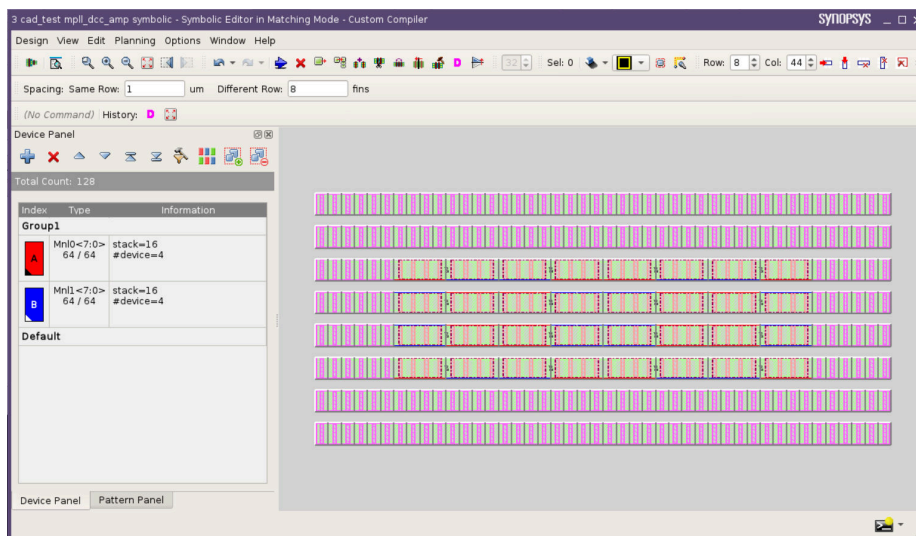


All the newly added dummies are now associated with **Group 1**.

3. Select **None** from the **Associate Spare Dummy** box if you do not want to associate the dummies to any group.

Here is an example of a differential pair, where two devices are implemented with 64 counts each, and there 224 dummies added in the neighborhood.

If you select all 224 dummies and **Group1**, then all the 224 dummies are associated with **Group1** exclusively to be potentially used either as A or B device.



Dummies associated with a group are cross highlighted whenever the device group is selected from the **Device Panel**. If cross selection is enabled, then the devices and

dummies are also cross selected on the symbolic canvas. If you want to select just the active devices in the group, you should select all devices under the group from the **Device Panel**. If you select the group name, the tool selects all devices and associated dummies, but if you select a device, the tool selects only the active devices. If cross selection is enabled, then if you click a group, the tool selects all its devices and associated dummies. You can also move them around on the symbolic canvas. This allows you to move groups around for better floor planning.

In a Symbolic Editor session with multiple groups, you can associate dummies to any group. Typically, spare dummies are added to create capacity. When the size of active devices change in a group, the dummies in physical proximity are associated to the group. In this way, the active devices and the associated dummies with that group are all placed within a rectangular area on the symbolic canvas. In a well-organized Symbolic Editor placement, different groups (devices plus associated spare dummies) occupy different rectangular regions in symbolic canvas. You can select already associated spare dummies and change its association to some other group, or with no group.

Note:

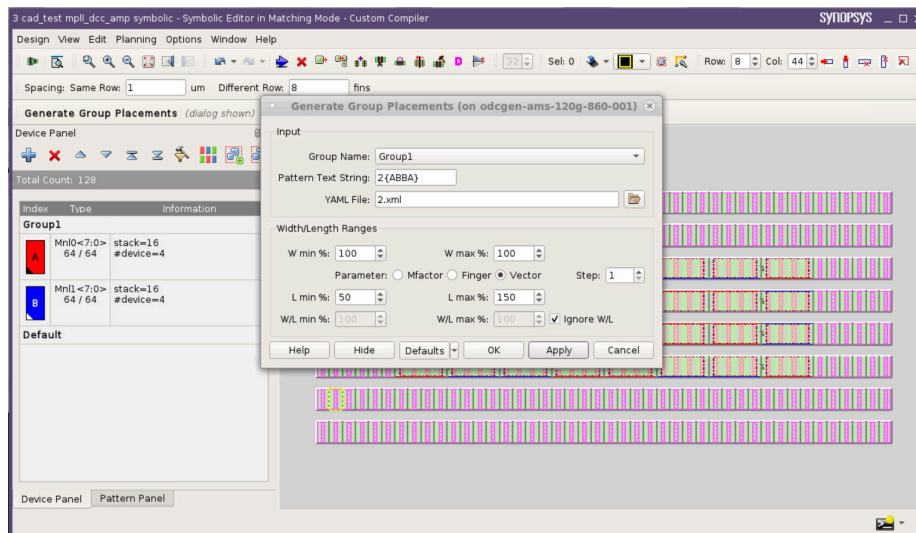
Association of dummies is always with the group. If the devices under the group are rearranged across different groups, then the dummies continue to remain associated with the same group.

Generating Group Placements

You can explore various device sizes and the corresponding placement possibilities after adding dummies and associating them with a Symbolic Editor group. You can also find out about different device sizes (total transistor width and length) that are possible for the active devices in the group, given the associated spare dummies with that group. It is assumed that the active devices and their associated spare dummies occupy a fully populated rectangular $m \times n$ tiles in the symbolic canvas.

To generate placements for groups:

1. In the **Symbolic Editor**, choose **Planning > Generate Group Placements**.

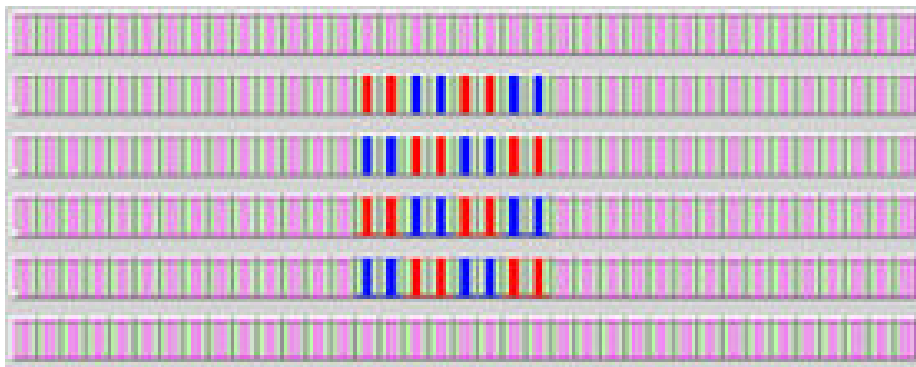


2. In the **Input** region, specify **Group Name**, **Pattern Text String**, and **YAML File** as input values for the placement.
3. In the **Width/Length Ranges** region, specify the width range **W min % W max %** for the active devices.
4. Select one of the **Parameter - Mfactor, Finger, or Vector** to modify the width of the device.
5. Specify the length range **L min % L max %**.
6. Specify the width/length ratio range **W/L min % W/L max %**.

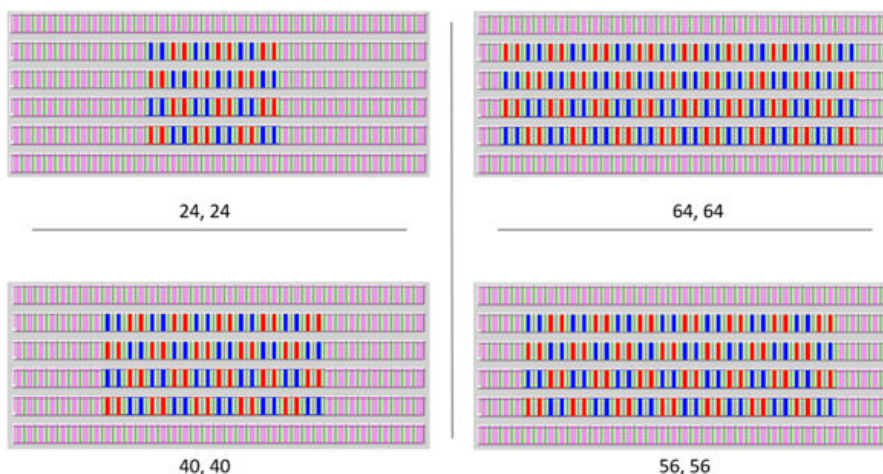
With these inputs and a Symbolic Editor group with an associated spare dummy, the tool determines what values of W and L are possible to be implemented. It writes out all the size options and placements to each option to a user-specified `yaml` file. If the file does not exist, then Symbolic Editor creates a new one. If such a file exists, then the Symbolic Editor appends a new `optVar` description into it, which contains device sizes and placement options for the current group. If an existing `yaml` file contains the same `optVar` description, then the Symbolic Editor overwrites that with the new size and placement options. A single `yaml` file can have group placements for different SED groups across different `instMatching` constraints. This `yaml` file can be specified to the Design Centering / Optimization tool so that it only explores these W and L values, as these are the ones that can be implemented in the layout while retaining the footprint of the device group.

Here is an example of a placement for a two device SED group, implementing a differential pair.

The SED group placement occupies 16 tiles for both the active devices and 190 tiles for the associated dummies with this group. You can increase the width of both the active devices, replacing some of the associated dummies with active devices and ensuring that the overall placement of the SED group still occupies the same sized rectangular area.



Here are the placement options with both devices occupying 24, 40, 56, and 64 tiles each. In all cases, the total area of the SED group (devices plus associated spare dummies) remains the same.



The `ilx_generateGroupPlacements` is an interactive command using which you can specify the W and L ranges of all devices in a group. The tool calculates what values of W and L are possible with the available spare dummies and these details are made available in a user specified `yaml` file.

Replacing Group Placements

You can change the group placements after the options of the device group are generated. These are stored in a `yaml` catalog file, which needs to be specified in the tool and the placement can be changed to one of those options, as needed.

The command `lx::replacePlacement` is used to replace the placement options.

```
lx::replacePlacement [-window <int>] [-verbosity <int>] -catalogFilename  
<string> -optVarName <string> -placementIndex <int>
```

The command reads the specified `yaml` catalog file of an SED session running in a specified window, and then replaces the current placement of an optimization variable (same as SED Group) with the one specified at an index position in the catalog file. This is essentially a placement swap in layout after the device sizes are changed in the schematic. This command is used in Design Centering optimization scripts that explore the design space by determining the device sizes to use next, updating the schematic with those parameters, changing the corresponding layout placements, and finally running post layout simulations.

The preference `lxSEDIgnoreModifiedDevicePlacement` is used to replace the placement options.

Note:

You need to set the preference before launching the **Symbolic Editor**.

A

Shortcut Keys Summary

This chapter describes the available shortcut keys when using the Symbolic Editor with Chaining Mode (default). For more information about customizing the binding set, see Customizing Keyboard Shortcuts in the Custom Compiler Environment User Guide.

This topic covers the following sections:

- [Symbolic Editor With Chaining Mode Keypad Shortcut Keys](#)
- [Symbolic Editor With Matching Mode Keypad Shortcut Keys](#)
- [Symbolic Editor Function Shortcut Keys](#)
- [Symbolic Editor Number Pad Shortcut Keys](#)

Symbolic Editor With Chaining Mode Keypad Shortcut Keys

The following table describes the available main keypad shortcut keys when using the Symbolic Editor with Chaining Mode (default).

Table 4 *Symbolic Editor With Chaining Mode Keypad Shortcut Keys*

Shortcut Key	Description
[	Zoom In
]	Zoom Out
A	Align specific gates
Alt+1	Show all connections of flight lines
Alt+2	Show gate to gate flight lines
Alt+3	Show S/D to S/D flight lines
Alt+4	Show all flight lines connected to gates
Alt+D	Delete tracks

Table 4 *Symbolic Editor With Chaining Mode Keypad Shortcut Keys (Continued)*

Shortcut Key	Description
Alt+I	Insert tracks
Alt+M	Move flight lines
C	Rechain specific objects
Ctrl+A	Select All
Ctrl+D	Deselect All
Ctrl+M	Merge S/D for the selected packs or gates
Ctrl+P	Preview Assistant
Ctrl+R	Redraw the canvas
Ctrl+S	Split the abutted objects
Ctrl+Space	Focus Console window
Ctrl+T	Zoom to fit selected objects
Ctrl+W	Close the design
Ctrl+Z	Zoom In
D	Measure distance
Delete	Delete selected objects
E	Open the Symbolic Design Options dialog box
Esc	Abort active command
F	Zoom to fit the whole design
H	Highlight flight lines
I	Swap Interdigitate for selected pair devices
M	Move selected objects to the specified position
Q	Modify flight line properties
S	Swap source/drain position for the specific objects
Shift+A	Unalign gates in the selected objects
Shift+K	Clear ruler

Table 4 *Symbolic Editor With Chaining Mode Keypad Shortcut Keys (Continued)*

Shortcut Key	Description
Shift+M	Merge selected gates
Shift+O	Fold selected gates
Shift +P	Add Peripheral Dummies
Shift+U	Redo
Shift+W	Next View
Shift+Z	Zoom Out
U	Undo
W	Previous View
Z	Set Viewport

Symbolic Editor With Matching Mode Keypad Shortcut Keys

The following table describes the available main keypad shortcut keys when using the Symbolic Editor with Matching Mode (default).

Table 5 *Symbolic Editor With Matching Mode Shortcut Keys*

Shortcut Key	Description
[	Zoom In
]	Zoom Out
A	Add Device
Alt+A	Reset customized spacing
Alt+N	Reset individual alignment
C	Copy specific objects
Ctrl+A	Select All
Ctrl+D	Deselect All
Ctrl+M	Merge S/D (source/drain) for the selected packs or gates
Ctrl+P	Preview Assistant

Table 5 *Symbolic Editor With Matching Mode Shortcut Keys (Continued)*

Shortcut Key	Description
Ctrl+R	Redraw the canvas
Ctrl+S	Split the abutted objects
Ctrl+Space	Focus Console window
Ctrl+T	Zoom to fit selected objects
Ctrl+W	Close the design
Ctrl+Z	Zoom In
D	Measure distance
Delete	Delete selected objects
E	Open the Symbolic Design Options dialog box
Esc	Abort active command
F	Zoom to fit the whole design
G	Generate pattern
I	Swap Interdigitate for selected pair devices
L	Calculate grid size
M	Move selected objects to the specified position
O	Optimize OD sharing
Q	Modify dummy properties
S	Swap source/drain position for the specific objects
Shift+C	Customize spacing
Shift+C	Clear ruler
Shift+N	Align individual device
Shift+U	Redo
Shift+W	Next View
Shift+Z	Zoom Out
U	Undo

Table 5 Symbolic Editor With Matching Mode Shortcut Keys (Continued)

Shortcut Key	Description
W	Previous View
Z	Set Viewport

Symbolic Editor Function Shortcut Keys

The following table describes the available function shortcut keys when using the Symbolic Editor (default).

Table 6 Symbolic Editor Function Shortcut Keys

Shortcut Key	Description
F1	Help
F6	Redraw the canvas
F8	Clear Highlight
F10	Open the Display Resource Editor dialog box
F11	Open the General Options dialog box

Symbolic Editor Number Pad Shortcut Keys

The following table describes the available number pad (numpad) shortcut keys when using the Symbolic Editor.

Table 7 Symbolic Editor Numpad Shortcut Keys

Shortcut Key	Description
1	Pan downward to the left
2	Pan down
3	Pan downward to the right
4	Pan left
5	Move the viewport to the cursor location and center it
6	Pan right

Table 7 Symbolic Editor Numpad Shortcut Keys (Continued)

Shortcut Key	Description
7	Pan upward and to the left
8	Pan up
9	Pan upward and to the right